Microbiology and Infectious Disease

A Novel Rapid Host Cell Entry Pathway Determines Intracellular Fate of *Staphylococcus aureus*

Reviewed Preprint
v2 • July 8, 2025
Revised by authors

Reviewed Preprint
v1 • November 29, 2024

Marcel Rühling, Fabio Schmelz, Kim Ulbrich, Fabian Schumacher, Julia Wolf, Maximilian Pfefferle, Magdalena Priester, Adriana Moldovan, Nadine Knoch, Andreas Iwanowitsch, Christian Kappe, Kerstin Paprotka, Burkhard Kleuser, Christoph Arenz, Martin J Fraunholz

Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany • Institute of Pharmacy, Freie Universität Berlin, Berlin, Germany • Institut for Chemistry, Humboldt Universität zu Berlin, Berlin, Germany

eLife Assessment

This **valuable** work proposes a novel, rapid S. aureus entry mechanism via Ca^{2+}-dependent lysosomal exocytosis and acid sphingomyelinase release, which influences bacterial sub-cellular fate. However, reliance on chemical inhibitors and the absence of a knockout phenotype weakens the overall impact, making the study **incomplete**.

https://doi.org/10.7554/eLife.102810.2.sa2

Abstract

Staphylococcus aureus is an opportunistic pathogen causing severe diseases. Recently, *S. aureus* was recognized as intracellular pathogen, whereby the intracellular niche promotes immune evasion and antibiotic resistance. Interaction of *S. aureus* with versatile host cell receptors was described previously, suggesting that internalization of the pathogen can occur via several pathways. It remains elusive whether the pathway of internalization can affect the intracellular fate of the bacteria. Here, we identified a mechanism governing cellular uptake of *S. aureus* which relies on lysosomal Ca^{2+}, lysosomal exocytosis and occurs concurrently to other well-known entry pathways within the same host cell population. This internalization pathway is rapid and active within only few minutes after bacterial contact with host cells. Compared to slow bacterial internalization, the rapid pathway demonstrates altered phagosomal maturation as well as translocation of the pathogen to the host cytosol and ultimately results in different rates of intracellular bacterial replication and host cell death. We show that these alternative infection outcomes are caused by the mode of bacterial uptake.

Introduction

Sphingolipids are important components of eukaryotic membranes and also serve as bioactive signaling molecules (1). The most abundant sphingolipid, sphingomyelin (SM), resides in the extracellular leaflet of the plasma membrane and hence is exposed to the environment (2, 3). SM is a substrate for sphingomyelinases (SMases) that cleave off the phosphocholine head group to produce ceramide. Acid sphingomyelinase (ASM) is a lysosomal enzyme involved in recycling of SM (4, 5). Several studies demonstrated that ASM can be released by human cells either by Ca^{2+}-dependent liberation of lysosomal content via so called lysosomal exocytosis (6-8) or by channeling the enzyme through the Golgi secretory pathway (9, 10). Extracellular ASM is active on the plasma membrane, where it has important roles during repair of membrane wounds (6, 11) but has also been shown to act on the SM-containing low-density lipoprotein (12). The release of ASM through lysosomal exocytosis mediates host cell entry of several human pathogens (7, 8, 13-17).

Staphylococcus aureus is a Gram-positive human commensal that asymptomatically colonizes about one third of the human population (18). It is an opportunistic pathogen causing diseases ranging from soft tissue and skin infections (19) to lethal diseases (20, 21). *S. aureus* is notorious for acquiring antibiotic resistances thereby causing >100,000 annual (22). The pathogen possesses intracellular virulence strategies (23, 24) which contribute to antibiotic resistance (25, 26) and immune evasion (27). Internalization of *S. aureus* by host cells is mediated by several adhesins on the staphylococcal surface that bind a plethora of receptors on the host cell plasma membrane (13, 28-38). For instance, staphylococcal fibronectin-binding proteins that form fibronectin bridges to $\alpha_5\beta_1$ integrins on host cells to initiate internalization (28) or clumping factor B which can interact with the host receptor annexin A2 (31).

After internalization, *S. aureus* resides within a phagosome-like compartment that matures by acquiring proteins associated with early and then late endosomes as well as lysosomes (39-41). In epithelial and endothelial cells, the bacteria translocate to the host cytosol ["phagosomal escape", (39, 42, 43)]. Phagosomal escape is mediated by so-called phenol-soluble modulins (42), which comprise a family of helical amphiphilic peptides and are transcriptionally controlled by the accessory gene regulator *agr*, a staphylococcal quorum sensing system (44). *S. aureus* replicates in the host cytosol and cause host cell death for example by expression of a cysteine protease. (45).

The mechanism of pathogen entry into host cells has been suggested to dictate the outcome of infections (46-49). For instance, phagosomes formed during natural internalization of *Brucella abortus* by macrophages exhibited different characteristics compared to phagosomes that were artificially generated by phorbol myristate acetate (49). The protozoan parasite *Toxoplasma gondii* actively invades its host cells. Phagosomes generated during this active invasion differed from those that were generated during Fc receptor-mediated uptake of antibody-coated parasites (48). Since previous observation connecting host cell entry and intracellular fate of pathogens were based on artificial induction of internalization, it is unclear whether pathogens can enter host cells within the same population via different naturally occurring mechanisms and if the entry route has a direct influence on the outcome of an intracellular infection.

Here, we describe the internalization of *S. aureus* by tissue cells via a rapid pathway taking place within minutes after contact between bacteria and host cell surface. This pathway requires nicotinic acid adenosine dinucleotide phosphate (NAADP)-dependent mobilization of lysosomal

Ca^{2+}, followed by lysosomal exocytosis and thereby release of ASM. Since the rapid uptake is concurrent to previously described internalization pathways, it probably has been missed due to long infection times used in traditional infection protocols [e.g., (28, 31, 35, 36)].

S. aureus bacteria, which enter host cells of the same cell population via the rapid pathway, cause a distinct infection outcome with altered phagosome maturation, bacterial translocation to host cytosol and eventually host cell death when compared to bacteria that enter host cells at later time points during infection. Thus, the outcome of an infection is decided at the single cell level during bacterial uptake at the host plasma membrane. Bacteria-containing phagosomes that were formed by the ASM-dependent rapid uptake pathway are delayed in their maturation and the bacteria do not escape these organelles efficiently, hence demonstrating a direct link between concurrently active modes of bacterial uptake and the resulting outcomes of intracellular *S. aureus* infection.

Results

S. aureus triggers lysosomal Ca^{2+} mobilization for host cell invasion

We previously showed that internalization of *S. aureus* by host cells is associated with cellular Ca^{2+} signaling (50). To validate these findings, we treated human microvascular endothelial cells (HuLEC, **Figure 1, A**) or HeLa (**Supp. Figure 1, A**) with the cell permeant Ca^{2+} chelator BAPTA-AM (51) to interfere with host cell Ca^{2+} signaling and subsequently infected the cells with *S. aureus*. We measured the number of internalized *S. aureus* and determined the invasion efficiency by normalizing bacteria numbers detected in treated samples to untreated controls (set to 100%). Invasion efficiency was reduced by BAPTA-AM in a concentration-dependent manner, indicating an involvement of Ca^{2+} in the host cell entry process of *S. aureus*.

The cytosolic Ca^{2+} levels, which are crucial for signaling processes, can be elevated by Ca^{2+} influx from the extracellular space. To test whether *S. aureus* internalization is mediated by Ca^{2+} influx, we measured invasion efficiency, in absence of Ca^{2+} from the cell culture medium (**Figure 1, A** and **Supp. Figure 1, A**). Internalization of *S. aureus* by host cells was not affected by absence of Ca^{2+} ("Ca^{2+}-free, 0 μM BAPTA-AM"), thereby excluding a Ca^{2+} influx as crucial for *S. aureus* host cell entry. The pore-forming staphylococcal α-toxin was shown to mediate Ca^{2+} influx into host cells (11). *S. aureus* invasion into host cells was largely independent of the production of α-toxin (**Supp. Figure 1, B**), which further supported that the internalization is independent from extracellular Ca^{2+}.

Since intracellular Ca^{2+} stores, such as the ER or lysosomes, can serve as alternative source of Ca^{2+} (52), we next blocked Ca^{2+} liberation either from the ER using 2-APB (53) or 8-Bromo-cyclic ADP ribose (8-Bromo-cADPR; (54)) or from lysosomes using *trans*-Ned19 (55). Interference with lysosomal but not ER Ca^{2+} release reduced internalization of *S. aureus* in HuLEC (**Figure 1, B**) or HeLa (**Supp. Figure 1, C, D**) at concentrations which do not affect bacterial viability (**Supp. Figure 1, E and F**).

Ned19 antagonizes NAADP, a second messenger that mediates opening of two pore channels (TPCs), which transport Ca^{2+} from the endo-lysosomal compartment into the cytosol (55). Hence, a HeLa cell pool depleted of TPC1 (**Supp. Figure 1, G**) showed substantially reduced invasion when compared to wildtype HeLa cells. This was particularly pronounced after a 10 min infection pulse, suggesting that Ca^{2+}-dependent invasion is important early in infection (**Figure 1, C**).

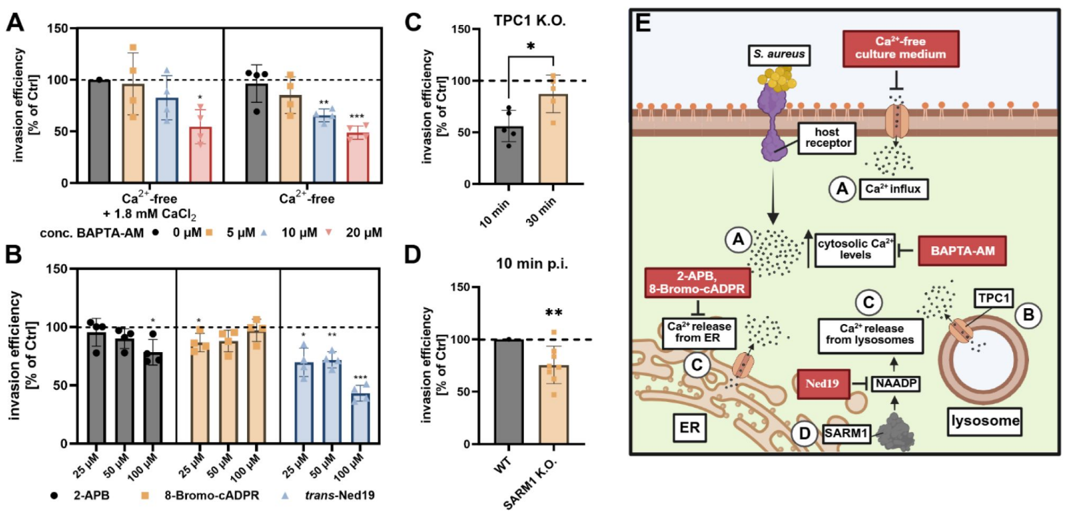


Figure 1

S. aureus invasion is dependent on Ca^{2+} liberation from lysosomal stores.

Treatment with, BAPTA-AM (A, n≥4), *trans*-Ned19 (B, n≥4), but not 2-APB and 8-Bromo-cADPR (B, n≥4) reduce *S. aureus* internalization by host cells. Genetic ablation of TPC1 (C, n=5) and SARM1 (D, n=8) reduced invasion of *S. aureus* 10 min p.i. HuLEC (A, B), HeLa WT (C, D), HeLa TPC1 K.O. (C) or HeLa SARM1 K.O. (D) cells were treated with the respective substance and were subsequently infected with *S. aureus* JE2 for 30 min if not indicated otherwise. Extracellular bacteria were removed by lysostaphin, and the number of intracellular bacteria was determined by CFU counting. Results were normalized to untreated controls to obtain invasion efficiency (in percent of control). (F) Scheme of host cell Ca^{2+} signaling and interfering agents. Letters indicate figure panels that supported the respective conclusion. Statistics: one sample t-test (A, B,D,), unpaired Student's t-test (C). Bars represent means ± SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.

CD38 is a well-known producer of NAADP in immune cells (56-58). Thus, we tested *S. aureus* invasion in HeLa cells treated with the specific CD38 inhibitor 78c, even though expression of CD38 in these cells is predicted to be very low (*proteinatlas.org*). We observed no effect on *S. aureus* invasion by CD38 inhibition (**Supp. Figure 1, H**). An alternative NAADP producer, with higher expression in our infection model, is *Sterile Alpha and TIR Motif Containing 1* [SARM1, (59)]. We measured a slightly decreased invasion efficiency 10 min p.i. in a cell pool lacking SARM1 when compared to wildtype cells (**Figure 1, D**; **Supp. Figure 1, I**), suggesting that SARM1 might be involved in NAADP production during internalization of *S. aureus* by host cells. However, there might be other enzymes that produce NAADP (60) during *S. aureus* infection.

Altogether, our findings show that interference with lysosomal Ca^{2+} mobilization limits *S. aureus* invasion particularly early in infection.

S. aureus invasion requires lysosomal exocytosis, ASM and its substrate sphingomyelin on the host cell surface

Since our data suggest that lysosomal Ca^{2+} release facilitates the internalization of *S. aureus* by host cells, we speculated that lysosomal exocytosis is involved in bacterial invasion. Lysosomal exocytosis is a Ca^{2+}-dependent process during which lysosomes fuse with the plasma membranes and release their content. The internalization of several human pathogens has been shown to depend on lysosomal exocytosis (8, 16, 61-63).

To visualize lysosomal exocytosis, we developed an assay that makes use of the split NanoLuc luciferase system [**Figure 2, A**; (64, 65)]. We here engineered HeLa cells to express the high-affinity NanoLuc peptide HiBiT between the signal peptide and the mature chain of lysosomal-associated membrane protein 1 (LAMP1). Transient localization of LAMP1 to the cell surface of mammalian cells is a hall mark for lysosomal exocytosis (66). To measure the release of the HiBiT-tagged LAMP1, we added LgBit together with the NanoLuc substrate Furimazine to the cell culture medium and measured luminescence of reconstituted NanoLuc in a microplate reader. Maximum luminescence was detected ~10-15 min upon starting the assay and decline afterwards, likely due to exhaustion of the luciferase substrate (**Figure 2, B**). Treatment with the Ca^{2+} ionophore ionomycin, a known activator of lysosomal exocytosis (11, 67), increased the luminescence, whereas Vacuolin-1, an inhibitor of lysosomal exocytosis (68), decreased luminescence when compared to an untreated control. Thus, we concluded that our assay successfully quantified exposure of LAMP1 on the host cell surface.

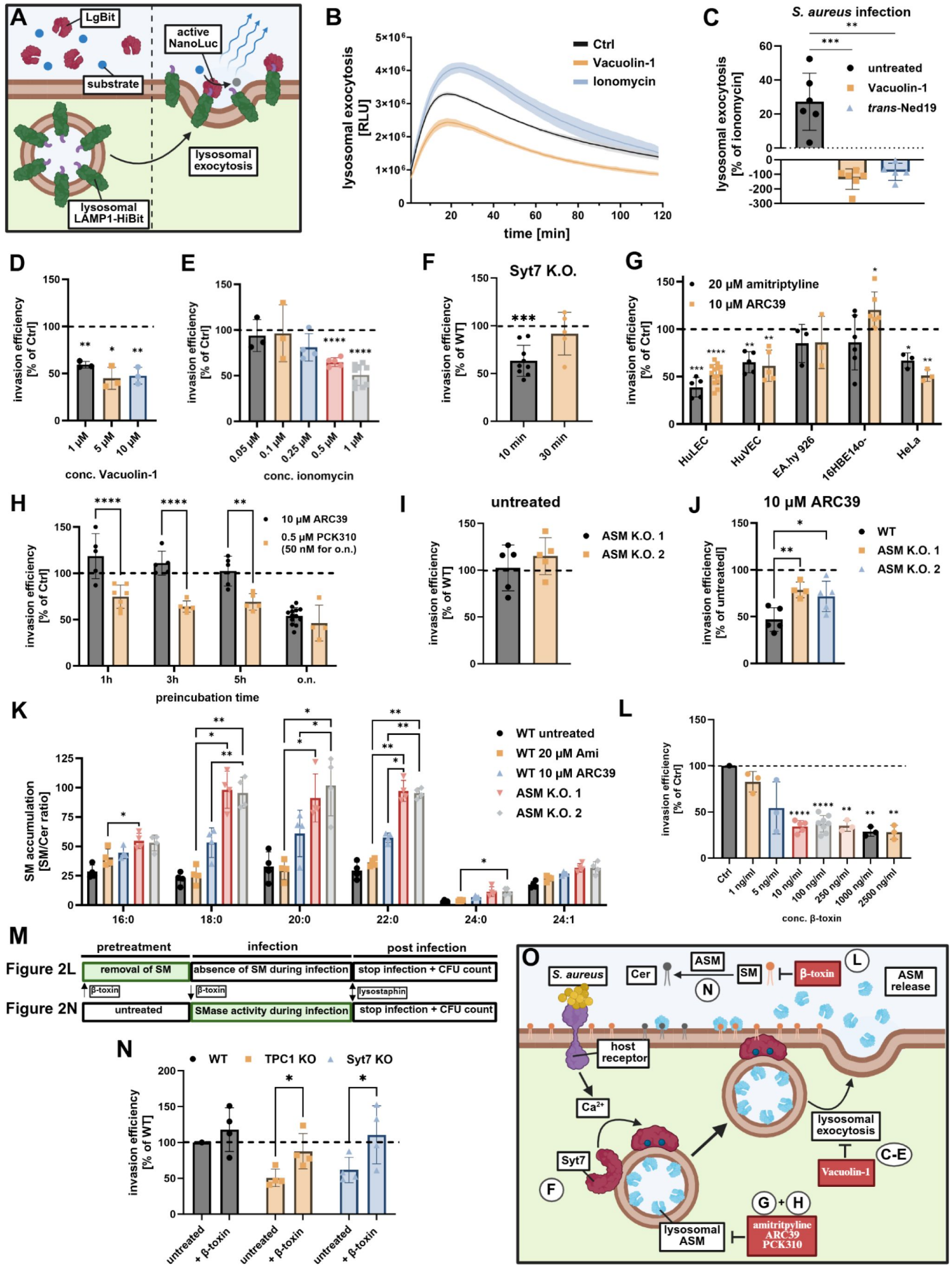


Figure 2

S. aureus invasion requires lysosomal exocytosis, ASM and plasma membrane sphingomyelin.

(A, B) Luminescence-based lysosomal exocytosis assay. HeLa cells expressing LAMP1-HiBit were treated with 1 µM ionomycin (immediately before the measurement) or 1 µM Vacuolin-1 (75 min before the measurement) in presence of LgBit and NanoLuc substrate. Lysosomal exocytosis was monitored by measuring chemiluminescence in a Tecan microplate reader. n=3 **(C) *S. aureus* triggers lysosomal exocytosis during infection.** HeLa cells expressing LAMP1-HiBit were pretreated for 75 min with 1 µM Vacuolin-1 or 200 µM *trans*-Ned19. Subsequently, cells were infected with *S. aureus* JE2 (MOI10), treated with 1 μM ionomycin or left untreated and lysosomal exocytosis was determined by measuring chemiluminescence in a Tecan microplate reader 10 min p.i. Luminescence detected in infected samples was scaled to untreated controls (0 %) and ionomycin (100%) to determine the proportion of all "releasable" lysosomes liberated during infection. n=6. **(D-F) *S. aureus* invasion requires Syt7-dependent lysosomal exocytosis.** HuLEC were treated with the lysosomal exocytosis inhibitor Vacuolin-1 or the lysosomal exocytosis inducer ionomycin and invasion efficiency of *S. aureus* was determined 30 min p.i. (D, E, n≥3). Invasion efficiency of *S. aureus* JE2 was determined in HeLa Syt7 K.O. cells 10 min and 30 min p.i. (F, n≥5; data normalized to HeLa wildtype). **(G, H) *S. aureus* internalization is dependent on ASM activity in endothelial and epithelial cells.** Indicated cell lines were treated with the ASM inhibitors amitriptyline or ARC39 (G) and invasion efficiency of *S. aureus* was determined (n≥3). (G) HuLEC were treated with PCK310 or ARC39 for the indicated periods. Subsequently, cells were infected with *S. aureus* for 30 min and invasion efficiency was determined by CFU counting (H). n≥4. **(I, J) Genetic ablation of ASM renders *S. aureus* invasion insensitive towards ARC39.** HeLa ASM K.O. cells were treated with 10 μM ARC39 or were left untreated and the number of invaded bacteria was determined 30 min p.i.. Invasion efficiency was determined by normalization to wildtype HeLa cells (I) or to corresponding untreated samples (J, n≥4), **(K) Genetic ablation of ASM is accompanied by massive alteration in cellular sphingolipid profiles.** HeLa wildtype and ASM K.O. cells were treated with 20 μM amitriptyline (75 min) and 10 μM ARC39 (22 h) or left untreated, as indicated. Whole cell sphingolipid profiles were determined by HPLC-MS/MS and the ratios of SM vs. Cer were calculated for lipid species with varying acyl side chains (chain length indicated on the x-axis). n= 4 **(L) *S. aureus* invasion requires SM on host cell plasma membranes.** HuLEC were treated with the bacterial SMase β-toxin for 75 min and were subsequently infected with *S. aureus* JE2 for 30 min (n≥3). **(M) Experimental design for β-toxin treatment of host cells during *S. aureus* infection.** Host cells were either pretreated with β-toxin to remove SM from the plasma membrane prior to infection (upper panel; **Figure 2, L**). Alternatively, β-toxin was added together with the bacteria to rescue the absence of ASM (lower panel, **Figure 2, N**). **(N) Presence of extracellular SMase activity restores the invasion defect in TPC1 and Syt7 K.O. cell lines.** HeLa wild type as well as TPC1 or Syt7 KO cell lines were infected with *S. aureus* JE2 in presence of 100 ng/ml of the bacterial SMase β-toxin and invasion efficiency 10 min p.i. was determined (n=4). In each experiment the numbers of intracellular *S. aureus* JE2 were determined by lysostaphin protection assay and CFU counting. Results obtained for the tested condition were normalized to the wildtype cell line or the mock-treated control. (**O) Scheme of lysosomal exocytosis and ASM release as well as interfering agents.** Cer: ceramide, SM: sphingomyelin, ASM: acid sphingomyelinase. xStatistics: one sample t-test (D-G, L), one-way ANOVA and Dunnett's multiple comparisons test (C, J). Mixed-effects model (REML) and Šídák's multiple comparisons (H). Two-way ANOVA and Tukey's multiple comparison (K). Two-way RM ANOVA and Šídák's multiple comparison (N) Bars represent means ± SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.

Next, we monitored LAMP1 surface levels during *S. aureus* infection. We assumed that ionomycin would trigger exocytosis of all "releasable" lysosomes. Hence, the detected luminescence signal during infection was scaled to an untreated control (=0 %) as well as ionomycin (=100%), to express LAMP1 surface levels as percentage of all "releasable" lysosomes. We detected that ~30% of all "releasable" lysosomes were liberated from the reporter cells 10 min after infection with *S. aureus* (**Figure 2, C**). Pretreatment of host cells with Vacuolin-1 or Ned-19 strongly reduced LAMP1 surface levels, suggesting that *S. aureus* triggers lysosomal exocytosis in an NAADP-dependent manner during invasion.

To test whether lysosomal exocytosis is required for *S. aureus* internalization by host cells, we measured invasion efficiency in HuLEC (**Figure 2, D**) and HeLa cells (**Supp. Figure 2, A**) pretreated with Vacuolin-1. Invasion was strongly reduced in inhibitor-treated samples, while the survival of the bacteria was not affected by the compound (**Supp. Figure 2, B**).

Next, we triggered lysosomal exocytosis in HuLEC by addition of ionomycin prior to infection, thereby depleting the pool of "releasable" lysosomes and thus preventing lysosomal exocytosis during infection. Ionomycin pretreatment reduced invasion efficiency in a concentration-dependent manner (**Figure 2, E**). Bacterial growth and survival as well as host cell integrity were not affected at the concentrations used (1 μM, **Supp. Figure 2, C-E**).

Synaptotagmin 7 (Syt7) is known to support fusion of lysosomes with the plasma membrane in a Ca^{2+}-dependent manner (69). Accordingly, Syt7-depleted HeLa cells demonstrated lower bacterial invasion (**Figure 2, F**). As already observed for TPC1 (**Figure 1, C**), this was most pronounced early in infection. Taken together, our results support the hypothesis that lysosomal exocytosis is important for invasion of host cells by *S. aureus* particularly early during infection.

Lysosomal exocytosis results in release of ASM (6, 11), an enzyme which has previously been associated with the internalization of several bacterial and viral pathogens (8, 13, 16, 62, 63, 70) and thus, we tested if ASM activity is also required for the uptake of *S. aureus* by host cells. Therefore, we treated different cell lines prior to infection with amitriptyline, a functional inhibitor of ASM (FIASMA), or the competitive ASM inhibitor ARC39 (71) and infected host cells with *S. aureus*. While invasion in HuLEC, human umbilical vein endothelial cells (HuVEC) and HeLa was reduced by both inhibitors, invasion in bronchial epithelial 16HBE14o$^-$ and the endothelial-epithelial hybrid cell line EA.hy926 was not or only marginally altered (**Figure 2, G**). The used inhibitor concentrations did not affect bacterial survival (**Supp. Figure 2, F, G**). Furthermore, we excluded that differences among cell lines arose from differential ASM activity or sensitivity to the inhibitors by monitoring enzymatic activity [**Supp. Figure 2, H**; (72) and **Supp. Figure 2, I**; (73)].

Next, we treated HuLEC with 10 µM ARC39 or 0.5 µM PCK310, a fast-acting ASM inhibitor. Already after 1h preincubation, PCK310 reduced invasion of *S. aureus*, whereas ARC39 required overnight treatment for a similar reduction (**Figure 2, H**). We confirmed the reduction of internalized colony forming units (CFU) by amitriptyline and PCK310 treatment with a microscopy-based approach, where we determined the number of invaded fluorescent bacteria per host cell (**Supp. Figure 2, J**). Taking together, our results show that ASM is involved in *S. aureus* invasion of host cells in a cell-type specific manner.

Next, we generated two ASM-depleted HeLa cell pools using CRISPR/Cas9 (74) with two distinct small guide RNAs (sgRNAs) and determined successful ablation of cellular ASM activity with a visible range ASM FRET probe [(73), **Supp. Figure 2, K**]. Surprisingly, we did not detect a reduced invasion efficiency in ASM K.O. cells, when compared to wildtype (**Figure 2, I**). To test if unspecific side effects of the ASM inhibitors could have contributed to reduced invasion in our previous experiments, we treated wildtype and ASM K.O.s with ARC39 and tested for *S. aureus* invasion efficiency (**Figure 2, K**). In wildtype cells, bacterial invasion again was strongly reduced after ARC39 treatment, whereas only a slight reduction was observed in the K.O.s, which may represent unspecific side effects of the inhibitor. Since the reducing effect of ARC39 on invasion is almost completely lost in ASM K.O. cells, the reduced invasion in wildtype cells upon ARC39 treatment predominantly must be caused by the inhibition of ASM.

Absence of ASM results in cellular accumulation of SM and usually is accompanied by the neurodegenerative diseases, Niemann-Pick syndrome A and B (75). Hence, we compared the sphingolipidomes of inhibitor-treated wildtype and ASM K.O. cells by high-pressure liquid chromatography coupled to tandem mass spectrometry (HPLC-MS/MS). The SM accumulation is described as ratio of cellular SM vs. ceramide concentrations (ASM educt vs. product) for lipid species with different acyl chain lengths (**Figure 2, K**). Whereas we did observe a moderately increased amount of most SM species upon ARC39 treatment, cells exposed to amitriptyline only showed a slight increase in SM 16:0, which can be explained by the shorter preincubation time used for amitriptyline (75 min vs. 22h). However, in both ASM K.O. cell lines we observed markedly increased SM levels when compared to either untreated or inhibitor-treated wildtype cells. The relatively low increase of SM levels after inhibitor treatment likely arises from short inhibitor pulses (in the range of minutes or hours), whereas SM in ASM K.O. cell lines accumulates over the entire culture period (days or weeks since gene ablation). We hypothesize that in the latter situation the host cell invasion of *S. aureus* is generally increased, while the ASM-dependent uptake pathway is absent. This is supported by the high residual internalization of *S. aureus* by

ASM K.O. cells upon ARC39 treatment. Previously, it was shown that *S. aureus* host cell entry is limited by caveolin-1, a protein associated with lipid microdomains (76, 77). Absence of ASM was demonstrated to interfere with caveolin-1-associateded endocytosis (78, 79) and thus, the increased invasion in ASM K.O. cells may be caused by dysfunctional caveolin-1.

Release of ASM by host cells results in cleavage of SM on the host cell surface, a process that previously was suggested to facilitate cell entry of human pathogens (7, 8, 13-17). To test whether *S. aureus* invasion requires SMase activity on the host cell surface, we removed SM, the substrate of ASM, from plasma membranes of HuLEC (**Figure 2, L**) and HeLa (**Supp. Figure 2, L**) by pretreatment with the bacterial SMase β-toxin (see **Figure 2, M**). We confirmed β-toxin-dependent SM conversion in living cells by detecting the ceramide metabolites of fluorescent SM analogs [**Supp. Figure 2, M**; (81)]. Plasma membrane pretreatment with bacterial SMase reduced *S. aureus* invasion efficiency in a concentration-dependent manner in both cell lines. 10 ng/ml β-toxin were sufficient to decrease invasion to ~30% of that measured in untreated control cells. Interestingly, a 250-fold higher concentration did not lead to further reduction, suggesting that 10 ng/ml β-toxin were sufficient to quantitatively convert SM to ceramide at the cell surface. In line with our previous experiments, this suggests that multiple *S. aureus* invasion pathways exist within the same host cell population, of which at least one requires SM and its enzymatic breakdown on the plasma membrane.

Since we detected enhanced Ned19-sensitive lysosomal exocytosis during *S. aureus* infection (**Figure 2, C**), we hypothesized that absence of TPC1 or Syt7 should impair lysosomal exocytosis and thereby release of ASM. Therefore, we infected wildtype as well as TPC1 or Syt7 K.O. cells with *S. aureus* in presence (but without pretreatment, see **Figure 2, M**) of 100 ng/ml β-toxin and determined the number of intracellular bacteria after 10 min (**Figure 2, N**). The addition of β-toxin to the culture medium completely rescued the invasion defect of the K.O. cells. This indicates that reduced bacterial internalization in absence of TPC1 and Syt7 results from a lack in ASM delivery to the cell surface by Ca^{2+}-dependent lysosomal exocytosis and that absence of ASM from the host cell surface, can be compensated by exogenous addition of the bacterial SMase β-toxin.

We want to emphasize the difference in experimental application of β-toxin (**Figure 2, M**) where we i) pretreat host cells with the bSMase to remove SM from the plasma membrane (**Figure 2, L** and **Supp. Figure 2, L**) and ii) rescue the absence of exocytosed ASM from the host cell surface by exogenous addition of β-toxin during the infection (**Figure 2, N**). In the former case, pretreatment leads to removal of SM from the plasma membrane thus preventing SM conversion by ASM during infection and resulting in reduced invasion. The latter leads to SM conversion by the bacterial SMase during infection thereby increasing invasion.

In summary, our data suggest the existence of a *S. aureus* invasion pathway that depends on the metabolic breakdown of SM by ASM on the host cell surface (**Figure 2, O**).

ASM- and Ca^{2+}-mediated invasion is rapid

To study the role of SM in *S. aureus* uptake kinetics, we incubated HuLEC with fluorescent sphingolipid analogs and recorded *S. aureus* infection by time lapse imaging [**Figure 3, A**; **Supp. Video 1**; **Supp. Figure 3**; **Supp. Video 2**; (73)]. We observed that the bacteria were rapidly engulfed by SM-containing membrane compartments. While association of bacteria with BODIPY-FL-C12-SM increased over time during the first 30 min of infection, this was not observed for BODIPY-FL-C12-Ceramide, suggesting that *S. aureus* invasion specifically relies on interaction with SM in host cell plasma membranes (**Figure 3, B**).

Next, we determined the invasion efficiency of *S. aureus* in a time-dependent manner and blocked invasion by either ASM inhibitors, ionomycin, BAPTA-AM or β-toxin pretreatment. We found a time-dependent increase in intracellular bacteria, whereby all treatment conditions reduced

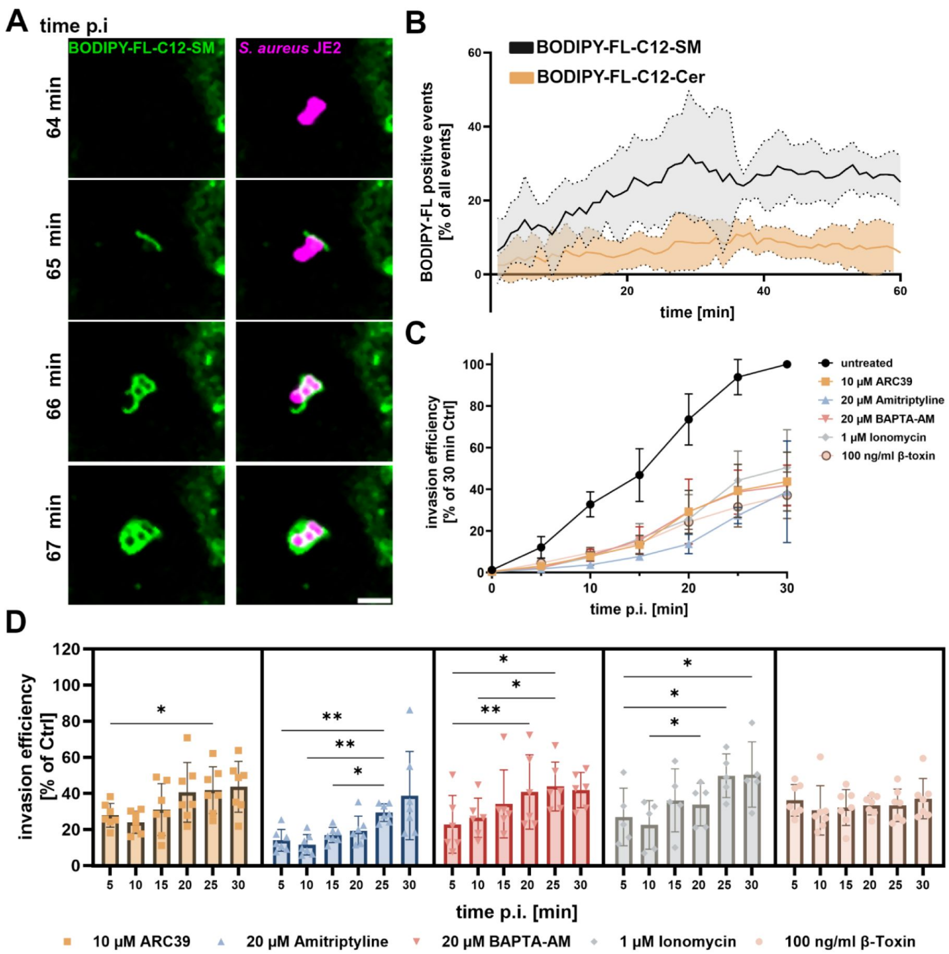


Figure 3

ASM- and Ca^{2+}-dependent uptake is rapid and predominantly mediates invasion early in infection.

(A, B) *S. aureus* associates with SM early during invasion. HuLEC were pretreated with BODIPY-FL-C12-SM or BODIPY-FL-C12-ceramide and infected with red-fluorescent *S. aureus.* Scale bar: 5 µm. Infection was monitored by live cell imaging and the proportion of bacteria that associated with the lipid analogs was quantified in each individual frame (SM: n=5, Cer: n=3). (**C, D) ASM and Ca^{2+}-dependent invasion is rapid.** HuLEC were treated with ARC39, amitriptyline, BAPTA-AM, ionomycin, or the bacterial SMase β-toxin. Then, cells were infected with *S. aureus* and the number of invaded bacteria was determined after different time periods. The results were either normalized to the 30 min time point of untreated controls (C) or to the corresponding time points of the untreated controls (D). (n ≥5). Statistics: Mixed effect analysis and Tukey's multiple comparison. Graphs represent means ± SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.

bacterial invasion at every time point with the untreated control at 30 min set to 100% (**Figure 3, C**). We next normalized the data to the untreated control of the corresponding time point. Except for the β-toxin treatment, the effect of all other treatments was most pronounced early (5-10 min) after infection (**Figure 3, D**). For instance, ionomycin treatment massively reduced *S. aureus* invasion to ~20% of untreated controls when we infected for 10 min. We concluded that in untreated controls 80% of the bacteria entered host cells via the pathway that depends on lysosomal exocytosis, Ca^{2+} and ASM, which is blocked by ionomycin treatment. Consequently, bacteria that were able to invade host cells in ionomycin-treated samples (the residual ~20%) must have employed other pathways. By contrast, in samples infected for 30 min a higher proportion of bacteria was able to enter host cells despite ionomycin treatment (~50% compared to untreated controls). This suggests that only 50% of all invaded bacteria were internalized in an ASM- and Ca^{2+}-dependent fashion upon the longer infection pulse. Together with the observed invasion defects in host cells with gene deletions in TPC1 (**Figure 1, C**) and Syt7 (**Figure 2, F**), which we exclusively detected during short infection times, our data suggest that early during infection bacteria enter host cells predominantly in an ASM- and Ca^{2+}-dependent manner. Thus, we concluded that the internalization pathway that depends on lysosomal exocytosis, ASM and Ca^{2+} is rapid when compared to other host cell entry pathways.

The mode of S. aureus invasion affects infection outcome

Next, we tested whether the pathway used for host cell entry also affects the intracellular fate of *S. aureus*. After invasion, the bacteria reside within Rab5-positive early and subsequently, in Rab7-positive late phagoendosomes (39). To trace phagosome maturation, we generated a HeLa cell line stably expressing mCherry-Rab5 as well as YFP-Rab7 and infected with *S. aureus*. We observed that the bacteria transiently associate with Rab5 for a few minutes, before they become positive for Rab7 (Supp. Video 3).

We next blocked the ASM-dependent invasion pathway by amitriptyline treatment and infected with *S. aureus* for different time periods. We determined the proportion of bacteria that was associated with Rab5 and/or Rab7 (**Figure 4, A**; see **Supp. Figure 4** for bacteria proportion that associates with Rab7). While ASM inhibition only marginally affected *S. aureus* association with Rab5-positive phagosomes, the proportion of bacteria residing in Rab7-positive vesicles was significantly reduced when compared to the untreated control. This observation was restricted to shorter infection pulses (5-45 min) and was not observed for long infection periods (60 min).

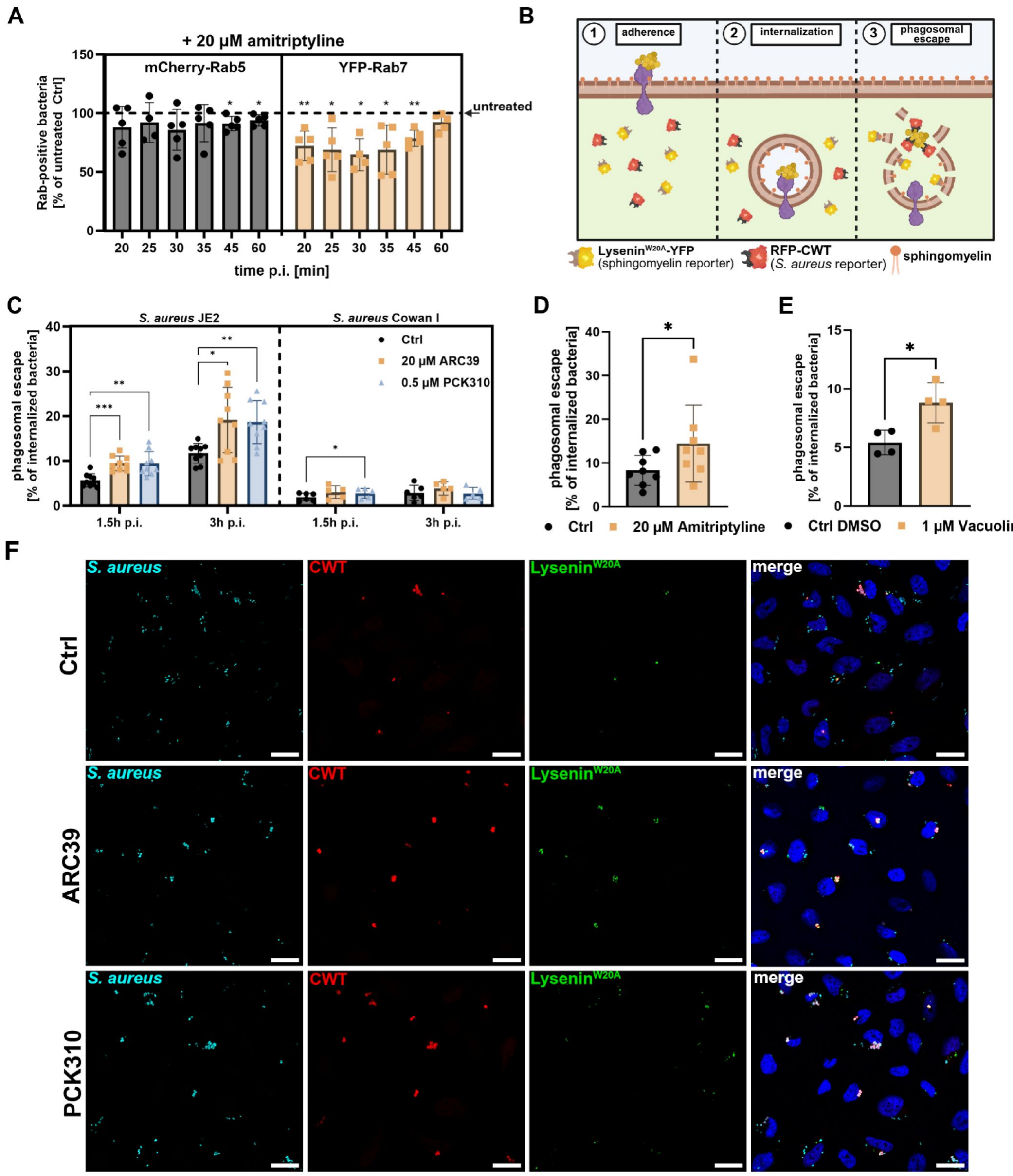


Figure 4

Blocking ASM-dependent invasion affects phagosomal maturation and escape during *S. aureus* infection

(A) Inhibition of ASM delays formation of Rab7-positive phagosomes. HeLa cells expressing mCherry-Rab5 and YFP-Rab7 were either treated with amitriptyline or left untreated. Then, cells were infected with *S. aureus* JE2 for indicated periods. Extracellular bacteria were removed and percentage of intracellular bacteria which associated with Rab5 or Rab7 was determined by CLSM. Results obtained for amitriptyline-treated samples were normalized to untreated controls (n=5). **(B) Detection of phagosomal SM and phagosomal escape by a reporter cell line expressing Lysenin^{W20A} -YFP and RFP-CWT.** After internalization, *S. aureus* resides in a phagosome preventing the recruitment of RFP-CWT to *S. aureus* cell wall and Lysenin^{W20A}-YFP to luminal SM, respectively. When the bacteria lyse the phagosomal membrane, luminal SM gets exposed to the cytosol and attracts Lysenin^{W20A} -YFP, while RFP-CWT is recruited to the staphylococcal surface. (**C-F) Blocking ASM-dependent internalization affects phagosomal escape.** RFP-CWT and Lysenin^{W20A} -YFP expressing HeLa were treated with PCK310,ARC39 (C, F), amitriptyline (D) or Vacuolin-1 (E) and infected with *S. aureus* strains JE2 or Cowan I. By CLSM, proportions of bacteria that recruited RFP-

CWT (phagosomal escape) were determined 3h p.i. if not indicated otherwise (n≥3). **Statistics:** one sample t-test (A), Mixed effects analysis (REML) and Tukey's multiple comparison (C,), paired Student's t-test (D, E) Graphs represent means ± SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.

We also observed a reduced proportion of bacteria associated with Rab7-positive membranes after treatment with PCK310 or bacterial SMase (**Supp. Figure 4, B, C**). Forty-five minutes p.i., about half of the bacteria were localized to Rab7-positive vesicles in untreated cells, whereas only ∼40% associated with Rab7 upon amitriptyline, PCK310 or β-toxin treatment (**Supp. Figure 4, C**). However, this was not caused by translocation of *S. aureus* to the host cytosol, which was excluded by expressing the fluorescence reporter RFP-CWT (42) in the YFP-Rab7 cell line. Accordingly, RFP-CWT-labelled bacteria in the host cytosol were excluded from the data set [**Supp. Figure 4, B-D**, (42)]. Hence, ASM-dependently generated phagoendosomes possess different maturation dynamics when compared to phagosomes formed in an ASM-independent fashion.

Next, we measured phagosomal escape while simultaneously detecting SM content of disrupted phagosomal membranes. Therefore, we used a HeLa reporter cell line that cytosolically expresses the phagosomal escape reporter RFP-CWT (42) as well as the SM reporter Lysenin^{W20A}-YFP (82), respectively (**Figure 4, B**). To validate the reporter system, we removed SM from the plasma membrane by β-toxin treatment prior to infection and monitored phagosomal escape by live cell imaging (**Supp. Figure 5**, **Supp. Video 4**). Bacterial translocation into the host cytosol is, again, indicated by recruitment of RFP-CWT. In untreated cells, phagosomal escape was accompanied by recruitment of Lysenin^{W20A}-YFP to vesicular membranes, suggesting that SM located at the luminal leaflet of phagosomes was exposed to the cytosol. By contrast, Lysenin^{W20A}-YFP was not recruited to escape events in β-toxin-treated cells, indicating that the SMase had removed SM from the membranes.

Next, we tested whether internalization of the bacteria via the ASM-dependent pathway would affect phagosomal escape of *S. aureus*. We first confirmed that invasion of *S. aureus* is sensitive to ASM inhibitors and β-toxin treatment, despite reporter gene expression in the cell line (**Supp. Figure 4, E**). Then reporter cells either were left untreated or were pre-exposed to ASM inhibitors and were infected with *S. aureus* strain JE2. In addition, *S. aureus* Cowan I, a non-cytotoxic strain known to have low escape rates, was used for infection. The proportion of bacteria that escaped from the phagosome (**Figure 4, C, D**) as well as the proportion of escape events, which were additionally positive for cytosolic SM (**Supp. Figure 4, F**) were determined 1.5 h and 3 h p.i. (see also **Supp. Figure 6, A-D**). Whereas no differences for Lysenin-positive escape events were detected (**Supp. Figure 4, F**), a higher proportion of bacteria escaped from phagosomes on abrogation of ASM activity regardless of the inhibitor used (**Figure 4, C, D**). A similar increase was observed when we blocked lysosomal exocytosis by Vacuolin-1 ((**Figure 4, E**), suggesting that the lack of ASM delivery to the host cell surface during bacterial invasion affects phagosomal escape.

Since blocking of ASM-dependent uptake reduced invasion efficiency yet increased phagosomal escape rates, we hypothesized that the presence of SM on the plasma membrane affects downstream events during *S. aureus* infection. To address this, we pretreated host cells with the bacterial SMase β-toxin to block SM-dependent invasion (**Figure 5, A**; 2a, 2b) and either infected the cells in presence (**Figure 5, A**; 3a) or after removal of the toxin (3b). β-toxin pretreatment, resulted in enhanced phagosomal escape rates of *S. aureus* JE2, independent of the presence of β-toxin during infection (**Figure 5, B**; 4a, 4b). This was not observed for infections with the Cowan I strain, indicating that removal of SM by β-toxin did not affect general phagosomal membrane stability (**Supp. Figure 4, G**). Moreover, β-toxin treatment strongly reduced recruitment of Lysenin^{W20A}, demonstrating effective removal of SM from membranes by β-toxin (**Figure 5, C** and **Supp. Figure 6, E-J**).

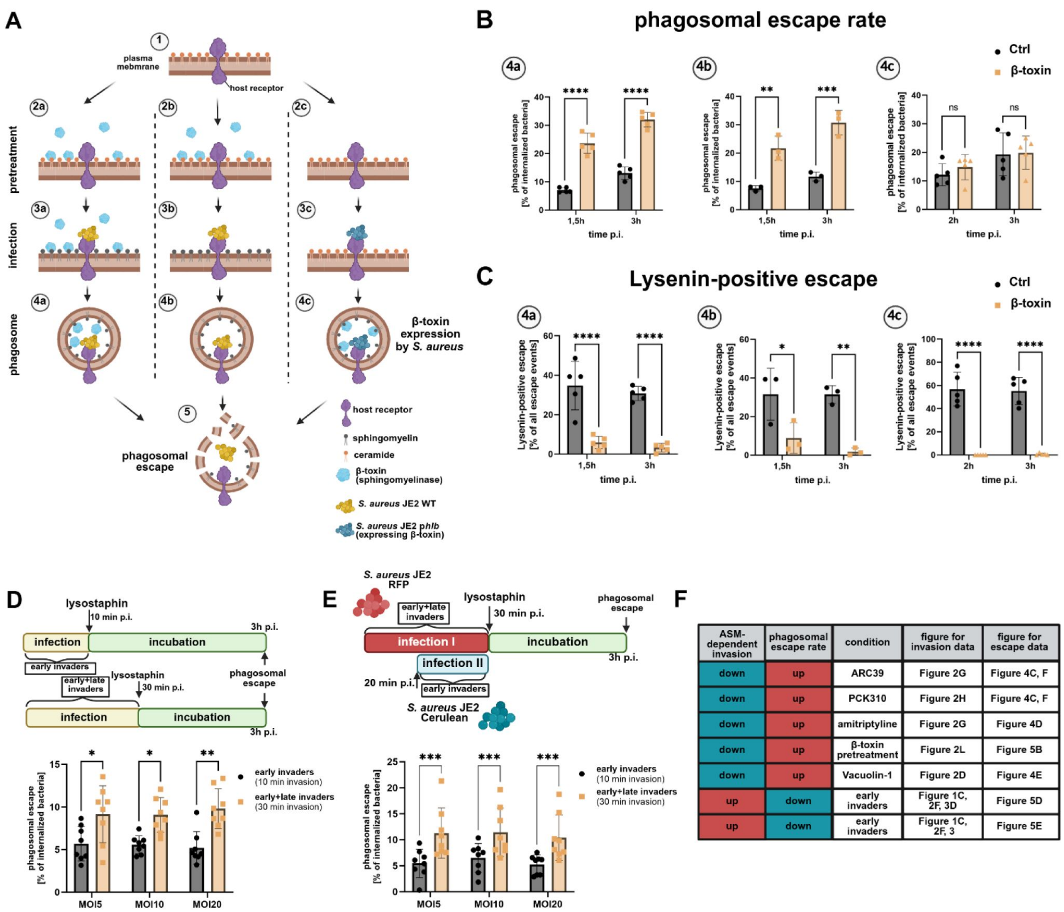


Figure 5

The intracellular fate of *S. aureus* is determined by host cell entry

(A-C) Phagosomal escape depends on presence of plasma membrane SM during invasion, but not presence of SM within phagosomal membranes HeLa RFP-CWT Lysenin^{W20A} -YFP were pretreated with β-toxin to remove surface SM (2a, 2b) or left untreated (2c). Then, cells were infected with *S. aureus* JE2 in presence (3a) or absence (3b) of β-toxin. Untreated samples (3c) were infected with *S. aureus* JE2 harboring a plasmid either encoding β-toxin and the fluorescence protein Cerulean (pCer+*hlb*) or solely Cerulean (pCer). Proportion of bacteria that recruited RFP-CWT (phagosomal escape, B) and the percentage of phagosomal escape events that additionally were positive for Lysenin^{W20A}-YFP (C) were determined at indicated time points p.i. (n=5). **(D, E) Early ASM-dependent invaders possess lower escape rates than late invaders.** HeLa cells expressing RFP-CWT were infected with indicated MOIs of *S. aureus* JE2 either for 10 min (early invaders) or 30 min (early+late invaders). Phagosomal escape rates were determined 3h p.i. (D). HeLa reporter cells expressing YFP-CWT were infected with an MOI=5 of *S. aureus* JE2 expressing a fluorescent protein (e.g. RFP) for 30 min (early+late invaders). After 20 min, the same samples were infected with *S. aureus* JE2 expressing another fluorophore (e.g. Cerulean) for 10 min (early invaders) and phagosomal escape was determined 3 h p.i. (E). **(F) Summary of experiments analyzing the influence of invasion pathway on phagosomal escape.** The measured effects of different conditions on rapid ASM-dependent invasion and accompanied alterations in phagosomal escape rates are summarized. red = increased, blue=decreased. **Statistics:** Two-way ANOVA and Šídák's multiple comparisons test (B-E). Graphs represent means ± SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.

To investigate if removal of SM was important at the cell surface or within the phagosome, we infected the cells with transgenic *S. aureus* JE2 strain either collinearly overexpressing β-toxin and cyan-fluorescent reporter Cerulean or solely expressing Cerulean. We did not observe increased phagosomal escape upon overexpression of β-toxin (**Figure 5, B**; 4c), although Lysenin recruitment to phagosomal membranes was reduced to levels of β-toxin-pretreated samples (**Figure 5, C**). Moreover, we did not detect substantial differences in escape rates of *S. aureus* 6850, a strain naturally producing β-toxin (**Supp. Figure 4, H**), and its isogenic β-toxin mutant. Again, escape of *S. aureus* 6850 was enhanced upon pretreatment with β-toxin (**Supp. Figure 4, I**) accompanied by absence of SM from phagosomal membranes (**Supp. Figure 4, J**).

Phagosomal escape of *S. aureus* thus is independent of SM levels within the infected phagosomes but rather reflects the presence of SM and ASM at the plasma membrane during host cell entry.

We previously showed that *S. aureus* predominantly enters host cells via the rapid ASM- and Ca^{2+}-dependent pathway early upon host cell contact (**Figure 1, C**; **Figure 2, F**; **Figure 3, D**), while other pathways are likely employed later during infection. Consequently, the proportion of pathways (ASM-dependent vs. ASM-independent) that is used for host cell entry relies on the infection time. To demonstrate that phagosomal escape is dependent on the pathway of cell entry, we infected reporter cells with *S. aureus* for either a 10 min ("early invaders") or a 30 min infection pulse (representing "early and late invaders") and at 3h p.i. we determined phagosomal escape rates (**Figure 5, D**) as well as the number of invaded bacteria (**Supp. Figure 4, K**). Independent of the multiplicity of infection (MOI) used, bacteria taken up within the first 10 min demonstrated significantly lower escape rates when compared to bacteria that were cocultured with host cells for 30 min.

This was corroborated by HeLa YFP-CWT infected with *S. aureus* expressing a fluorescent protein (e.g. mRFP) for 30 min ("early and late invaders"). After 20 min, we initiated a second infection pulse for 10 min ("early invaders") with a *S. aureus* strain expressing another fluorescence protein (e.g. Cerulean) and determined phagosomal escape rates of both bacterial recombinants (**Figure 5, E**). Again, early invaders demonstrated lower escape rates when compared to the 30 min control. This was independent of the fluorescent marker expressed by the strain that was used for each infection pulse.

Taken together, we observed higher escape rates 3h p.i., whenever we blocked the rapid ASM-dependent invasion pathways. Consistently, escape rates of bacteria predominantly internalized via the ASM-dependent route ("early invaders") were decreased (**Figure 5, F**). Hence, we concluded that the pathway of host cell entry directly affects phagosomal escape.

We next monitored phagosomal escape over 10 h in cells pretreated with ASM inhibitors (**Figure 6, A**; **Supp. Video 5**; see **Supp. Figure 7**, for absolute bacteria numbers of individual replicates). ASM inhibition enhanced escape rates 3 h p.i. when compared to untreated controls. In contrast, 6 h p.i. escape rates were found decreased upon ASM inhibition.

Taken together, the efficiency of phagosomal escape of *S. aureus* is directly affected by the mode of cell entry, whereby the rapid SM/ASM-dependent pathway results in delayed phagosomal escape, while bacteria that entered host cells ASM-independently escape earlier during infection.

Since translocation to the host cytosol is a prerequisite for *S. aureus* replication in non-professional phagocytes, we recorded intracellular growth of *S. aureus.* In untreated controls replication started about 5 h p.i., and thus shortly after phagosomal escape (3-4 h p.i., **Figure 6, B**). By contrast, we did not observe significant bacterial replication in cells treated with ASM inhibitors (**Figure 6, B**).

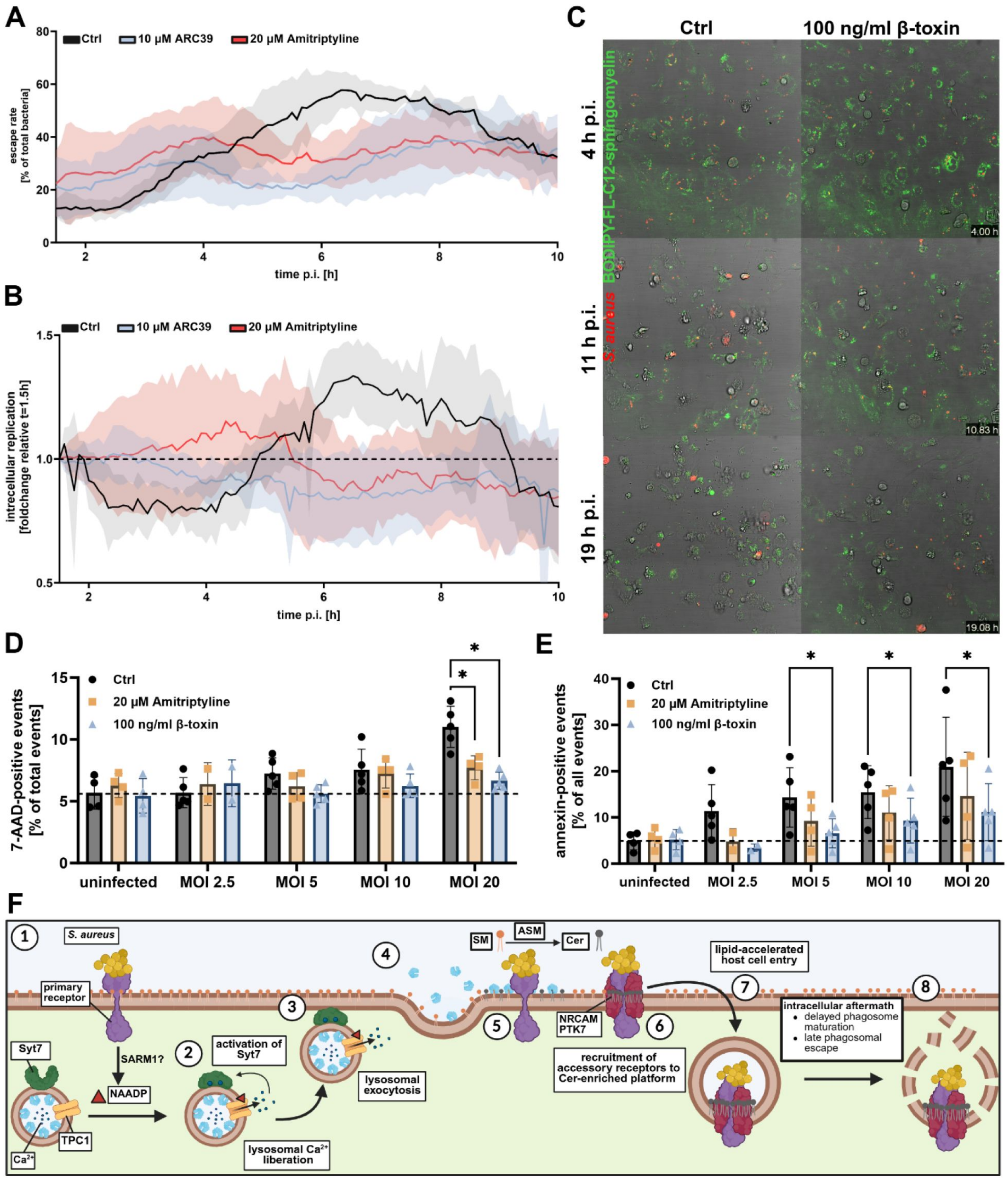


Figure 6

Blocking ASM-dependent internalization affects intracellular replication and host cell survival.

(A, B) HeLa cells expressing RFP-CWT and Lysenin^{W20A}-YFP were treated with amitriptyline or ARC39 or were left untreated (Ctrl). The cells were subsequently infected with *S. aureus* and infection was monitored for 10 h by CLSM. Proportion of escaped bacteria (A) or the intracellular replication (B) was determined (n= 5). **(C)** HuLEC were stained with BODIPY-FL-C12-SM (green), and were either pretreated with 100 ng/ml of β-toxin or were left untreated. Then, cells were infected with *S. aureus* (red) and infection was monitored by CLSM for 19 h. **(D, E)** HuLEC were treated with amitriptyline or β-toxin and subsequently infected with *S. aureus*. After 21 h, plasma membrane integrity was measured by 7-AAD staining (D), and proportion of apoptotic cells was determined by annexin V staining (E) (n=5). See also **Supp. Figure 8**. (F) Model: *S. aureus* interacts with an unknown primary receptor (1) that triggers production of NAADP, which in turn activates TPC1 to mediate lysosomal Ca^{2+} release and activation of Syt7 (2). Syt7-dependent lysosomal exocytosis (3) leads to release of ASM (4) and production of Cer on the plasma membrane (5) thereby resulting in the recruitment of co-receptors [6, (83)] and rapid internalization of *S. aureus* by host cells (7). Bacteria, which enter the host cells via the rapid pathway, experience a different intracellular fate than bacteria that employed other pathways. **Statistics:** Mixed effects analysis (REML) and Tukey's multiple comparison test. Graphs represent mean ± SD. *p≤0.05. Created in *BioRender*.

We next treated HuLEC with β-toxin, stained the cells with BODIPY-FL-C12-SM and infected the cells with fluorescent *S. aureus*. Live cell imaging demonstrated that most untreated host cells died after 11 h p.i., whereas β-toxin-treated samples showed higher survival rates (**Figure 6, C**; **Supp. Video 6**). This was also observed by cell death assays measuring i) membrane integrity (7-AAD staining, **Figure 6, D**), ii) apoptosis (Annexin V staining, **Figure 6, E**), iii) number of host cells that remained attached to the substratum (**Supp. Figure 8, A**) and iv) host cell lysis by LDH release (**Supp. Figure 8, B**) at 21 h p.i..

Discussion

ASM has been previously reported to be implicated in the invasion of several viral, bacterial, and eukaryotic pathogens (7, 8, 13-17). However, the dynamics of ASM-mediated host cell entry, the underlying receptors as well as effects on post-invasion events are only poorly understood.

We here show that ASM is involved in the uptake of *S. aureus* by certain human cells within minutes after the pathogen contacts the host cell and this pathway is concurrent with ASM-independent uptake pathways. We demonstrate that the infection outcome of bacteria that enter the host cells via this rapid pathway is distinct from other entry mechanisms.

We observed a reduction of bacterial invasiveness upon treatment with different ASM inhibitors (**Figure 2, G**) or removal of the enzyme substrate SM from the plasma membrane (**Figure 2, L**) confirming that ASM activity on the host cell surface is crucial for pathogen internalization. Surprisingly, the uptake of *S. aureus* by ASM K.O. cells was similar to wildtype cells (**Figure 2, I**). However, in ASM K.O.s *S. aureus* invasion was only slightly affected by the ASM inhibitor ARC39, while ARC39-treated wildtype cells showed a stronger reduction (**Figure 2, J**). We concluded that reduced invasion in wildtype cells upon ARC39 treatment is mostly due to inhibition of ASM and consistently, the effect of ARC39 is strongly attenuated in ASM K.O.s. Hence, we postulate that other ASM-independent host cell pathways are upregulated in ASM K.O.s, which might be caused by the strongly altered sphingolipid profile detected in ASM K.O.s (**Figure 2, K**). For instance, *S. aureus* internalization by host cells is strongly restricted by caveolin-1, a protein associated with lipid membrane domains (76, 77). Accordingly, alterations in membrane composition might affect distribution of caveolin-1 on the plasma membrane, thereby fostering uptake of *S. aureus*. In consistence, previous studies reported dysfunction of caveolin-1-dependent endocytic processes in ASM-deficient cells (78, 79).

Release of ASM is caused by lysosomal exocytosis (6). Conventional methods for detection of lysosomal exocytosis often depend on the usage of anti-LAMP1 antibodies that are added to the cell culture medium for monitoring LAMP1 exposure on the plasma membrane (66). However, most antibodies are non-specifically bound by staphylococcal protein A (84) and hence, these protocols are not compatible with *S. aureus* infection studies. Therefore, we developed a novel approach to measure lysosomal exocytosis via a split luciferase-based assay that demonstrated lysosomal exocytosis during *S. aureus* infection (**Figure 2, A-C**). Consequently, depleting host cells from lysosomes that can undergo Ca^{2+}-dependent exocytosis by ionomycin pretreatment, reduced invasion, even though ionomycin can influence general host cells Ca^{2+} homeostasis (85). Additionally, the lysosomal exocytosis inhibitor Vacuolin-1 resulted in drastically reduced internalization of bacteria (**Figure 2, D**), although the substance is known for side effects such as inhibition of PIKfyve thereby affecting the endo-lysosomal compartment (86). However, we confirmed the role of lysosomal exocytosis for *S. aureus* invasion of host cells by genetic ablation of Syt7, a key protein in lysosomal exocytosis (69).) It remains elusive, which other factors are involved in delivery of lysosomes to and fusion of lysosomes with the plasma membrane. Next to Syt7, there might be other proteins such as Synaptotagmin 1 with overlapping function that could compensate for the absence of Syt7 (87).

Elevation of cytosolic Ca^{2+} levels lead to recruitment of lysosomes to the cell surface (67). Whereas most studies implicated Ca^{2+} influx from the extracellular milieu (7, 16), we here demonstrate that *S. aureus* mainly triggers lysosomal Ca^{2+} mobilization as was shown by treatment with the inhibitor *trans*-Ned19 as well as a genetic K.O. of the endo-lysosomal Ca^{2+} channel TPC1 (**Figure 1, B and E**; **Supp. Figure 1, C**). There are two TPCs (TPC1 and TPC2) that mediate lysosomal Ca^{2+} liberation (88), which have previously been associated with regulation of phagocytic processes (89) and host cell entry of several viruses such as Ebola virus (90), Middle East respiratory syndrome coronavirus [MERS-CoV, (91)] or SARS-CoV-II (92). In line with our study, absence of one of the two TPCs severely interfered with infection (90)(91).

TPC-dependent Ca^{2+} liberation was shown to induce exocytosis, for instance, during release of cytolytic granules from T cells (93) or during secretion of insulin (94). Our observations suggest that TPC1 is also involved in exocytosis of lysosomes, since exogenous addition of a bacterial SMase immediately before infection rescued the invasion defect in TPC1 and Syt7 K.O. cells (**Figure 2, M**) indicating that reduced invasion results from a decreased delivery of ASM to the host cell surface. Thus, NAADP-dependent Ca^{2+} liberation from lysosomal stores mediates Syt7-dependent fusion of lysosomes with the plasma membrane resulting in the release of ASM (**Figure 6, F**). While ionomycin treatment led to exocytosis of all "releasable" lysosomes, infection with *S. aureus* led to lysosomal exocytosis of only ~30% of the ionomycin control (**Figure 2, C**), suggesting that *S. aureus* likely triggers local lysosomal exocytosis predominantly at host-bacteria contact sites, whereas ionomycin acts globally on all cells. A similar scenario has been observed for exocytosis of cytolytic granules in T lymphocytes, which differed between TPC/NAADP- and ionomycin-induction (93).

The conversion of SM to ceramide by ASM has been associated with the generation of ceramide-enriched platforms. These represent lipid microdomains which facilitate the recruitment of certain protein receptors (95, 96). We previously observed the interaction of *S. aureus* with several host cell surface receptors such as NRCAM or PTK7. The interaction of *S. aureus* with several host receptors was reduced upon blocking ASM-dependent internalization of *S. aureus* (83). It is thus tempting to speculate that accessory receptors identified in this study may be recruited to bacteria-host contact sites to facilitate rapid pathogen internalization.

Inhibition of CD38, a known NAADP producer (56, 57), did not influence uptake of staphylococci (**Supp. Figure 1, H**). By contrast, we detected a decreased invasion in cells deficient for the Toll-like receptor adaptor protein SARM1, an alternative NAADP producer [**Figure 1, D**; (59)]. However, the role of SARM1 in lysosomal exocytosis and ASM release requires further investigation. The impact of SARM1 on internalization of *S. aureus* was rather moderate, suggesting the existence of other NAADP-generating enzymes, such as the recently identified NADPH oxidases DUOX1, DUOX2, NOX1 and NOX2(60).

One important characteristic of the ASM-dependent internalization pathway is its velocity - taking place within the initial ten minutes of an infection with *S. aureus*, during which treatment with small molecules (**Figure 3, D**), as well as genetic ablation of TPC1 (**Figure 1, C**) and Syt7 (**Figure 2, F**) resulted in reduction of intracellular bacteria. This was accompanied by the quick formation of *S. aureus*-containing phagosomes decorated with fluorescent SM analogs (**Figure 3, A**; **Supp. Figure 3**, **Supp. Video 1**, **Supp. Video 2**).

Neither treatment with inhibitors or enzymes nor genetic ablation blocking the rapid internalization pathway abolished the bacterial invasion of host cells completely. This supports our hypothesis that several concomitant host cell entry pathways exist, which is corroborated by a large number of host cell receptors interacting with *S. aureus* (13, 28, 31, 33-38). This pathway exists only in certain host cell types, since among the cell types, as is -for instance- illustrated by our experiments in which internalization of *S. aureus* was only sensitive to ASM inhibition in endothelial (HuLEC and HuVEC) and HeLa cells (**Figure 2, G**). The proportion of bacteria taken up by rapid internalization is dependent on the duration of infection and is higher in early (~50-80% of all internalized bacteria 10 min p.i.) compared to late infection (~15-50% of all internalized bacteria 30 min p.i.).

For simplicity we describe the pathway investigated in this study as “rapid” or “ASM-dependent”, however a clear cut between this and other pathways cannot be easily delineated. Additionally, internalization factors such as host cell receptors might possess overlapping function and could be involved in several pathways. In this context, ASM might simply accelerate certain internalization pathways, which could also take place in a slower fashion without involvement of ASM.

When we blocked the rapid internalization pathway by ASM inhibitors (**Figure 4, A, C**; **Supp. Figure 4, B**), Vacuolin-1 (**Figure 4, E**, **Supp. Figure 4**) or by removal of surface SM (**Figure 5, B**; **Supp. Figure 4, B, C**), we observed changes in both, maturation of bacteria-containing vesicles and phagosomal escape. A delayed maturation of phagosomes may affect acidification of the vesicles, a property that is sensed by *S. aureus* leading to expression of a different subset of virulence factors (97) and hence may alter intracellular pathogenicity.

The bacterial SMase β-toxin originates from *S. aureus*, however, most human pathogenic *S. aureus* strains including JE2, the strain we use in the present study, do not produce β-toxin, due to a genomic integration of a prophage (80). Phagosomal escape of *S. aureus* only was dependent on SM on the plasma membrane during invasion but not on the presence of SM within the phago-endosome (**Figure 5, A-C**; **Supp. Figure 4, I, J**). This not only corroborated that phagosomal escape of clinically relevant *S. aureus* strains is independent of the bacterial SMase β-toxin (42), but also demonstrated that the outcome of phagosomal escape is influenced by processes at the plasma membrane during cell entry.

To exclude that inhibitor or bacterial SMase treatment caused changes in phagosomal escape by affecting cellular processes other than bacterial uptake, we infected untreated wildtype cells with *S. aureus* for 10 or 30 min, respectively, and monitored phagosomal escape. Bacteria that invaded within the first 10 minutes after contacting host cells (“early invaders”) are predominantly ASM-dependent (**Figure 3**) and demonstrated significantly less phagosomal escape (**Figure 5, G**, H). Consistently, blocking the ASM-dependent pathway led to increased proportions of ASM-independent invaders, which in turn resulted in higher escape rates (**Figure 4, C-E**; **Supp. Figure 4, G**). Intracellular *S. aureus* infections thus are directly influenced by the mode of internalization. This ultimately resulted in lower host cytotoxicity (**Figure 6, C-E**; **Supp. Figure 8**) and bacterial replication (**Figure 6, B**) when we blocked the ASM-dependent invasion.

An uptake-related intracellular fate has been suggested for *Mycobacterium bovis* (46), *Toxoplasma gondii* (48) or *Brucella abortus* (49). However, in these studies the host cells were either depleted from cholesterol, which causes a variety of side effects (47), or internalization was artificially induced by chemicals or ectopic expression of receptors, respectively.

Taken together, we here describe a rapid internalization pathway for *S. aureus* into human epithelial and endothelial cells. Bacteria that enter the host cells via this infection route face a distinct intracellular fate. Here we show, to our knowledge, for the first time that host cells within the same population can be invaded by a pathogen via distinct routes and that this results in a differential infection outcome. Rapid internalization may prove beneficial for pathogens in an *in vivo* setting since internalization by host cells shortens the exposure to the innate immune system within a host. Interestingly, the absence of ASM in *Smpd1*$^{-/-}$ mice enhanced the potency of antibiotics to clear *S. aureus* sepsis (98). It is tempting to speculate that reduced host cell invasion in absence of ASM caused a prolonged exposure of the pathogen to antibiotics. Several FIASMAs are already approved drugs (99), and may have future use in infection treatment.

Materials and Methods

Cell culture

HeLa (ATCC CCL-2^{TM}), 16HBE14o-(kindly provided by Prof. Jan-Peter Hildebrandt, University of Greifswald) and EA.hy 926 (103) were cultivated in RPMI+GlutaMAX^{TM} medium (Gibco^{TM}, Cat. No. 72400054) supplemented with 10% (v/v) heat-inactivated (56°C at 30 min) fetal bovine serum (FBS, Sigma Aldrich, Cat. No. F7524) and 1 mM sodium pyruvate (Gibco^{TM}, Cat.No. 11360088).

HuLEC-5a (ATCC CRL-3244^{TM}) and HuVEC (Gibco^{TM}, Cat. No.C01510C) were cultured in MCDB131 medium (Gibco^{TM}, Cat. No. 10372019) supplemented with microvascular growth supplement (Gibco^{TM}, Cat. No. S00525), 2 mM GlutaMAX^{TM} (Gibco^{TM}, 35050061), 5 % (v/v) heat-inactivated (56°C at 30 min) FBS, 2.76 µM hydrocortisone (Sigma Aldrich, Cat. No. H0888), 0.01 ng/ml human epidermal growth factor (Pep Rotech, AF-100-15) and 1x Penicillin-Streptomycin (Gibco^{TM}, Cat. No. 15140122).

HeLa

HuLEC and HuVEC were detached by StemPro^{TM} Accutase^{TM} (Gibco^{TM}, Cat. No. A1110501) and seeded at the indicated density two days prior to the experiment, whereas HeLa, 16HBE14o- and EA.hy 926 were detached with TrypLE^{TM} (Gibco^{TM}, Cat. No. 12604013) and seeded one day prior to the experiment.

Generation of HeLa KO cell lines

Generation of K.O. cell lines is based on a previous protocol (74). sgRNAs (**Table 1**) were designed with the CHOPCHOP online tool (chopchop.cbu.uib.no) and synthesized as primer pairs (Sigma Aldrich). After phosphorylation with T4 nucleotide kinase (ThermoFisher, Cat. No. ER0031), the sgRNAs were cloned into the BbsI sites of pSpCas9(BB)-p2A-GFP (AddGene #48138, (74)) via forced ligation by BbsI (ThermoFisher, Cat. No. ER1012) and T4 Nulceotide ligase (ThermoFisher, Cat. No. EL0011). The constructs were then transformed in *E. coli* DH5α (ThermoFisher, Cat. No EC0112) and subsequently sequence verified.

HeLa cells were transfected with 1 µg each of two distinct sgRNA constructs using JetPrime (Polyplus) following the manufacturer's instructions. After 36-48h, cells were sorted for strong GFP expression in a FACS Aria III (BD Bioscience).To isolate ASM K.O. cell lines, we included a second sorting step. Therefore, the cells were incubated with 10 µM FRET probe (73) for 2 h at 37°C/5% CO_2 and sorted for cells negative in FITC emission, which correlates with the cellular ASM activity. The resulting cell pools were cultivated and tested for loss of expression of the respective sgRNA target by Western Blotting or ASM activity assays.

The following antibodies were used: mouse anti-PTK7 monoclonal IgG1 (bio-techne, Cat. No. MAB4499) with a horse radish peroxidase (HRP)-conjugated anti-mouse antibody (SantaCruz, Cat. No. sc-516102) and rabbit anti-TPC1 polyclonal (Thermo Fisher, Cat. No. PA5-41048) or rabbit anti-SARM1 monoclonal antibody (Cell Signaling Technology, Cat. No. 13022) with an HRP-conjugated anti-rabbit antibody (Biozol, Cat. No. 111-035-144).

Generation of reporter constructs

pLV-mRFP-CWT. mRFP and the cell wall-targeting domain of lysostaphin CWT were amplified using primers MP-mRFP-f and euk_mRFP-r, as well as infus-CWT-f and infus-CWT-r, respectively. As PCR templates we used pmRFP-LC3 (PMID 17534139) and pLV-YFP-CWT (104). The resulting

Purpose	Name	sequence (5´→ 3´)
Gene deletion	Syt7-sgRNA2 rv	AAACATGACGTCATTGCGGCTGAG
	Syt7-sgRNA2 fw	CACCCTCAGCCGCAATGACGTCAT
	Syt7-sgRNA1 rv	AAACCACGATGAGTCCGACCGCCG
	Syt7-sgRNA1 fw	CACCCGGCGGTCGGACTCATCGTG
	TPCN1-sgRNA2 rv	AAACAGGACGGCGCTCCTCATCTT
	TPCN1-sgRNA2 fw	CACCAAGATGAGGAGCGCCGTCCT
	TPCN1-sgRNA1 rv	AAACTTCAGACCCCGTAAGTAATC
	TPCN1-sgRNA1 fw	CACCGATTACTTACGGGGTCTGAA
	SARM1-sgRNA1 rv	AAACTAGAGTCGAGCAACGGCACG
	SARM1-sgRNA1 fw	CACCCGTGCCGTTGCTCGACTCTA
	SARM1-sgRN2 rv	AAACGCACGTAGAATAGTTGGCGA
	SARM1-sgRNA2 fw	CACCTCGCCAACTATTCTACGTGC
	SMPD1-sgRNA1 fw	CACCAGCCAGGGTGTTCTCATACC
	SMPD1-sgRNA1 rv	AAACGGTATGAGAACACCCTGGCT
	SMPD1-sgRNA2 fw	CACCGTTCTTTGGCCACACTCATG
	SMPD1-sgRNA2 rv	AAACCATGAGTGTGGCCAAAGAAC
Cloning	MP-mRFP-f	GAGACTAGCCTCGAGGTTTAAACCACCATGGCCTCCTCCGAGGACGTC
	euk_mRFP-r	GGAATGCCCGGGCGCGCCGGTGGAGTGGCGGC
	infus-CWT-f	ACCGGCGCGCCCGGGCATTCCACGCCCAATACAGGTTGG
	infus-CWT-r	GAGGTTGATTATCATATGACTAGTTCACTTTATAGTTCCCCAAAG
	eYFP-rev	CTTGTTGATGGGCTCGAGCCCTTGTACAGCTCGTCCATGCCGAG
	Pm-eYFP-inf	GACTAGCCTCGAGGTTTAAACCACCATGGTGAGCAAGGGCGAGGAG
	hLysenin-infus-r	GAGGTTGATTATCATATGACTAGTTCAGCCCACCACCTCCAG
	InfReamp-f	GGCTCGAGCCCATCAACAAG
	cYFP-f	ATGGTGAGCAAGGGCGAGGAGCTG
	his1-Flag-f	CATCATCATCATCACGGAGCGGACTACAAGGATGACGACGATAAGG
	his2-Inf-f	GACTAGCCTCGAGGTTTAAACCACCATGGCACATCATCATCATCATCA CGGAGCG
	Apex-YFP-infus-r	CGCCCTTGCTCACCATCTCGAGCCGTCCAGGGTCAGGCGCTCC
	Lamp1-as	GATAGTCTGGTAGCCTGCGTGACTCC
	EF1a_fwd	TCAAGCCTCAGACAGTGGTTC
	LV-WPRE-seq	GTATCCACATAGCGTAAAAGGAGC
	IRES-eYFP-rev	GCCCTTGCTCACCATGGTTGTGGCGCGCCATTATCATCGTGTTTTTC
	LampSP-Hibit-r	GCTAATCTTCTTGAACAGCCGCCAGCCGCTCACTGCTGACGCACAATG CATGAGGC
	Inf-Hibit-LAMP-f	GTTCAAGAAGATTAGCGCAATGTTTATGGTGAAAAATGGC
	Lamp-IRES-f	GGCTACCAGACTATCTAATGAACTAGGGGTAAATTCCGCC
	LgBit-LVPme-fwd	GAGACTAGCCTCGAGGTTTAAACCACCATGGTCTTCACACTCGAAGAT TTCG
	LgBit-LVSpe-rev	GAGGTTGATTATCATATGACTAGTTTAACTGTTGATGGTTACTCGG
	PmeInfus-SPLamp-fwd	GAGACTAGCCTCGAGGTTTAAACCACCATGGCGGCCCCCGGCAGCGC
	LV-Spe-rev	GAGGTTGATTATCATATGACTAG
	cYFP-f	ATGGTGAGCAAGGGCGAGGAGCTG
	Lamp1-p2a-f	GGCTACCAGACTATCGGCAGTGGAGAGGGCAGAGGAAG
	p2a-eYFP-r	CTCGCCCTTGCTCACTGGGCCAGGATTCTCCTC

Table 1

Oligonucleotides used in this study

plasmid was termed pLV-mRFP-CWT. The PCR products were subsequently cloned into pLVTHM (101) restricted with PmeI and SpeI by InFusion cloning following the manufacturer's recommendations.

pLV-YFP-Lysenin. The sequence of the sphingomyelin-binding but not pore-forming earthworm toxin mutant Lysenin^{W20A} (82) was synthesized by GeneArt (ThermoFisher) adapted to human codon usage and with a 5' flanking region encoding a glycine-rich linker sequence. The synthesized fragment served as template for subsequent PCRs.

Lysenin^{W20} was amplified using oligonucleotides InfReamp-f and hLysenin-infus-r (**Table 1**). YFP was amplified with Pm-eYFP-inf and eYFP-rev from template pLVTHM-YFP-CWT (104). Both PCR products were joined with pLVTHM vector restricted with PmeI and SpeI by InFusion cloning following the manuacturer's recommendations, thereby creating pLV-YFP-Lysenin.

Constructs were transformed into *E. coli* DH5α and sequences were validated. Plasmid preparations were performed using standard laboratory procedures. VSV-G-pseudotyped Lentiviral particles were generated by transfecting with each lentiviral vector as well as the plasmids pMD2.G (VSV G) and psPAX2 following the protocol of (101) and target cells were transduced as described previously (104).

Bacteria culture

S. aureus liquid cultures were either grown in 37 g/L brain heart infusion (BHI) medium (Thermo Fisher, Cat. No. 237300) or in 30 g/l TSB medium (Sigma Aldrich, Cat. No. T8907). *E. coli* liquid cultures were grown in LB medium [10 g/l tryptone/peptone (Roth, Cat. No. 8952.4,) 5g/l yeast extract (Roth, Cat. No 2363.2), 10 g/l NaCl (Sigma Aldrich, Cat. No. S5886)].

For agar plates, 15 g/L agar (Otto Norwald, Cat. No. 257353), was added to either TSB (*S. aureus*) or LB (E. coli).

Media and plates were supplemented with appropriate antibiotics [5 µg/ml erythromycin (Sigma Aldrich, Cat. No. E5389), 10 µg/ml chloramphenicol (Roth, Cat. No. 3886.2), 100 ng/ml Carbenicillin (Roth, Cat. No. 6344.2)].

Strains used in this study are listed in **Table 2**.

S. aureus growth curves

S. aureus overnight cultures were grown in BHI medium, and 1 ml was centrifuged at 14,000 for 1 min. Bacteria were washed thrice with DPBS and either resuspended in BHI (for amitriptyline and ARC39 treatment) or infection medium [MCDB131 medium (Gibco^{TM}, Cat. No. 10372019) complemented with 2 mM GlutaMAX^{TM} (Gibco^{TM}, 35050061) and 10 % (v/v) heat-inactivated (56°C at 30 min) FBS, for ionomycin treatment). Bacteria suspensions were diluted to OD_{600}=0.1 in the respective medium. Then, 400 µL per well of the suspension as well as respective blanks were transferred into a 48 well plate and OD_{600} was determined every 16 min in Tecan mPlex200 microplate reader.

S. aureus infections

Host cells were seeded in 6 well plates (2 x10^5 cells per well), 12 well plates (1 x10^5 cells per well) or 24 well plates (0.5x10^5 cell per well) either one day (HeLa, EA.hy 926, 16HBE14o$^-$) or two days (HuLEC, HuVEC) prior to the experiment. Host cells were washed thrice with DPBS and infection medium or Ca^{2+}-free infection medium [DMEM w/o calcium (Gibco^{TM}, Cat. No. 21068028) complemented with 10 % (v/v) heat-inactivated (56°C at 30 min) FBS and 200 µM BAPTA (Merck Millipore, 196418)] with or without 1.8 mM $CaCl_2$ (Roth, Cat. No. 5239.1) was added. If indicated,

Strain	Remarks	Source
S. aureus 6850		(105)
S. aureus 6850 Δ*hlb*	deficient in β-toxin production	(42)
S. aureus Cowan I	NCTC 8530, isolated from septic arthritis, agr dysfunction, low expression of toxins and proteases	ATCC 12598
S. aureus JE2	MRSA, USA300	(106)
S. aureus JE2 Δ*hla*	deficient in α-toxin production	(107)
S. aureus JE2 Δ*hlb*	deficient in β-toxin production	(11)
S. aureus JE2 SarAP1 Cerulean	Constitutive expression of the fluorescence protein Cerulean	This work
S. aureus JE2 SarAP1 mRFP	Constitutive expression of the fluorescence protein mRFP	(45)
S. aureus JE2 p*Cer*	Anhydrous tetracycline (AHT) inducible expression of Cerulean, construct published in (108) and was transduced into *S. aureus* JE2	This work
S. aureus JE2 p*Cer+hlb*	Anhydrous tetracycline (AHT) inducible expression of Cerulean and β-toxin, construct published in (108) and was transduced into *S. aureus* JE2	This work
S. aureus Cowan I *SarAP1 Cerulean*	Constitutive expression of the fluorescence protein Cerulean	This work
S. aureus 6850 SarAP1 Cerulean	Constitutive expression of the fluorescence protein Cerulean	(45)
S. aureus 6850 Δ*hlb* SarAP1 Cerulean	Constitutive expression of the fluorescence protein Cerulean	This work

Table 2

Bacterial Strains used in this study

cells were pretreated with compounds prior to infection (for details about treatments see **Table 1**). For experiments involving blocking of receptors by antibodies, respective solvent controls were implemented (final concentrations in infection medium: 0.002% (w/v) NaN_3 (Sigma Aldrich, Cat. No. S2002) 5% glycerol (Roth, Cat. No 3783.2) in 10% DPBS for 30 ng/ml anti-NRCAM antibody as well as well as 0.01 % (w/v) NaN_3 for 50 ng/ml anti-CD73 and 50 ng/ml anti-MELTF antibodies.

An *S. aureus* overnight culture grown in BHI containing the appropriate antibiotics was diluted to an OD_{600}=0.4 in the same medium. When inducible expression of genes was required for the strains *S. aureus* JE2 p*Cer* and *S. aureus* JE2 p*Cer+hlb,* 200 ng/ml anhydrous tetracycline (AHT, AcrosOrganics, 233131000) was added. The culture was grown to an OD_{600}= 0.6-1.0 and 1 ml bacterial suspension was harvested by centrifugation and washed twice with Dulbecco's Phosphate Buffered Saline (DPBS, Gibco^{TM}, Cat. No 14190169). The bacteria were resuspended in infection medium (or Ca^{2+}-free infection medium when infection was performed in absence of extracellular Ca^{2+}). The number of bacteria per ml in the suspension was determined with a Thoma counting chamber and the MOI was determined. If not indicated otherwise, an MOI=10 was used for infections.

The infection was synchronized by centrifugation 800xg/8 min/RT (end of the centrifugation: t=0). To determine bacterial invasion, the infection was stopped after 30 min (unless indicated otherwise) by removing extracellular bacteria with 20 µg/ml Lysostaphin (AMBI, Cat. No. AMBICINL) in infection medium for 30 min. Then, host cells were washed thrice with DPBS, lysed by addition of 1 ml/well (12 well plate) or 0.5 ml/well (24 well plate) Millipore water and the number of bacteria in lysates was determined by plating serial dilutions (10^{-1}, 10^{-2}, 10^{-3}) on TSB agar plates. Plates were incubated overnight at 37 °C and colony forming units (CFU) were enumerated. To determine invasion efficiency, the number of bacteria determined in tested samples were normalized to untreated controls (set to 100%). For measuring invasion dynamics, the number of bacteria was either normalized to the 30 min time point of untreated controls or to the corresponding time points of untreated controls.

If later time points in infection were investigated, infection medium containing 2 µg/ml Lysostaphin was added to the cells until the indicated time.

For testing bacterial susceptibility to inhibitors ("survival") or adherence, Lysostaphin treatment was omitted, and host cells were immediately lysed by addition of Millipore water or washed five times with DPBS and subsequently lysed with Millipore water, respectively. The number of bacteria in lysates was determined by CFU counting (dilutions 10^{-2}, 10^{-3}, 10^{-4}) on TSB agar plates.

Phagosomal escape assays

For phagosomal escape assays with amitriptyline, ARC39, PCK310 and β-toxin, HeLa cells expressing RFP-CWT and Lysenin^{W20A}-YFP were infected with the indicated *S. aureus* strain as described in the previous paragraph. Infections with *S. aureus* JE2 p*Cer* and *S. aureus* JE2 p*Cer+hlb* were carried out in presence of 200 ng/ml AHT.

For phagosomal escape assays, HeLa cells expressing RFP-CWT and Lysenin^{W20A}-YFP were infected for 10 min (early invaders) or 30 min (early and late invaders) at the indicated MOIs.

For the phagosomal escape assays with two *S. aureus* JE2 strains expressing different fluorescence proteins, HeLa cells expressing YFP-CWT were infected with an MOI=5 of an initial *S. aureus* JE2 strain (*S. aureus* JE2 SarAP1 Cerulean or *S. aureus* JE2 SarAP1 mRFP). After 12 min of infection, a second infection pulse was initiated by adding the second *S. aureus* JE2 strain (expressing the complementary fluorescence protein, *S. aureus* JE2 SarAP1 mRFP or *S. aureus* JE2 SarAP1 Cerulean, respectively) and centrifuging the bacteria on the host cells. To exclude that the type of fluorescence protein expressed by the *S. aureus* strains contributes to the experimental outcome, either strain was used for the first and second infection pulse (each combination: n=4).

Table 3

Host cell treatment with various compounds

compound	Supplier/Cat.No.	preincubation time	removal prior to infection?
2-APB (2-Aminoethoxydiphenylborate)	Sigma Aldrich/ D9754	75 min	removed prior to infection
8-Bromo-cADPR (8-Bromo-cyclic ADP ribose)	Biolog/065-005	75 min	present during infection
amitriptyline	Sigma Aldrich/ A8404	75 min	present during infection
ARC39	(71)	18 h, add fresh inhibitor for another 4h or as indicated	present during infection
BAPTA-AM	Merck Millipore/ 196419; (51)	25 min	removed prior to infection
BODIPY-FL™-C12-Ceramide	Santa Cruz/ sc-503923	90 min	present during infection
BODIPY-FL™-C12-Sphingomyelin	ThermoFisher/D7711	90 min	present during infection
CD38i/78c	Sigma Aldrich/ 5387630001	75 min	present during infection
Ionomycin	Sigma Aldrich/ I0634	75 min	present during infection
PCK310	To be published elsewhere, patent pending	4h or as indicated	present during infection
trans-Ned19	Sigma Aldrich/ SML2830; (55)	75 min	present during infection
Vacuolin-1	Biozol/ MCE-HY-118630	75 min	present during infection
β-toxin	Sigma Aldrich/ S8633	75 min	removed prior to infection if not indicated otherwise

Phagosomal maturation assays

To monitor phagosomal maturation, HeLa cells expressing mCherry-Rab5 and YFP-Rab7 or YFP-Rab7 and RFP-CWT were infected with *S. aureus* JE2 SarAP1 Cerulean.

After 3 h p.i. (unless indicated otherwise), cells were washed thrice with DPBS and fixed with 0.2% glutaraldehyde (Sigma Aldrich, Cat. No. 10333) / 4% paraformaldehyde in PBS (Morphisto, Cat. No 11762.01000) for 30 min at RT. Then, cells were washed thrice with DPBS, stained with 5µg/ml Hoechst 34580 (Thermo Fisher, Cat. No. H21486) and mounted in Mowiol [24g glycerol (Roth, Cat. No. 3783.2), 9.6 g Mowiol®4-88 (Roth, Cat. No 0713.2), 48 ml 0.2 M TRIS-HCl pH 8.5 (Sigma Aldrich, Cat. No T1503), 24 ml Millipore water].

Samples were imaged with a Leica TCS SP5 confocal microscope (Wetzlar, Germany; Software Leica LAS AF Version 2.7.3.9723) with a 40x immersion oil objective (NA1.3) and a resolution of 1024x1024 pixels [Cerulean (Ex. 458 nm/Em. 460-520), YFP (Ex. 514 nm/Em. 520-570nm), mRFP/mCherry (Ex. 561 nm/Em.571-635nm) and Hoechst 34580 (Ex. 405 nm/Em.410-460nm). At least 10 fields of view per samples were recorded.

Image analysis was performed in Fiji (109) using a previously described macro (108) that identifies and extracts individual bacteria from images as regions of interest (ROIs, see **Supp. Figure 6**). Subsequently, fluorescent intensity in every channel is measured to determine recruitment of fluorescence reporters to the bacteria. Results were exported as text files and proportion of bacteria that recruited individual reporters was determined with the Flowing 2 software (Turku Bioscience Center). Phagosomal escape rates were determined as the proportion of CWT-positive bacteria of the total number of intracellular bacteria. Similarly, the proportion of Lysenin^{W20A}-positive escape events, the ratio of Lysenin^{W20A}-/CWT-positive events and all CWT-positive events was calculated.

To measure the proportion of bacteria that were associated with Rab5 and/or Rab7, the proportion of Rab5- and/or Rab7-positive membranes around bacteria of all intracellular bacteria was determined. For experiments obtained with HeLa cells expressing YFP-Rab7 and RFP-CWT, bacteria that acquired RFP-CWT were removed from the dataset. Subsequently, the proportion of Rab7-positive bacteria in relation to all CWT-negative intracellular bacteria was determined.

Live cell imaging

The indicated host cells were seeded one (HeLa) or two days (HuLEC) prior to the experiments in µ-slide 8 well live cell chambers (ibidi, Cat. No 80826-90) with a density of 0.375×10^5 cells per well.

For monitoring host cell entry, cells were washed thrice with DPBS and were then, incubated for 90 min in infection medium with 1 µM BODIPY-FL-C_{12}-sphingomyelin (Thermo Fisher Cat. No. D7711), 10 µM BODIPY-FL-C_{12}-ceramide (Santa Cruz, Cat.No. sc-503923) or 10 µM visible-range FRET probe (73). Pretreatment with the FRET probe was performed in infection medium containing 1 % (v/v) FBS. Then, cells were washed thrice with DPBS and imaging medium [RPMI 1640 w/o phenol red (Gibco™, Cat. No. 11835030) containing 10% (v/v) heat-inactivated (56°C/30 min) FBS (Sigma Aldrich, Cat. No. F7524) and 30 mM HEPES (Gibco™, Cat.No. 15630080)] applied. Samples stained with the FRET probe were imaged in imaging medium containing 1% FBS and in presence of 10 µM FRET probe. Samples were infected with *S. aureus* JE2 SarAP1 Cerulean (FRET probe samples) or *S. aureus* JE2 SarAP1 mRFP (BODIPY-FL-C12-sphingomylein/-ceramide) at an MOI=50 without synchronization by centrifugation. Infection was monitored with a Leica TCS SP5 confocal microscope (Wetzlar, Germany) in intervals of 1 min by recording BODIPY-FL (Ex. 496 nm/Em. 500-535 nm), mRFP (Ex. 561 nm/Em.571) FITC (Ex. 488 nm/Em. 500-560), Cerulean (Ex. 458 nm/Em. 460-520), BODIPY-TR (Ex. 594 nm/Em. 610-680) and FRET (Ex.488 nm/ Em:610-680) channels. Cells were recorded with a 40x immersion oil objective and a resolution of 2048x2048 pixels. For samples stained with the FRET probe, 20 µg/ml Lysostaphin was added 40 min p.i. to remove extracellular bacteria.

For quantification of bacteria that associate with BODIPY-FL-C_{12}-sphingomyelin/-ceramide, individual bacteria in single frames were identified, extracted as ROIs and BODIPY-FL fluorescence in ROIs was measured in Fiji (109) as described before for phagosomal escape assays. Results were extracted as text files and the proportion of bacteria associating with BODIPY-FL was determined with Flowing2 (Turku Bioscience Center).

For monitoring intracellular *S. aureus* infection, host cells were pretreated and infected with *S. aureus* JE2 SarAP1 Cerulean (HeLa RFP-CWT/Lysenin^{W20A}-YFP) or *S. aureus* JE2 SarAP1 mRFP (HuLEC) as described in "*S. aureus* infection". HuLEC additionally were treated with 1 µM BODIPY-FL-C_{12}-sphingomyelin in infection medium for 90 min and washed thrice with DPBS before the indicated treatment was applied. After extracellular bacteria were removed with 20 µg/ml lysostaphin for 30 min, imaging medium containing 2 µg/ml lysostaphin was applied and infections were monitored in intervals of 5 min (HeLa RFP-CWT) or 20 min (HuLEC). Phagosomal escape was evaluated in Fiji as described above. To determine intracellular replication, the number of bacteria was determined for each individual frame as described before (108). The number of bacteria at each time point was then normalized to the number of bacteria detected in the first frame to calculate the relative replication.

ASM and NSM activity assays

Thin layer chromatography (TLC)-based ASM/NSM activity assays were adapted from previously published protocols (72). To determine cellular ASM activity, the indicated cell lines were seeded in 24 well plates with a density of $1x10^5$ cell per well one day (HeLa, EA.hy 926, 16HBE14o$^-$) or two days (HuLEC, HuVEC) prior to the experiment. Cells were either treated with 20 µM amitriptyline or with 10 µM ARC39 for 22 h (fresh inhibitor was applied after 18h) in infection medium. Then, cells were washed thrice and lysed by addition of 100 µL per well ASM lysis buffer [250 mM NaOAc pH 5 (Roth, Cat. No 6773.2), 0.1% Nonidet® P40 substitute (AppliChem. Cat. No. A1694,0250), 1.3 mM EDTA 1.3 mM (Roth, Cat. No. 8040.2), 1x protease inhibitor cocktail (Sigma Aldrich, 11873580001)] for 15 min/4°C. Cells were scraped from the substratum and protein concentration in the resulting lysates was measured with a Pierce™ bicinchoninic acid (BCA) assay kit (Thermo Fisher, Cat. No. 23227). 1 µg of protein was incubated with 100 µL ASM lysate assay buffer [200 mM NaOAc pH 5 (Roth, Cat. No 6773.2), 0.02 % Nonidet® P40 substitute (AppliChem. Cat. No. A1694,0250), 500 mM NaCl (VWR, Cat. No. 27810.364)] containing 0.58 µM BODIPY-FL-C_{12}-Sphingomyelin (Thermo Fisher, Cat. No. D7711) for 4h at 37°C and 300 rpm.

For measuring bacterial SMase/NSM activity in *S. aureus* cultures, an overnight culture of the indicated strain was grown in BHI medium, centrifuged 14.000xg and the supernatant was sterile filtered with 0.2 µm filter. 100 µL of sterile supernatant was incubated with 100 µL NSM assay buffer [200 mM HEPES, pH 7.0 (Roth, Cat. No. 6763.3), 200 mM $MgCl_2$ (Roth, Cat.No. 2189.2), 0.05% Nonidet P-40 (AppliChem. Cat. No. A1694,0250)] for 4h/37°C/300 rpm.

The reactions were stopped by the addition of 2:1 $CHCl_3$:MeOH (Roth, Cat. No. 3313.2 and Cat. No. 8388.6). Samples were vortexed, centrifuged at 13,000 x g for 3 min and 50-100 µL of the lower organic phase were transferred to a fresh tube. Samples were completely evaporated using a SpeedVac 5301 concentrator (Eppendorf), resuspended in 10 µL 2:1 $CHCl_3$/MeOH and spotted in 2.5 µL aliquots on a TLC plate (Alugram, Xtra Sil G/UV254, 0.2 mm/silica gel 60; VWR, Cat. No. 552-1006).

Plates were developed using 80:20 $CHCl_3$:MeOH and subsequently scanned with a Typhoon 9200 Scanner (Amersham). For quantification, intensities of the lower (SM) and the upper bands (Cer) were measured in Fiji (109) and activity was determined based on the reaction time, protein amount and SM/Cer ratios.

For the flow cytometry based read out, cells seeded in a density of $0.5x10^5$ cells per well in a 24 well plate either one day (HeLa, EA.hy 926, 16HBE14o$^-$) or two days (HuLEC, HuVEC) prior to the experiment. Cells were treated with ARC39 and amitriptyline as described above. Then, cells were washed thrice with DPBS and incubated with 10 µM FRET probe (73) for 2h in presence of the inhibitors. Subsequently, cells were washed thrice, detached with TrypLE™ (Gibco™, Cat. No. 12604013) and resuspended with 2% (v/v) FBS in DPBS. Samples were analyzed for FITC (Ex. 488 nm/Em. band pass 530/30 nm) and BODIPY-TR (Ex.: 561nm/ Em. band pass 695/40nm) fluorescence with Attune NxT flow cytometer (Thermo Fisher, Attune Cytometric Software v5.2.0). Cell populations were analyzed for FITC and BODIPY-TR mean fluorescence in Flowing 2 (Turku Bioscience Center) and FITC vs. BODIPY-TR ratios were calculated to determine arbitrary probe conversion.

Determination of β-toxin activity on living cells

HeLa cells were seeded in a 24 well plate with a density of $0.5x10^5$ cells per well. Cells were incubated with 1 µM BODIPY-FL-C_{12}-SM for 24h. Then, samples were washed thrice with DPBS and treated with 100 ng/ml β-toxin in infection medium or left untreated for 75 min. Subsequently, cells were washed thrice with DPBS, lysed with 200 µL MeOH and detached with a cell scraper. 400 µL $CHCl_3$ and samples were centrifuged for 5 min/25.000xg to remove cell debris. 100 µL of the sample were completely evaporated using a SpeedVac 5301 concentrator (Eppendorf), resuspended in 10 µL 2:1 $CHCl_3$/MeOH and spotted in 2.5 µL aliquots on a TLC plate (Alugram, Xtra Sil G/UV254, 0.2 mm/silica gel 60; VWR, Cat. No. 552-1006). TLC was developed with 80:20 $CHCl_3$:MeOH and scanned with a Typhoon RGB scanner (Amersham). Fluorescence intensities were evaluated in Fiji and ratios ceramide vs. SM were calculated.

Construction of HiBit-SP-LAMP1-p2A-eYFP

To visualize the transient surface exposure of lysosomal membranes on the plasma membrane of epithelial cells we made use of a split NanoLuc luciferase. We inserted a region encoding the 11 amino acid residues encompassing HiBiT peptide of the split NanoLuc (Promega) between the signal peptide and the N-terminus of mature LAMP1.

For that purpose, we amplified a 144 bp fragment encoding the signal peptide of LAMP1 (SP_{LAMP1}, PmeInfus-SPLamp-fwd & LAMP1SP-HiBiT-r), the 1.2 kb coding region of mature LAMP1 (mLAMP1, Inf-HiBiT-LAMP-f & LAMP1-as) and eYFP (cYFP-f & LV-Spe-rev) via PCR using the indicated primer pairs respectively and using LAMP1-YFP containing plasmid as DNA template (41). A p2A peptide encoding sequence was amplified with oligonucleotides Lamp1-p2a-f and p2a-eYFP-r using pspCas9 (74) as a template. As vector we used the PmeI/SpeI-restricted pLVTHM (101).

The resulting DNA Fragments contained overlapping ends of at least 16 bp, which were assembled by an InFusion enzyme mix (Takara Biotech) according to the manufacturer 's instructions. The resulting plasmid was transformed into chemically competent *E. coli* DH5α and plated on LB agar containing 100 µg/ml ampicillin. Colonies were inoculated, and plasmid were prepared from the resulting overnight cultures using the Qiagen Plasmid Mini kit. Sequences were determined using Sanger sequencing (Microsynth) thereby establishing the correct assembly of the vector. Pseudotyped lentivirus particles were generated in HEK293 cells and transduction of HeLa cells was conducted following a published protocol (101).

eYFP production in transgenic cells here served as reporter for transfection of the cells and production of the SP-HiBiT-LAMP1 proportion of the fusion protein. eYFP-positive cells were FACS-sorted for medium eYFP expression levels and the resulting cell pool was named HeLa SP_{LAMP}1-HiBit-LAMP1-p2A-EYFP.

Determination of externalization of lysosomal membranes using split NanoLuc biolumin-escence

For measurement of LAMP1 externalization, $4x10^4$ per well of HeLa SP_{LAMP}1-HiBit-LAMP1-p2A-EYFP, hereafter named HeLa HiBiT-LAMP1, were seeded in white flat-bottom 96 well plates (Nunc™ MicroWell™ 96 Wells, Nunclon Delta-treated, Cat. No. 136101) and grown overnight yielding $8x10^4$ cells at the next day. Bioluminescence was detected using the Promega Nano-Glo HiBiT Extracellular Detection Kit (Promega # N2420).

For infection with *S. aureus*, bacteria from an overnight culture were diluted to an OD600 of approximately 0.5 into 10 ml TSB and grown for 1 h. In case a preincubation with inhibitors was required (e.g. as for vacuolin-1), cells were washed twice with DPBS, and inhibitors were added in indicated concentrations in a final volume of 200 µl per well.

Extracellular detection reagent was prepared by mixing 4 ml extracellular buffer, 20 µl LgBit protein and 40 µl bioluminescent substrate [Promega, Cat. No. N2420]. 100 µl extracellular detection reagent was added per well and luminescence was measured five times in 2 min intervals on a TECAN Infinite Pro 200 plate reader (37 °C, attenuation: OD1, integration time 250 ms). Then, inhibitors or bacteria were added accordingly, and luminescence measurements were continued in 2 min intervals using the settings above. The relative luminescence units (RLUs) determined in samples infected with *S. aureus* were scaled to the untreated control (set to 0%) and cells treated with 1 µM ionomycin (set to 100 %). The results describe the LAMP1 externalization as proportion of all lysosomes that can be released from host cells by ionomycin treatment.

Determination of cellular sphingolipid profiles via HPLC-MS/MS

HeLa wildtype or ASM K.O. cells were seeded with a density of $3.5x10^6$ cells per well in a 6-well plate. wildtype cells were treated as described in **Table 3** with 10 µM ARC39 or 20 µM amitriptyline. Then, cells were washed thrice with DPBS, lysed with 250 µL cold MeOH and detached with a cell scraper on ice. Wells were rinsed with an additional 250 µL MeOH to ensure entire sample collection.

Cell suspensions were then subjected to lipid extraction using 1 mL MeOH/$CHCl_3$ (2:1, v:v) as described before (110). The extraction solvent contained C17 ceramide (C17 Cer) and d_{31}-C16 sphingomyelin (d_{31}-C16 SM) (both Avanti Polar Lipids, Alabaster, USA) as internal standards. Chromatographic separations were achieved on a 1290 Infinity II HPLC (Agilent Technologies, Waldbronn, Germany) equipped with a Poroshell 120 EC-C8 column (3.0 × 150 mm, 2.7 µm; Agilent Technologies). MS/MS analyses were carried out using a 6495C triple-quadrupole mass spectrometer (Agilent Technologies) operating in the positive electrospray ionization mode (ESI+). Cer and SM were quantified by multiple reaction monitoring (qualifier product ions in parentheses): $[M-H_2O+H]^+ \rightarrow m/z$ 264.3 (282.3) for all Cer and $[M+H]^+ \rightarrow m/z$ 184.1 (86.1) for all SM subspecies (C16, C18, C20, C22, C24 and C24:1) (110). Peak areas of Cer and SM subspecies, as determined with MassHunter Quantitative Analysis software (version 10.1, Agilent Technologies), were normalized to those of the internal standards (C17 Cer or d_{31}-C16 SM) followed by external calibration in the range of 1 fmol to 50 pmol on column.

Cytotoxicity Assays

For all cytotoxicity assays, HuLEC were seeded in 24 well plates with a density of $0.5x10^5$ cells per well two days prior to the experiment. Then, cells were infected with *S. aureus* JE2 with the indicated MOI as described in "*S. aureus* infection" and cytotoxicity was determined 21 h p.i. For cytotoxicity measurements upon ionomycin treatment, cells were incubated with the indicated concentration of ionomycin in infection medium for 75 min and then, cytotoxicity assays were conducted.

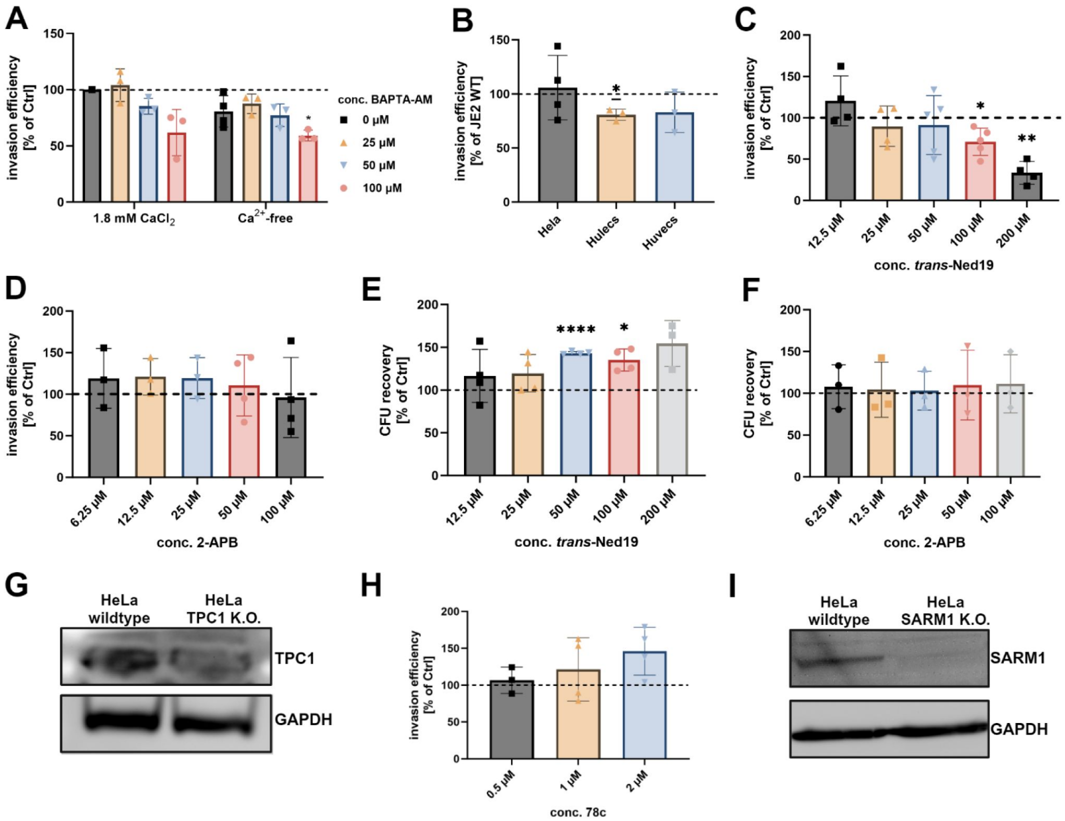
Lactate dehydrogenase (LDH) assay was performed with the Cytotoxicity Detection KitPLUS (LDH, Sigma Aldrich, Cat. No. 4744934001) according to manufacturer's instructions.

For annexin V and 7-Aminoactinomycin D (7-AAD) assays, cells were washed thrice with DPBS, detached with 250 µL per well trypsin and resuspended in 250 µL per well staining buffer [1.7% (v/v) APC Annexin V (BD Pharmingen^{TM}, Cat. No. 550475), 1.7% (v/v) 7-AAD (BD Pharmingen^{TM}, Cat No. 559925), 2% (v/v) heat-inactivated (56°C/30 min) FBS (Sigma Aldrich, Cat. No. F7524), 4 mM $CaCl_2$ (Roth, Cat. No. 5239.1) in DPBS]. After incubation for 10 min/RT, cells were analyzed with an Attune NxT flow cytometer for 7-AAD (Ex. 488 nm/Em. band pass 695/40 nm) and APC (Ex. 637 nm/Em. band pass 670/14 nm). Cell populations were analyzed with Flowing2 (Turku Biosciences Center) and gates were adjusted according to untreated control cells. To determine the proportion of cells that remained attached to the substratum during the infection, the number of cells was determined based on the number of detected single cells, flow rate and sample volume.

Statistical analysis

Statistical analysis was performed in GraphPad prism (V10.1.2). One-sample t-test was used for analysis of normalized data sets. Otherwise, one- or two-way ANOVA, dependent on the number of variables, was used in combination with appropriate multiple comparisons testing. Details about sample size and corresponding statistical analysis can be found in respective figure legends. All data are shown as mean ± standard deviation.

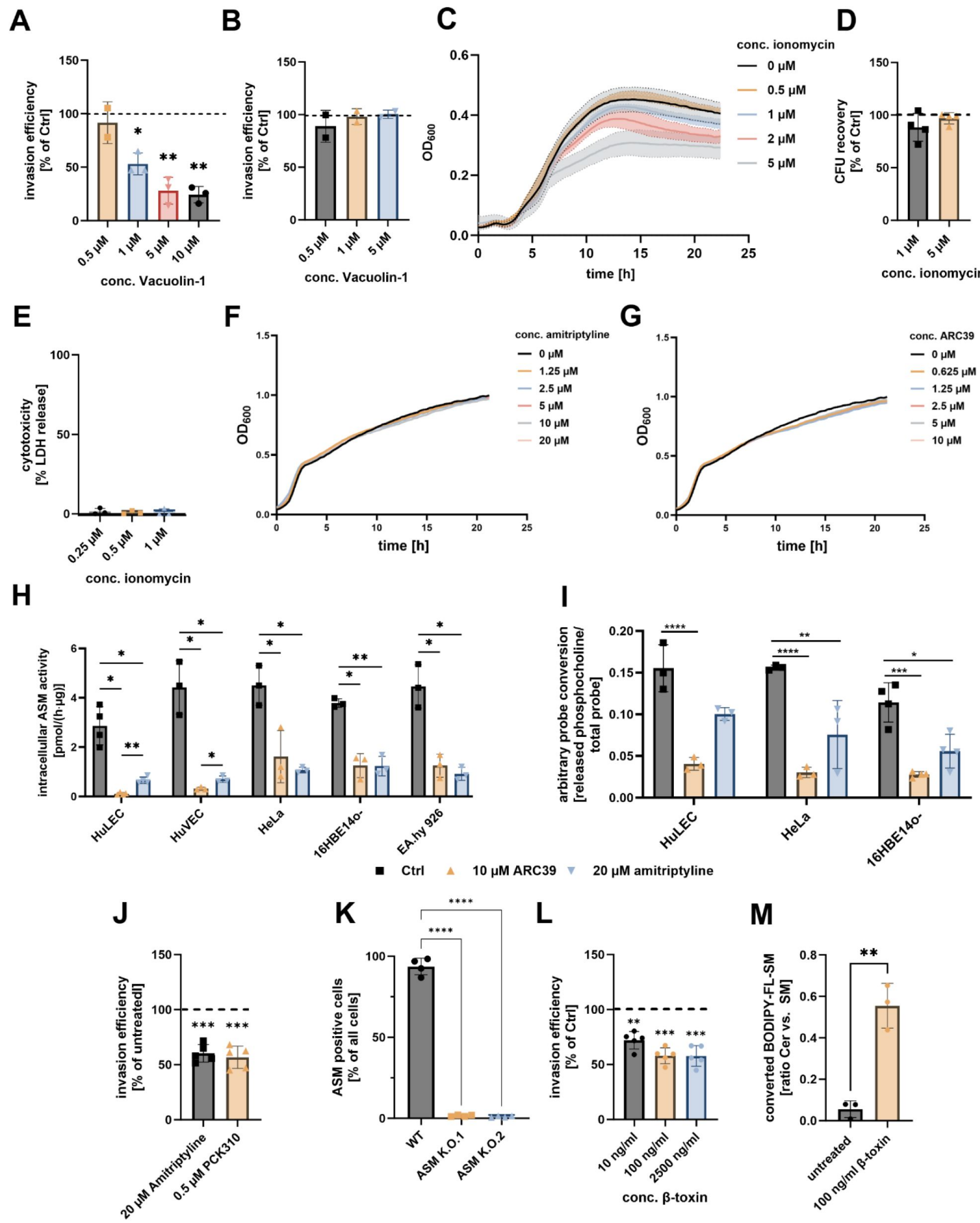
Supplementary Figures



Supp. Figure 1

Supporting information for Ca^{2+}-dependent internalization of *S. aureus*.

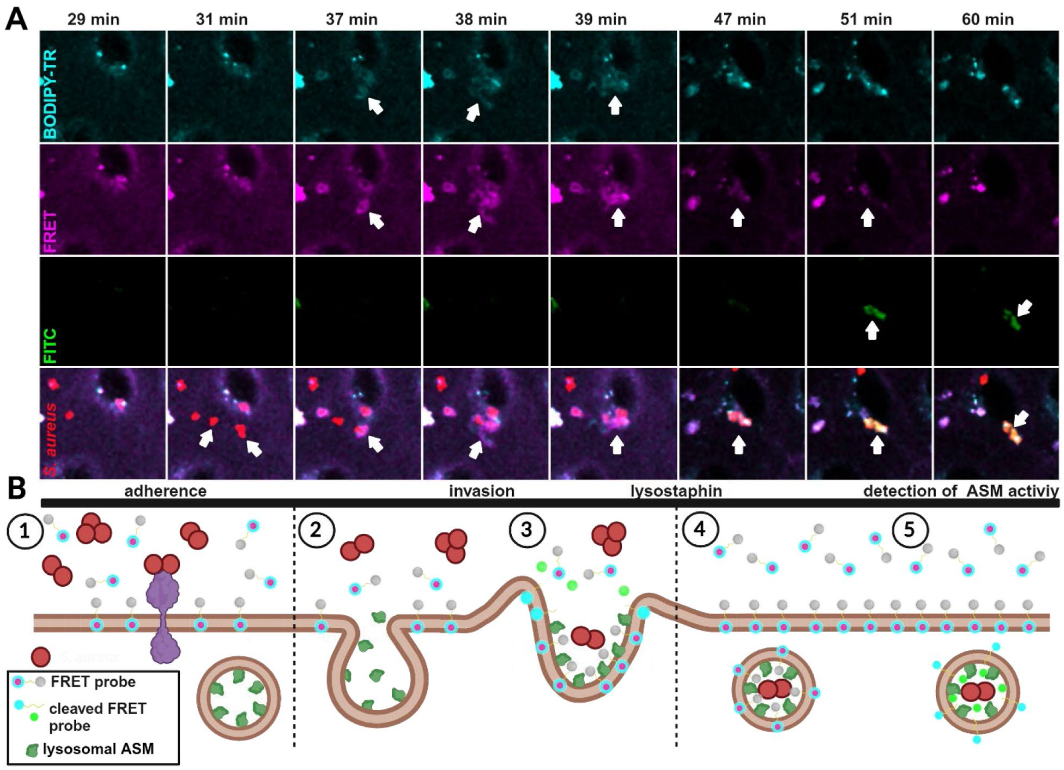
(A) Chelation of intracellular but not presence of extracellular Ca^{2+} affects *S. aureus* internalization by HeLa cells. HeLa cells were preloaded with varying concentrations of the cell-permeable Ca^{2+} chelator BAPTA-AM and then infected with *S. aureus* JE2 in presence (+1.8 mM CaCl2) or absence (Ca^{2+}-free) of Ca^{2+} in the cell culture medium. Invasion efficiency was determined by CFU plating. (**B) α-toxin plays only a subordinate role during *S. aureus* uptake by host cells.** HeLa, HuLEC or HuVEC were infected with *S. aureus* JE2 wildtype or a strain deficient in α-toxin production (Δ*hla*). Invasion efficiency was determined by CFU plating. The number of intracellular bacteria was normalized to samples infected with the wildtype. **(C, D) *trans*-Ned19 but not 2-APB affects invasion of *S. aureus* JE2 in HeLa.** HeLa cells were pre-treated with *trans*-Ned19 (**C**) or 2-APB (**D**). 2-APB was removed from the cells shortly before infection to avoid direct contact of inhibitor and bacteria (due to a bactericidal effect of 2-APB; data not shown). Then, cells were infected with *S. aureus* JE2 and invasion efficiency was determined. **(E, F) *trans*-Ned19 and 2-APB have no effect on *S. aureus* JE2 in our infection protocol.** HeLa cells were pre-treated with or 2-APB (**E**) or *trans*-Ned19 (**F**). 2-APB was removed from the cells shortly before infection. Then, cells were infected with *S. aureus* JE2 and, after 30 min, total bacteria (extra- and intracellular) were recovered by CFU plating. (**G) Reduction of TPC1 expression in a Cas9-treated cell pool (HeLa TPC1 K.O.).** TPC1 as well as GAPDH (loading control) was detected in lysates of wildtype and TPC1 K.O. cells via Western blot. **(H) CD38 is not involved in *S. aureus* internalization.** HeLa cells were pretreated with the CD38 inhibitor 78c and invasion efficiency of *S. aureus* JE2 was determined. (**I) Reduction of SARM1 expression in a Cas9-treated cell pool (HeLa SARM1 K.O.).** TPC1 as well as GAPDH (loading control) was detected in lysates of wildtype and SARM1 K.O. cells via Western blot. Statistics: one sample t-test, Bars represent mean +/-SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.



Supp. Figure 2

Supporting information for the involvement of lysosomal exocytosis and ASM in *S. aureus* invasion

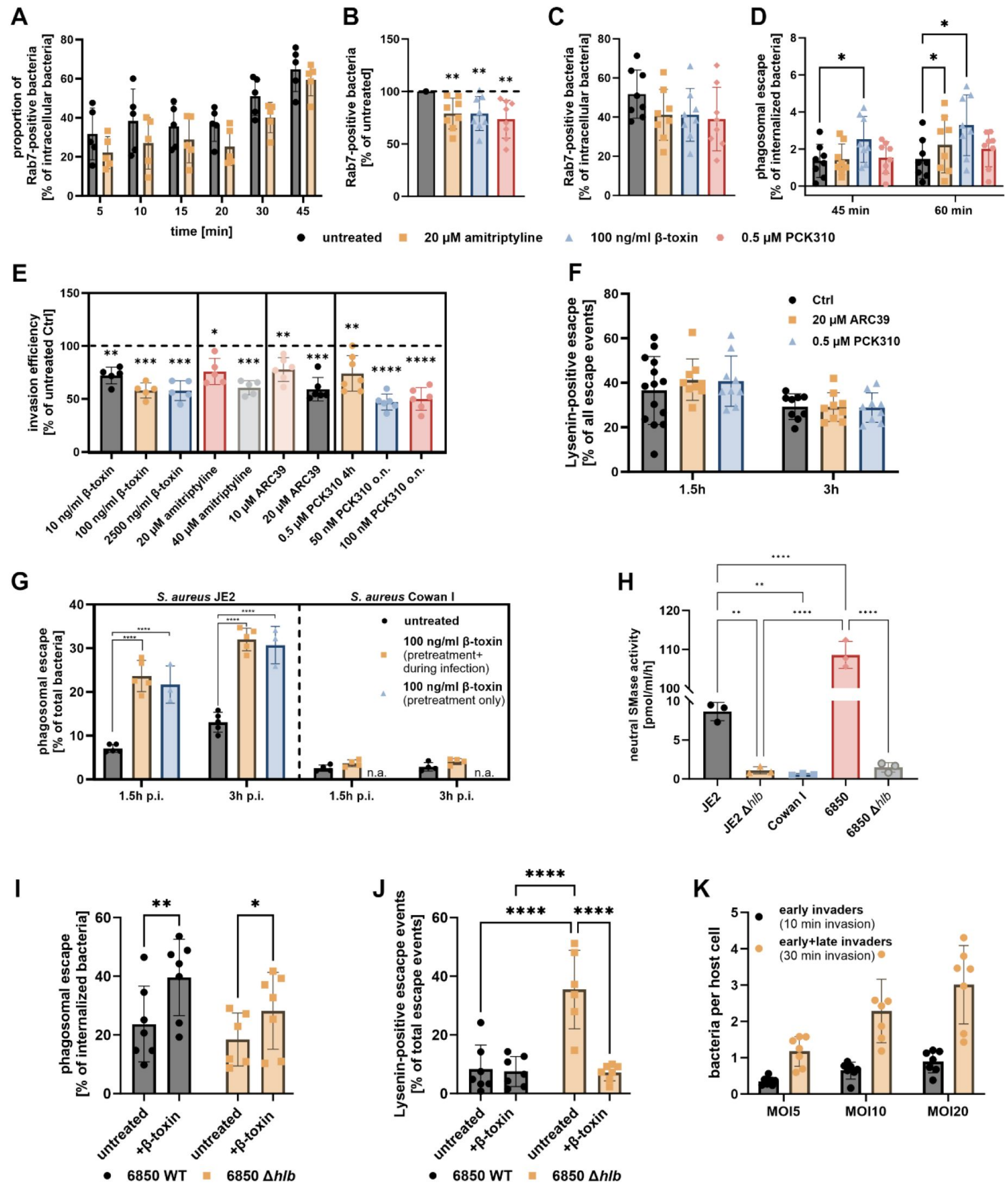
(A) High ionomycin concentrations interfere with growth of *S. aureus* JE2. *S. aureus* JE2 was grown in in presence of varying concentrations of ionomycin. Growth (OD600) was determined in a microplate reader. **(B) Ionomycin does not affect survival of *S. aureus* JE2 during infection of host cells.** HeLa cells were pretreated with varying concentrations of ionomycin and subsequently, infected with *S. aureus* JE2. After 30 min, cells were lysed and surviving extra- and intracellular bacteria were recovered by CFU count. (**C) Ionomycin does not affect host cell survival.** HeLa cells were treated with the indicated concentrations of ionomycin, and cytotoxicity was determined by measuring in lactate dehydrogenase (LDH) release. **(D) Blocking lysosomal exocytosis by Vacuolin-1 reduces *S. aureus* invasion in HeLa.** HeLa cells were treated with increasing concentrations of the lysosomal exocytosis inhibitor Vacuolin-1. The invasion efficiency of *S. aureus* JE2 was determined 30 min p.i. **(E) Vacuolin-1 has no bactericidal effect.** HeLa cells were treated with Vacuolin-1 and infected with *S. aureus* JE2. After 30 min, total bacteria (extra- and intracellular) were recovered by CFU plating and normalized to untreated controls. **(F, G) The ASM inhibitors amitriptyline and ARC39 have no effect on growth of *S. aureus*.***S. aureus* JE2 was grown in presence of varying concentrations of amitriptyline (**F**) or ARC39 (**G**) in BHI medium and growth was determined by measuring OD600 in a microplate reader. **(H, I) ASM activity and ASM inhibition by ARC39 and amitriptyline are similar among human cell lines.** Several cell lines were treated with amitriptyline or ARC39 and ASM activity within cell lysates was assessed by either thin layer chromatography-based ASM activity assays, detecting the conversion of BODIPY-C12-SM (**H**), or by a flow cytometry-based ASM activity assay and conversion of visible-range FRET probe (**I**). **(J) Microscopy-based measurement of invasion efficiency in amitriptyline- and PCK310-treated Hela cells.** Hela cells were treated with 20 μM amitriptyline (75 min) or 0.5 μM PCK310 (4 h) and infected with *S. aureus* JE2 expressing Cerulean. Number of intracellular bacteria per host cell was determined by CLSM. n=5. **(K) Validation of ASM K.O. cell pools.** ASM activity in HeLa wildytpe or ASM K.O.s was determined with the visible-range FRET probe and flow cytometry. The proportion of ASM-positive cells in the cell population was determined. n=4. **(L) Removal of SM from the plasma membrane by β-toxin affects *S. aureus* host cell entry in HeLa.** HeLa cells were pretreated with the indicated concentrations of β-toxin and subsequently the invasion efficiency of *S. aureus* JE2 was determined. **(M) Treatment of HeLa with the bacterial SMases β-toxin results in generation of ceramide.** HeLa cells were preloaded with 1 μM BODIPY-FL-C12-SM and treated with β-toxin for 75 min. Quantities of ceramide and SM were determined by TLC and ratio of ceramide vs. SM were calculated. Statistics: one sample t-test (D, J, L), mixed-effects model (REML) and Tukey's multiple comparison (H, I), one-way ANOVA and Dunnett's multiple comparison (K), unpaired Student's t-test (M). Bars represent mean +/-SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.



Supp. Figure 3

S. aureus associates with a visible-range SM probe during host cell invasion.

(A) Live cell imaging of *S. aureus* infection reveals association of the bacteria with a SM analog during cell entry. HuLEC were treated with 10 µM of a visible range FRET probe in presence of 1% FBS. Then, cells were infected with *S. aureus* JE2 in presence of the probe and infection was monitored by live cell confocal imaging. After 40 min, lysostaphin was added to remove extracellular bacteria. The BODIPY-TR signal demonstrates the localization of the probe, the FRET signal indicates an intact, non-metabolized probe, whereas FITC fluorescence indicates probe cleavage by ASM. Bacteria of interest (white arrows) adhere to host cell between 29 and 31 min p.i. **(B) Hypothetical model of *S. aureus* invasion.** (1). The interaction of bacteria with the host cell surface triggers lysosomal exocytosis, release of ASM (2) and the uptake of the bacteria (cmp. to A: at 37-39 min p.i.) by ASM-dependent membrane remodeling (3). ASM is co-internalized together with bacteria (4) and subsequently, cleaves the probe within the *S. aureus*-containing phagosome (5) (cmp: to A: increasing FITC signal starting at 47 min p.i.). Created in *BioRender*.

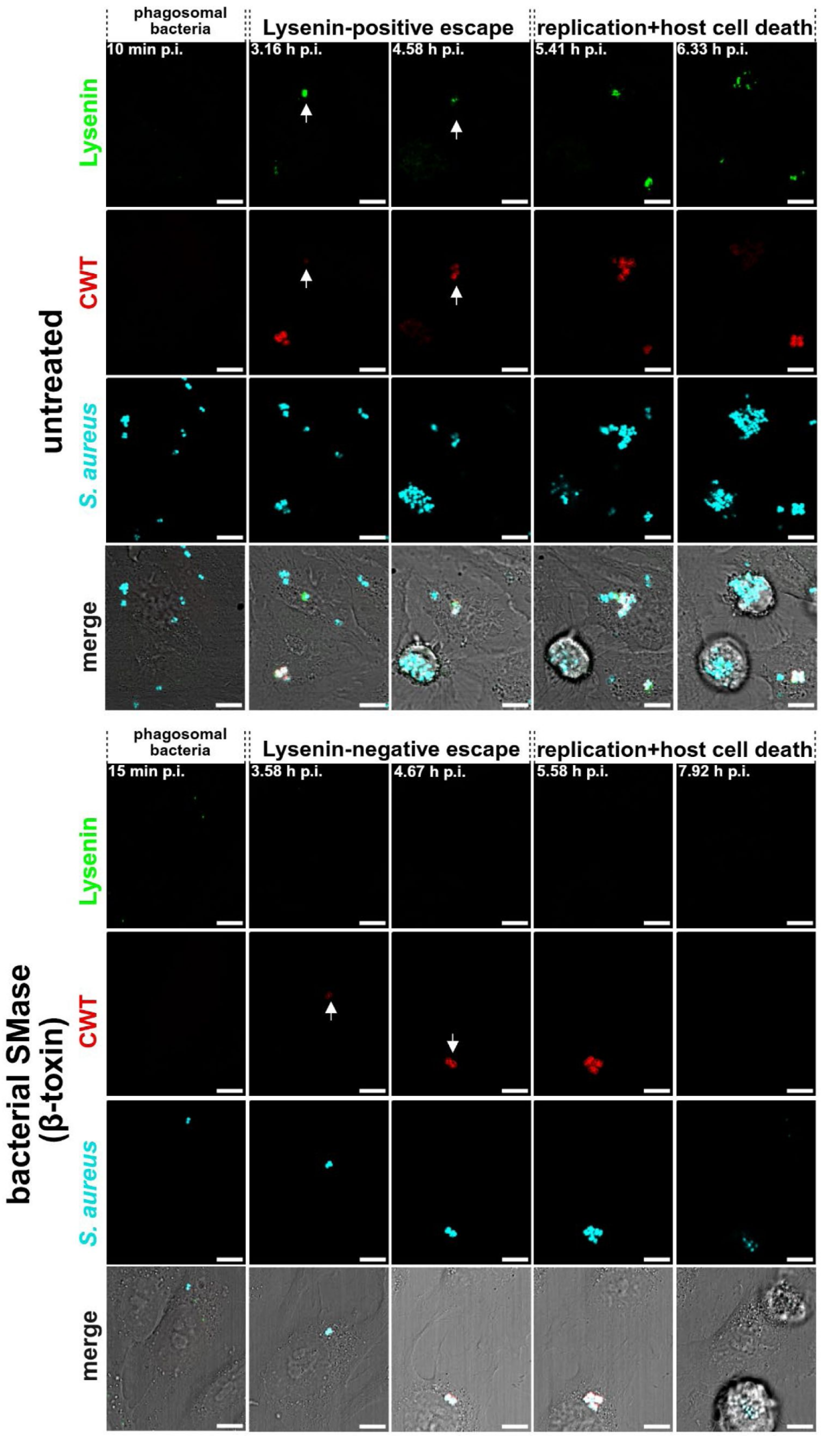


Supp. Figure 4

Supporting information for phagosomal maturation and phagosomal escape measurements.

(A) amitriptyline treatment reduces the proportion of Rab7-positive bacteria. HeLa cells expressing YFP-Rab7 and mCherry-Rab5 were infected with *S. aureus* JE2 for the indicated time point and extracellular bacteria were removed with lysostaphin for 15 min. Then, samples were fixed and imaged by CLSM. Proportion of bacteria associated with YFP-Rab7 was determined. n=5. **(B, C) Reduced association of *S. aureus* with Rab7 is not due to changes in phagosomal escape.** HeLa cells expressing YFP-Rab7 and the phagosomal escape marker RFP-CWT were treated with amitriptyline, PCK310 or β-toxin and subsequently, infected with *S. aureus* JE2 for 45 min and analyzed by CLSMs. Bacteria that escaped from phagosomes were subtracted from the data set and the proportion of residual bacteria associated with Rab7 was determined (C). Results of treated samples were normalized to untreated controls (B, n=7). **(D) Phagosomal escape assay in early infection.** A HeLa reporter cell line expressing YFP-Rab7 and RFP-CWT was treated with amitriptyline, β-toxin and PCK310 and subsequently infected with *S. aureus* JE2 for 30 min.

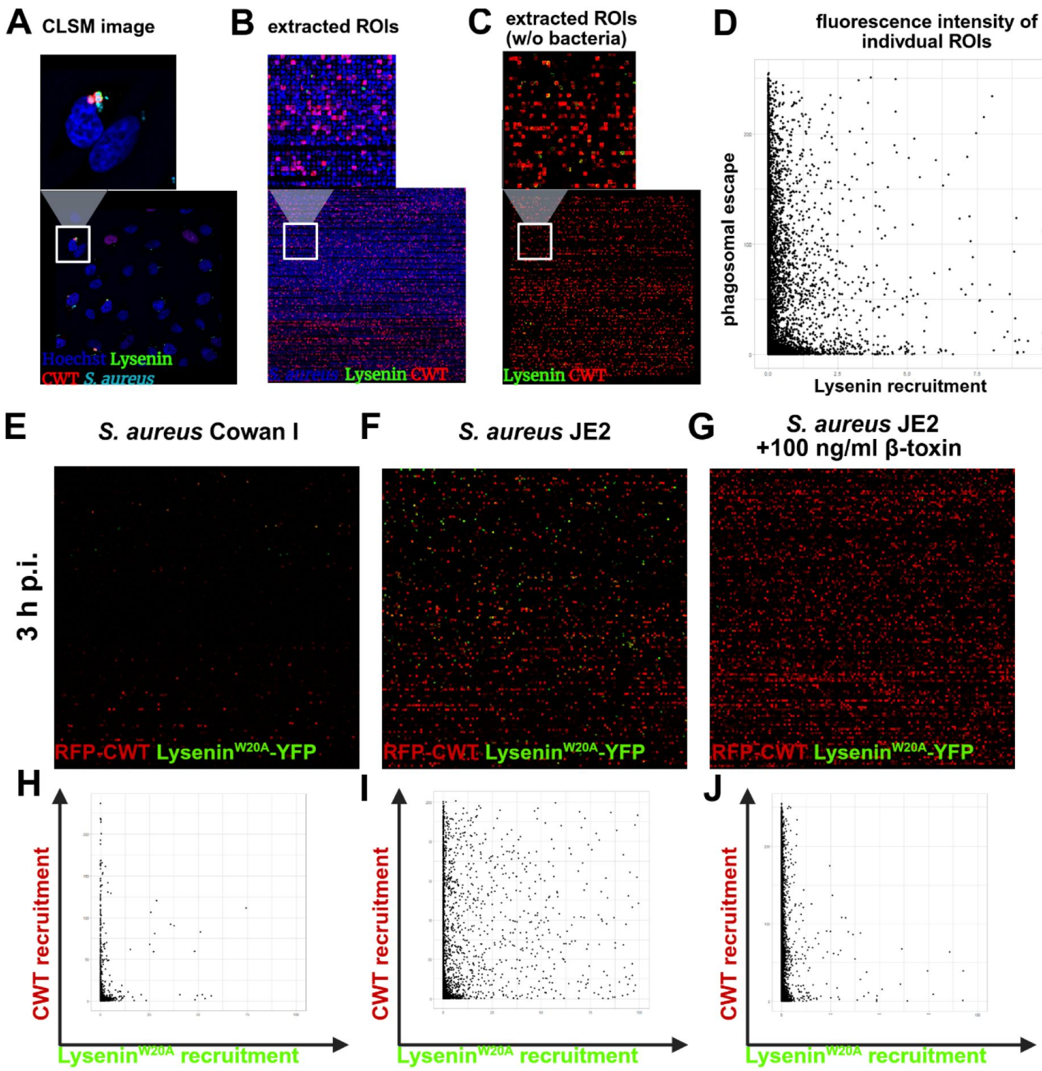
Extracellular bacteria were removed by lysostaphin and the proportion of bacteria that acquired the phagosomal escape marker RFP-CWT was determined 45 and 60 min p.i. (n=8). **(E) β-toxin treatment and ASM inhibition reduced invasion efficiency of *S. aureus* JE2 in a dual reporter HeLa cell line.** A HeLa reporter cell line expressing RFP-CWT and Lysenin^{W20A}-YFP was treated with the indicated concentrations of β-toxin and amitriptyline for 75 min, the indicated concentrations of ARC39 for 22 h, 0.5 µM PCK310 for 4h or 50/100 nM PCK310 for 22h (o.n.) and invasion efficiency of *S. aureus* JE2 was determined. (n≥5) **(F) Inhibition of ASM has no influence on the proportion of Lysenin^{W20A}-positive escape events.** The dual reporter cell line was infected with *S. aureus* JE2. Extracellular bacteria were removed and the proportion of bacteria that acquired the phagosomal escape marker RFP-CWT as well as the SM reporter Lysenin^{W20A}-YFP was determined by CLSM. The proportion of escaped (RFP-CWT-positive) bacteria that additionally acquired Lysenin^{W20A}-YFP was calculated. (n=9). **(G) β-toxin pretreatment increases phagosomal escape of *S. aureus* JE2 but not *S. aureus* Cowan I.** RFP-CWT and Lysenin^{W20A} -YFP expressing HeLa were treated with 100 ng/ml β-toxin infected with *S. aureus* strains JE2 or Cowan I. Thereby, β-toxin was either removed prior to infection (pretreatment only) or was present during the whole experiment (pretreatment + during infection). By CLSM, proportions of bacteria that recruited RFP-CWT (phagosomal escape) were determined (n≥3). **(H) *S. aureus* strains vary in β-toxin expression.** The neutral SMase activity in culture supernatants of the indicated *S. aureus* strains or isogenic β-toxin mutants (Δ*hlb*) was determined by a TLC-based approach (n=3). **(I, J) β-toxin pretreatment of host cells but not β-toxin expression of endocytosed *S. aureus* affects phagosomal escape.** HeLa RFP-CWT/Lysenin^{W20A} -YFP were infected with *S. aureus* 6850 or an isogenic strain deficient in β-toxin production (*Δhlb*). The proportion of bacteria that escaped from the phagosome (I) as well as the percentage of phagosomal escape events that were Lysenin^{W20A} -positive (J) were determined 3h p.i. (n=6). **(K) Increasing the MOI compensates for lower numbers of invading bacteria at shorter infection times.** HeLa cells were either infected for 10 min or 30 min with varying MOIs. Then, the number of intracellular bacteria per host cell was determined 3h p.i. by CLSM. (n=7). Statistics: One samples t-test (B, E), mixed effects analysis (REML) and Dunnett's multiple comparison (D), Two-way ANOVA with Tukey's multiple comparison (G), One-way ANOVA with Tukey's multiple comparison (H), mixed effects analysis (REML) and uncorrected Fisher's LSD (I), mixed effects analysis (REML) with Šídák's multiple comparison (J).Bars represent mean +/-SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.



Supp. Figure 5

Validation of Lysenin^{W20A}-YFP and RFP-CWT reporter cell line

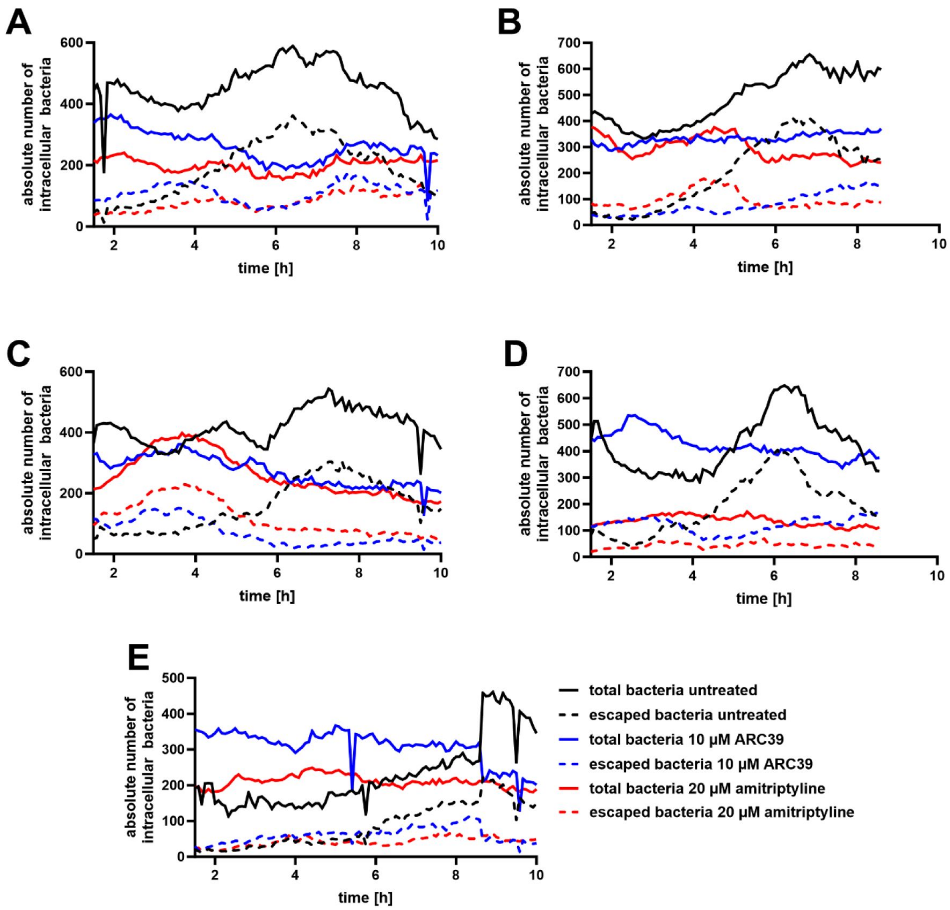
Hela cells expressing the SM reporter Lysenin$^{W20A\text{-}YFP}$(green) and the phagosomal escape reporter RFP-CWT (red) were either pretreated with 100 ng/ml bacterial SMase (A) or left untreated (B). Cells were infected with Cerulean-expressing *S. aureus* JE2 (cyan), extracellular bacteria were removed, and intracellular infection was monitored by time-lapse imaging at a Leica TCS SP5 microscope in 5 min intervals. White arrows indicate Lysenin-positive (A)or Lysenin-negative (B) escape events. Scale bars: 10 μm. Created in *BioRender*.



Supp. Figure 6

Evaluation of phagosomal escape and Lysenin^{W20A} recruitment during *S. aureus* infection.

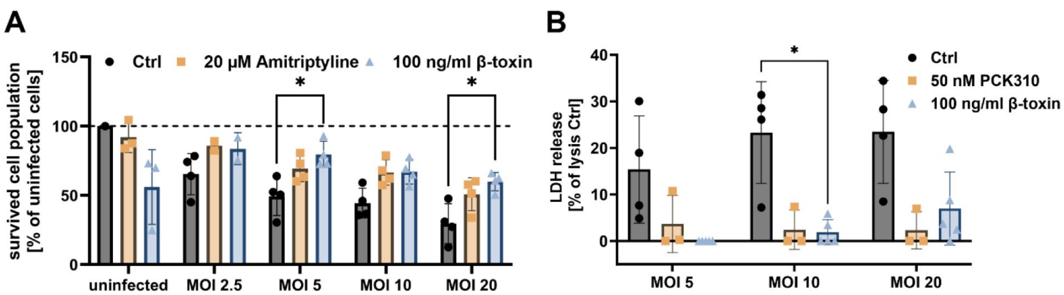
HeLa reporter cells expressing RFP-CWT and Lysenin^{W20A}-YFP were infected with *S. aureus* JE2, fixed and imaged by CLSM (**A**). Individual bacteria within the images were identified as regions of interest (ROIs) and extracted in each color channel. All extracted ROIs are depicted in a montage (**B**). If bacteria signals are omitted, the montage of extracted ROIs shows the proportion of reporter recruitment. The average fluorescence intensity of both reporter fluorophores is measured within each individual ROIs is plotted as dot plots (**D**). From these results, the proportion of bacteria that recruited the markers (CWT-RFP = phagosomal escape and Lysenin^{W20A} = SM-rich-phagosome) is calculated. **(E-J) Exemplary escape assay.** Montages of extracted ROIs (without bacterial fluorescence) as well as the resulting intensity measurement in individual ROIs 3 h p.i. are depicted for infections of *S. aureus* Cowan I (E, H) and JE2 (F, I) as well as an infection with *S. aureus* JE2 where host cells were treated with 100 ng/ml β-toxin prior to infection (G, J). Data are quantified in **Figure 4, C**. Created in *BioRender*.



Supp. Figure 7

Individual replicates of time-dependent phagosomal escape and replication assays

HeLa cells expressing RFP-CWT were pretreated with 20 µM amitriptyline or 10 µM ARC39, infected with *S. aureus* JE2 and infection was monitored by live cell imaging (for further details see **Figure 6, A and B**). The absolute numbers of all intracellular bacteria and bacteria that escaped from the phagosome, which were detected during live cell imaging, are depicted.



Supp. Figure 8

Inhibition of ASM and β-toxin treatment increase number of host cells remaining attached to the substratum and reduce host cell lysis during infection. HuLEC were treated with amitriptyline or β-toxin

(**A**) or PCK310 and β-toxin (**B**) and were then infected with *S. aureus* JE2. After 30 min, extracellular bacteria were removed and the number of cells remaining attached to the substratum was determined (**A**) or the proportion of host cells that were lysed during the infection was measured by lactate dehydrogenase (LDH) release (**B**). Statistics: Mixed-effects analysis and Dunnett's multiple comparison (A, B). Bars represent mean +/-SD. *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. Created in *BioRender*.

Acknowledgements

This work was supported by the Deutsche Forschungsgemeinschaft (DFG; *https://www.dfg.de*) within the research training group RTG2581 (M.R., F.S.). C.K. and C.A. are grateful to the DFG AR 376/22-1. M.R. was supported by funds of the Bavarian State Ministry of Science and the Arts and the University of Würzburg to the Graduate School of Life Sciences (GSLS), University of Würzburg. The DFG funded the Leica TCS SP5 CLSM under project code 116162193 and the BD FACSAriaIII cell sorter under project code 206080318.

We thank Sibylle Schneider-Schaulies and Thomas Rudel for valuable discussions and critically reading the manuscript, and Nadine Vollmuth as well as Christian Stigloher for scientific advice and intense discussions. We further thank Daniel Herrmann for assistance with HPLC-MS/MS measurements of sphingolipids. We are indebted to Kathrin Stelzner for conducting FACS of cell lines and generation of *S. aureus* strains. TPC1/2 double-knock out HeLa cells were kindly provided by Norbert Klugbauer (Freiburg, Germany). pmRFP-LC3 was a gift from Tamotsu Yoshimori (Addgene plasmid # 21075; *http://n2t.net/addgene:21075*; RRID:Addgene_21075, (100)). pLVTHM (101), pMD2.G and psPAX2 (unpublished) were a gift from Didier Trono (Addgene plasmids # 12247, 12259 and 12260, respectively). pLX304-Flag-APEX2-NES was a gift from Alice Ting (Addgene plasmid # 92158, (102)) and pSpCas9(BB)-p2A-GFP (PX458) (74) was a gift from Feng Zhang (Addgene plasmid # 48138; *http://n2t.net/addgene:48138*). Figures were created in BioRender.

Additional information

Author Contributions

M.R., M.F. conceptualized the work, designed the methodology and wrote the original draft manuscript. M.F., B.K. acquired funding. M.R. F.S., K.U., M.P., A.M., N.K., J.W., A.I., C.K., K.P., F. Schu performed the experiments; M.R., M.F. supervised the experiments. M.R., C.A., M.F. analyzed the data; all authors edited and revised the manuscript.

Funding

Deutsche Forschungsgemeinschaft (RTG2581)

Deutsche Forschungsgemeinschaft (AR 376/22-1)

Bavarian State Ministry for Science and Art

Graduate School of Life Sciences

Additional files

Supplemental Video 1. *S. aureus associates with BODIPy-FL-C_{12}-sphingomyelin during invasion For further details, see Figure 3, A.*

Supplemental Video 2. *S. aureus associates with a visible-range FRET probe during invasion For further details, see Supp. Figure 3.*

Supplemental Video 3. *S. aureus associates with Rab5 and Rab7 upon host cell entry Hela cells expressing mCherry-Rab5 (red) and YFP-Rab7 (green) were infected with fluorescent S. aureus JE2 (cyan). Time-laps imaging was performed at a Leica TCS SP5 microscope in intervals of 45 s. Scale bar: 10 µm.*

Supplemental Video 4. *Validation of Lysenin and CWT reporter cell line. For further details see Supp. Figure 5.*

Supplemental Video 5. *Supp. Video 5 Phagosomal Escape In Long Term Infection For further details see Figure 6, a.*

Supplemental Video 6. *β-toxin pretreatment protects HuLEC from S. aureus infection For further details see Figure 6, c.*

Author response video 1.

References

1. Hannun Y. A., Obeid L. M. (2018) **Sphingolipids and their metabolism in physiology and disease** *Nature Reviews Molecular Cell Biology* **19**:175–191 Google Scholar

2. van Meer G., Voelker D. R., Feigenson G. W. (2008) **Membrane lipids: where they are and how they behave** *Nat Rev Mol Cell Biol* **9**:112–124 Google Scholar

3. Segawa K., Suzuki J., Nagata S. (2014) **Flippases and scramblases in the plasma membrane** *Cell Cycle* **13**:2990–2991 Google Scholar

4. Brady R. O., Kanfer J. N., Mock M. B., Fredrickson D. S. (1966) **The metabolism of sphingomyelin** *II. Evidence of an enzymatic deficiency in Niemann-Pick diseae. Proc Natl Acad Sci U S A* **55**:366–369 Google Scholar

5. Breiden B., Sandhoff K. (2021) **Acid Sphingomyelinase, a Lysosomal and Secretory Phospholipase C, Is Key for Cellular Phospholipid Catabolism** *Int J Mol Sci* **22** Google Scholar

6. Tam C., et al. (2010) **Exocytosis of acid sphingomyelinase by wounded cells promotes endocytosis and plasma membrane repair** *J Cell Biol* **189**:1027–1038 Google Scholar

7. Fernandes M. C., et al. (2011) **Trypanosoma cruzi subverts the sphingomyelinase-mediated plasma membrane repair pathway for cell invasion** *J Exp Med* **208**:909–921 Google Scholar

8. Peters S., et al. (2019) **Neisseria meningitidis Type IV Pili Trigger Ca(2+)-Dependent Lysosomal Traffcking of the Acid Sphingomyelinase To Enhance Surface Ceramide Levels** *Infect Immun* **87** Google Scholar

9. Jenkins R. W., et al. (2010) **Regulated Secretion of Acid Sphingomyelinase: IMPLICATIONS FOR SELECTIVITY OF CERAMIDE FORMATION*** *Journal of Biological Chemistry* **285**:35706–35718 Google Scholar

10. Kornhuber J., Rhein C., Muller C. P., Muhle C. (2015) **Secretory sphingomyelinase in health and disease** *Biol Chem* **396**:707–736 Google Scholar

11. Krones & Rühling, et al. (2021) **Staphylococcus aureus alpha-Toxin Induces Acid Sphingomyelinase Release From a Human Endothelial Cell Line** *Front Microbiol* **12**:694489 Google Scholar

12. Schissel S. L., et al. (1998) **Secretory sphingomyelinase, a product of the acid sphingomyelinase gene, can hydrolyze atherogenic lipoproteins at neutral pH. Implications for atherosclerotic lesion development** *J Biol Chem* **273**:2738–2746 Google Scholar

13. Li C., et al. (2018) **Regulation of Staphylococcus aureus Infection of Macrophages by CD44, Reactive Oxygen Species, and Acid Sphingomyelinase** *Antioxid Redox Signal* **28**:916–934 Google Scholar

14. Grassme H., et al. (1997) **Acidic sphingomyelinase mediates entry of N. gonorrhoeae into nonphagocytic cells** *Cell* **91**:605–615 Google Scholar

15. Carpinteiro A., et al. (2020) **Pharmacological Inhibition of Acid Sphingomyelinase Prevents Uptake of SARS-CoV-2 by Epithelial Cells** *Cell Rep Med* **1**:100142 Google Scholar

16. Luisoni S., et al. (2015) **Co-option of Membrane Wounding Enables Virus Penetration into Cells** *Cell Host & Microbe* **18**:75–85 Google Scholar

17. Avota E., Gulbins E., Schneider-Schaulies S. (2011) **DC-SIGN mediated sphingomyelinase-activation and ceramide generation is essential for enhancement of viral uptake in dendritic cells** *PLoS Pathog* **7**:e1001290 Google Scholar

18. Sakr A., Brégeon F., Mège J. L., Rolain J. M., Blin O. (2018) **Staphylococcus aureus Nasal Colonization: An Update on Mechanisms, Epidemiology, Risk Factors, and Subsequent Infections** *Front Microbiol* **9**:2419 Google Scholar

19. McCaig L. F., McDonald L. C., Mandal S., Jernigan D. B. (2006) **Staphylococcus aureus-associated skin and soft tissue infections in ambulatory care** *Emerg Infect Dis* **12**:1715–1723 Google Scholar

20. Liesenborghs L., et al. (2019) **Staphylococcus aureus endocarditis: distinct mechanisms of bacterial adhesion to damaged and inflamed heart valves** *Eur Heart J* **40**:3248–3259 Google Scholar

21. Silversides J. A., Lappin E., Ferguson A. J. (2010) **Staphylococcal toxic shock syndrome: mechanisms and management** *Curr Infect Dis Rep* **12**:392–400 Google Scholar

22. Antimicrobial Resistance C. (2022) **Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis** *Lancet* **399**:629–655 Google Scholar

23. Fraunholz M., Sinha B. (2012) **Intracellular Staphylococcus aureus: live-in and let die** *Front Cell Infect Microbiol* **2**:43 Google Scholar

24. Rodrigues Lopes I., et al. (2022) **Microscopy-based phenotypic profiling of infection by Staphylococcus aureus clinical isolates reveals intracellular lifestyle as a prevalent feature** *Nature Communications* **13**:7174 Google Scholar

25. Hommes J. W., Surewaard B. G. J. (2022) **Intracellular Habitation of Staphylococcus aureus: Molecular Mechanisms and Prospects for Antimicrobial Therapy** *Biomedicines* **10**:1804 Google Scholar

26. Qazi S. N., Harrison S. E., Self T., Williams P., Hill P. J. (2004) **Real-time monitoring of intracellular Staphylococcus aureus replication** *J Bacteriol* **186**:1065–1077 Google Scholar

27. Abel J., et al. (2011) **Staphylococcus aureus evades the extracellular antimicrobial activity of mast cells by promoting its own uptake** *J Innate Immun* **3**:495–507 Google Scholar

28. Sinha B., et al. (1999) **Fibronectin-binding protein acts as Staphylococcus aureus invasin via fibronectin bridging to integrin alpha5beta1** *Cell Microbiol* **1**:101–117 Google Scholar

29. Agerer F., et al. (2005) **Cellular invasion by Staphylococcus aureus reveals a functional link between focal adhesion kinase and cortactin in integrin-mediated internalisation** *J Cell Sci* **118**:2189–2200 Google Scholar

30. Agerer F., Michel A., Ohlsen K., Hauck C. R. (2003) **Integrin-mediated invasion of Staphylococcus aureus into human cells requires Src family protein-tyrosine kinases** *J Biol*

Chem **278**:42524–42531 Google Scholar

31. Ying Y. T., et al. (2021) **Annexin A2-Mediated Internalization of Staphylococcus aureus into Bovine Mammary Epithelial Cells Requires Its Interaction with Clumping Factor B** *Microorganisms* **9** Google Scholar

32. Stroschein-Stevenson S. L., Foley E., O'Farrell P. H., Johnson A. D. (2006) **Identification of Drosophila gene products required for phagocytosis of Candida albicans** *PLoS Biol* **4**:e4 Google Scholar

33. Towell A. M., et al. (2021) **Staphylococcus aureus binds to the N-terminal region of corneodesmosin to adhere to the stratum corneum in atopic dermatitis** *Proceedings of the National Academy of Sciences* **118**:e2014444118 Google Scholar

34. Askarian F., et al. (2016) **The interaction between Staphylococcus aureus SdrD and desmoglein 1 is important for adhesion to host cells** *Scientific Reports* **6**:22134 Google Scholar

35. Bravo-Santano N., et al. (2019) **Host-directed kinase inhibitors act as novel therapies against intracellular Staphylococcus aureus** *Scientific Reports* **9**:4876 Google Scholar

36. Tribelli P. M., et al. (2020) **Staphylococcus aureus Lpl protein triggers human host cell invasion via activation of Hsp90 receptor** *Cell Microbiol* **22**:e13111 Google Scholar

37. Hirschhausen N., et al. (2010) **A novel staphylococcal internalization mechanism involves the major autolysin Atl and heat shock cognate protein Hsc70 as host cell receptor** *Cell Microbiol* **12**:1746–1764 Google Scholar

38. Dziewanowska K., et al. (2000) **Staphylococcal fibronectin binding protein interacts with heat shock protein 60 and integrins: role in internalization by epithelial cells** *Infect Immun* **68**:6321–6328 Google Scholar

39. Moldovan A., Fraunholz M. J. (2019) **In or out: Phagosomal escape of Staphylococcus aureus** *Cell Microbiol* **21**:e12997 Google Scholar

40. Jauregui C. E., Mansell J. P., Jepson M. A., Jenkinson H. F. (2013) **Differential interactions of Streptococcus gordonii and Staphylococcus aureus with cultured osteoblasts** *Mol Oral Microbiol* **28**:250–266 Google Scholar

41. Giese B., et al. (2009) **Staphylococcal alpha-toxin is not suffcient to mediate escape from phagolysosomes in upper-airway epithelial cells** *Infect Immun* **77**:3611–3625 Google Scholar

42. Grosz M., et al. (2014) **Cytoplasmic replication of Staphylococcus aureus upon phagosomal escape triggered by phenol-soluble modulin alpha** *Cell Microbiol* **16**:451–465 Google Scholar

43. Horn J., Stelzner K., Rudel T., Fraunholz M. (2018) **Inside job: Staphylococcus aureus host-pathogen interactions** *Int J Med Microbiol* **308**:607–624 Google Scholar

44. Wang R., et al. (2007) **Identification of novel cytolytic peptides as key virulence determinants for community-associated MRSA** *Nat Med* **13**:1510–1514 Google Scholar

45. Stelzner K., et al. (2021) **Intracellular Staphylococcus aureus employs the cysteine protease staphopain A to induce host cell death in epithelial cells** *PLoS Pathog* **17**:e1009874 Google Scholar

46. Gatfield J., Pieters J. (2000) **Essential role for cholesterol in entry of mycobacteria into macrophages** *Science* **288**:1647–1650 Google Scholar

47. Zidovetzki R., Levitan I. (2007) **Use of cyclodextrins to manipulate plasma membrane cholesterol content: evidence, misconceptions and control strategies** *Biochim Biophys Acta* **1768**:1311–1324 Google Scholar

48. Joiner K. A., Fuhrman S. A., Miettinen H. M., Kasper L. H., Mellman I. (1990) **Toxoplasma gondii: Fusion Competence of Parasitophorous Vacuoles in Fc Receptor-Transfected Fibroblasts** *Science* **249**:641–646 Google Scholar

49. Kim S., Watarai M., Makino S.-i., Shirahata T. (2002) **Membrane sorting during swimming internalization of Brucella is required for phagosome traffcking decisions** *Microbial Pathogenesis* **33**:225–237 Google Scholar

50. Stelzner K., et al. (2020) **Intracellular Staphylococcus aureus Perturbs the Host Cell Ca(2+) Homeostasis To Promote Cell Death** *mBio* **11** Google Scholar

51. Tsien R. Y. (1981) **A non-disruptive technique for loading calcium buffers and indicators into cells** *Nature* **290**:527–528 Google Scholar

52. Bootman M. D., Bultynck G. (2020) **Fundamentals of Cellular Calcium Signaling: A Primer** *Cold Spring Harb Perspect Biol* **12** Google Scholar

53. Maruyama T., Kanaji T., Nakade S., Kanno T., Mikoshiba K. (1997) **2APB, 2-aminoethoxydiphenyl borate, a membrane-penetrable modulator of Ins(1,4,5)P3-induced Ca2+ release** *J Biochem* **122**:498–505 Google Scholar

54. Walseth T. F., Lee H. C. (1993) **Synthesis and characterization of antagonists of cyclic-ADP-ribose-induced Ca2+ release** *Biochim Biophys Acta* **1178**:235–242 Google Scholar

55. Naylor E., et al. (2009) **Identification of a chemical probe for NAADP by virtual screening** *Nature Chemical Biology* **5**:220–226 Google Scholar

56. Kim S. Y., Cho B. H., Kim U. H. (2010) **CD38-mediated Ca2+ signaling contributes to angiotensin II-induced activation of hepatic stellate cells: attenuation of hepatic fibrosis by CD38 ablation** *J Biol Chem* **285**:576–582 Google Scholar

57. Rah S. Y., Mushtaq M., Nam T. S., Kim S. H., Kim U. H. (2010) **Generation of cyclic ADP-ribose and nicotinic acid adenine dinucleotide phosphate by CD38 for Ca2+ signaling in interleukin-8-treated lymphokine-activated killer cells** *J Biol Chem* **285**:21877–21887 Google Scholar

58. Piedra-Quintero Z. L., Wilson Z., Nava P., Guerau-de-Arellano M. (2020) **CD38: An Immunomodulatory Molecule in Inflammation and Autoimmunity** *Front Immunol* **11**:597959 Google Scholar

59. Angeletti C., et al. (2022) **SARM1 is a multi-functional NAD(P)ase with prominent base exchange activity, all regulated bymultiple physiologically relevant NAD metabolites** *iScience* **25**:103812 Google Scholar

60. Gu F., et al. (2021) **Dual NADPH oxidases DUOX1 and DUOX2 synthesize NAADP and are necessary for Ca(2+) signaling during T cell activation** *Sci Signal* **14**:eabe3800 Google Scholar

eLife

61. Caler E. V., Chakrabarti S., Fowler K. T., Rao S., Andrews N. W. (2001) **The Exocytosis-regulatory protein synaptotagmin VII mediates cell invasion by Trypanosoma cruzi** *J Exp Med* **193**:1097–1104 Google Scholar

62. Grassmé H., Riehle A., Wilker B., Gulbins E. (2005) **Rhinoviruses infect human epithelial cells via ceramide-enriched membrane platforms** *J Biol Chem* **280**:26256–26262 Google Scholar

63. Miller M. E., Adhikary S., Kolokoltsov A. A., Davey R. A. (2012) **Ebolavirus Requires Acid Sphingomyelinase Activity and Plasma Membrane Sphingomyelin for Infection** *Journal of Virology* **86**:7473–7483 Google Scholar

64. Hall M. P., et al. (2012) **Engineered Luciferase Reporter from a Deep Sea Shrimp Utilizing a Novel Imidazopyrazinone Substrate** *ACS Chemical Biology* **7**:1848–1857 Google Scholar

65. Dixon A. S., et al. (2016) **NanoLuc Complementation Reporter Optimized for Accurate Measurement of Protein Interactions in Cells** *ACS Chemical Biology* **11**:400–408 Google Scholar

66. Andrews N. W. (2017) **Detection of Lysosomal Exocytosis by Surface Exposure of Lamp1 Luminal Epitopes** *Lysosomes: Methods and Protocols* :205–211 https://doi.org/10.1007/978-1-4939-6934-0_13 | Google Scholar

67. Rodriguez A., Webster P., Ortego J., Andrews N. W. (1997) **Lysosomes behave as Ca2+-regulated exocytic vesicles in fibroblasts and epithelial cells** *J Cell Biol* **137**:93–104 Google Scholar

68. Cerny J., et al. (2004) **The small chemical vacuolin-1 inhibits Ca(2+)-dependent lysosomal exocytosis but not cell resealing** *EMBO Rep* **5**:883–888 Google Scholar

69. Reddy A., Caler E. V., Andrews N. W. (2001) **Plasma membrane repair is mediated by Ca(2+)-regulated exocytosis of lysosomes** *Cell* **106**:157–169 Google Scholar

70. Grassmé H., et al. (1997) **Acidic sphingomyelinase mediates entry of N. gonorrhoeae into nonphagocytic cells** *Cell* **91**:605–615 Google Scholar

71. Roth A. G., et al. (2009) **Potent and selective inhibition of acid sphingomyelinase by bisphosphonates** *Angew Chem Int Ed Engl* **48**:7560–7563 Google Scholar

72. Muhle C., Kornhuber J. (2017) **Assay to measure sphingomyelinase and ceramidase activities effciently and safely** *J Chromatogr A* **1481**:137–144 Google Scholar

73. Kappe C., Mohamed Z. H., Naser E., Carpinteiro A., Arenz C. (2020) **A Novel Visible Range FRET Probe for Monitoring Acid Sphingomyelinase Activity in Living Cells** *Chemistry* **26**:5780–5783 Google Scholar

74. Ran F. A., et al. (2013) **Genome engineering using the CRISPR-Cas9 system** *Nature Protocols* **8**:2281–2308 Google Scholar

75. Schuchman E. H., Desnick R. J. (2017) **Types A and B Niemann-Pick disease** *Mol Genet Metab* **120**:27–33 https://doi.org/10.1016/j.ymgme.2016.12.008 | Google Scholar

76. Hoffmann C., et al. (2010) **Caveolin limits membrane microdomain mobility and integrin-mediated uptake of fibronectin-binding pathogens** *J Cell Sci* **123**:4280–4291 Google Scholar

77. Tricou L.-P., et al. (2024) **Staphylococcus aureus can use an alternative pathway to be internalized by osteoblasts in absence of β1 integrins** *Scientific Reports* **14**:28643 Google Scholar

78. Rappaport J., Garnacho C., Muro S. (2014) **Clathrin-mediated endocytosis is impaired in type A-B Niemann-Pick disease model cells and can be restored by ICAM-1-mediated enzyme replacement** *Mol Pharm* **11**:2887–2895 Google Scholar

79. Rappaport J., Manthe R. L., Garnacho C., Muro S. (2015) **Altered Clathrin-Independent Endocytosis in Type A Niemann-Pick Disease Cells and Rescue by ICAM-1-Targeted Enzyme Delivery** *Mol Pharm* **12**:1366–1376 Google Scholar

80. Pantůcek R., et al. (2004) **Identification of bacteriophage types and their carriage in Staphylococcus aureus** *Arch Virol* **149**:1689–1703 Google Scholar

81. Rühling M., et al. (2024) **Trifunctional sphingomyelin derivatives enable nanoscale resolution of sphingomyelin turnover in physiological and infection processes via expansion microscopy** *Nature Communications* **15**:7456 Google Scholar

82. Ellison C. J., Kukulski W., Boyle K. B., Munro S., Randow F. (2020) **Transbilayer Movement of Sphingomyelin Precedes Catastrophic Breakage of Enterobacteria-Containing Vacuoles** *Curr Biol* **30**:2974–2983 Google Scholar

83. Rühling M., Schmelz F., Kempf A., Paprotka K., Fraunholz Martin J. (2025) **Identification of the Staphylococcus aureus endothelial cell surface interactome by proximity labeling** *mBio* **0**:e03654–3624 Google Scholar

84. Falugi F., Kim H. K., Missiakas D. M., Schneewind O. (2013) **Role of protein A in the evasion of host adaptive immune responses by Staphylococcus aureus** *mBio* **4**:e00575–513 Google Scholar

85. Morgan A. J., Jacob R. (1994) **Ionomycin enhances Ca2+ influx by stimulating store-regulated cation entry and not by a direct action at the plasma membrane** *Biochem J* **300**:665–672 Google Scholar

86. Sano O., et al. (2016) **Vacuolin-1 inhibits autophagy by impairing lysosomal maturation via PIKfyve inhibition** *FEBS Letters* **590**:1576–1585 Google Scholar

87. Schonn J.-S., Maximov A., Lao Y., Südhof T. C., Sørensen J. B. (2008) **Synaptotagmin-1 and -7 are functionally overlapping Ca2+ sensors for exocytosis in adrenal chromaffn cells** *Proceedings of the National Academy of Sciences* **105**:3998–4003 Google Scholar

88. Morgan A. J., Platt F. M., Lloyd-Evans E., Galione A. (2011) **Molecular mechanisms of endolysosomal Ca2+ signalling in health and disease** *Biochemical Journal* **439**:349–378 Google Scholar

89. Davis L. C., Morgan A. J., Galione A. (2020) **NAADP-regulated two-pore channels drive phagocytosis through endo-lysosomal Ca(2+) nanodomains, calcineurin and dynamin** *EMBO J* **39**:e104058 Google Scholar

90. Sakurai Y., et al. (2015) **Ebola virus. Two-pore channels control Ebola virus host cell entry and are drug targets for disease treatment** *Science* **347**:995–998 Google Scholar

91. Gunaratne G. S., Yang Y., Li F., Walseth T. F., Marchant J. S. (2018) **NAADP-dependent Ca(2+) signaling regulates Middle East respiratory syndrome-coronavirus pseudovirus translocation through the endolysosomal system** *Cell Calcium* **75**:30–41 Google Scholar

92. Ou X., et al. (2020) **Characterization of spike glycoprotein of SARS-CoV-2 on virus entry and its immune cross-reactivity with SARS-CoV** *Nature Communications* **11**:1620 Google Scholar

93. Davis L. C., et al. (2012) **NAADP activates two-pore channels on T cell cytolytic granules to stimulate exocytosis and killing** *Curr Biol* **22**:2331–2337 Google Scholar

94. Arredouani A., et al. (2015) **Nicotinic Acid Adenine Dinucleotide Phosphate (NAADP) and Endolysosomal Two-pore Channels Modulate Membrane Excitability and Stimulus-Secretion Coupling in Mouse Pancreatic β Cells** *J Biol Chem* **290**:21376–21392 Google Scholar

95. Bollinger C. R., Teichgräber V., Gulbins E. (2005) **Ceramide-enriched membrane domains** *Biochimica et Biophysica Acta (BBA) - Molecular Cell Research* **1746**:284–294 Google Scholar

96. Zhang Y., Li X., Becker K. A., Gulbins E. (2009) **Ceramide-enriched membrane domains—Structure and function** *Biochimica et Biophysica Acta (BBA) - Biomembranes* **1788**:178–183 Google Scholar

97. Flannagan R. S., Kuiack R. C., McGavin M. J., Heinrichs D. E. (2018) **Staphylococcus aureus Uses the GraXRS Regulatory System To Sense and Adapt to the Acidified Phagolysosome in Macrophages** *mBio* **9** Google Scholar

98. Peng H., et al. (2015) **Acid sphingomyelinase inhibition protects mice from lung edema and lethal Staphylococcus aureus sepsis** *J Mol Med* **93**:675–689 Google Scholar

99. Kornhuber J., et al. (2010) **Functional Inhibitors of Acid Sphingomyelinase (FIASMAs): a novel pharmacological group of drugs with broad clinical applications** *Cell Physiol Biochem* **26**:9–20 Google Scholar

100. Kimura S., Noda T., Yoshimori T. (2007) **Dissection of the autophagosome maturation process by a novel reporter protein, tandem fluorescent-tagged LC3** *Autophagy* **3**:452–460 Google Scholar

101. Wiznerowicz M., Trono D. (2003) **Conditional suppression of cellular genes: lentivirus vector-mediated drug-inducible RNA interference** *J Virol* **77**:8957–8961 Google Scholar

102. Hung V., et al. (2017) **Proteomic mapping of cytosol-facing outer mitochondrial and ER membranes in living human cells by proximity biotinylation** *eLife* **6**:e24463 https://doi.org/10.7554/eLife.24463 | Google Scholar

103. Edgell C. J., McDonald C. C., Graham J. B. (1983) **Permanent cell line expressing human factor VIII-related antigen established by hybridization** *Proceedings of the National Academy of Sciences* **80**:3734–3737 Google Scholar

104. Blättner S., et al. (2016) **Staphylococcus aureus Exploits a Non-ribosomal Cyclic Dipeptide to Modulate Survival within Epithelial Cells and Phagocytes** *PLOS Pathogens* **12**:e1005857 Google Scholar

105. Vann J. M., Proctor R. A. (1987) **Ingestion of Staphylococcus aureus by bovine endothelial cells results in time- and inoculum-dependent damage to endothelial cell monolayers** *Infect Immun* **55**:2155–2163 Google Scholar

106. Fey P. D., et al. (2013) **A genetic resource for rapid and comprehensive phenotype screening of nonessential Staphylococcus aureus genes** *MBio* **4**:e00537–512 Google Scholar

107. Becker K. A., et al. (2018) **Staphylococcus aureus Alpha-Toxin Disrupts Endothelial-Cell Tight Junctions via Acid Sphingomyelinase and Ceramide** *Infect Immun* **86** Google Scholar

108. Giese B., et al. (2011) **Expression of δ-toxin by Staphylococcus aureus mediates escape from phago-endosomes of human epithelial and endothelial cells in the presence of β-toxin** *Cell Microbiol* **13**:316–329 Google Scholar

109. Schindelin J., et al. (2012) **Fiji: an open-source platform for biological-image analysis** *Nat Methods* **9**:676–682 Google Scholar

110. Prell A., Wigger D., Huwiler A., Schumacher F., Kleuser B. (2024) **The sphingosine kinase 2 inhibitors ABC294640 and K145 elevate (dihydro)sphingosine 1-phosphate levels in various cells** *J Lipid Res* **65**:100631 Google Scholar

Author information

Marcel Rühling
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany
ORCID iD: 0009-0006-4757-5014

Fabio Schmelz
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany

Kim Ulbrich
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany

Fabian Schumacher
Institute of Pharmacy, Freie Universität Berlin, Berlin, Germany

Julia Wolf
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany

Maximilian Pfefferle
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany

Magdalena Priester
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany

Adriana Moldovan
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany

Nadine Knoch
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany

Andreas Iwanowitsch
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany

Christian Kappe
Institut for Chemistry, Humboldt Universität zu Berlin, Berlin, Germany

Kerstin Paprotka
Institute of Pharmacy, Freie Universität Berlin, Berlin, Germany

Burkhard Kleuser
Institute of Pharmacy, Freie Universität Berlin, Berlin, Germany
ORCID iD: 0000-0002-1888-9595

Christoph Arenz
Institut for Chemistry, Humboldt Universität zu Berlin, Berlin, Germany

Martin J Fraunholz
Chair of Microbiology, Julius-Maximilians-University Würzburg, Würzburg, Germany
ORCID iD: 0000-0002-4581-6244

For correspondence: martin.fraunholz@uni-wuerzburg.de

Editors

Reviewing Editor
Shaeri Mukherjee
University of California, San Francisco, San Francisco, United States of America

Senior Editor
Dominique Soldati-Favre
University of Geneva, Geneva, Switzerland

Reviewer #2 (Public review):

In the manuscript, Ruhling et al propose a rapid uptake pathway that is dependent on lysosomal exocytosis, lysosomal Ca2+ and acid sphingomyelinase, and further suggest that the intracellular trafficking and fate of the pathogen is dictated by the mode of entry. Overall, this is manuscript argues for an important mechanism of a 'rapid' cellular entry pathway of S.aureus that is dependent on lysosomal exocytosis and acid sphingomyelinase and links the intracellular fate of bacterium including phagosomal dynamics, cytosolic replication and host cell death to different modes of uptake.

Key strength is the nature of the idea proposed, while continued reliance on inhibitor treatment combined with lack of phenotype for genetic knock out is a major weakness. While the authors argue a role for undetectable nano-scale Cer platforms on the cell surface caused by ASM activity, results do not rule out a SM independent role in the cellular uptake phenotype of ASM inhibitors.

The authors have attempted to address many of the points raised in the previous revision. While the new data presented provide partial evidence, the reliance on chemical inhibitors and lack of clear results directly documenting release of lysosomal Ca2+, or single bacterial tracking, or clear distinction between ASM dependent and independent processes dampen the enthusiasm.

I acknowledge the author's argument of different ASM inhibitors showing similar phenotypes across different assays as pointing to a role for ASM, but the lack of phenotype in ASM KO cells is concerning. The author's argument that altered lipid composition in ASM KO cells could be overcoming the ASM-mediated infection effects by other ASM-independent mechanisms is speculative, as they acknowledge, and moderates the importance of ASM-dependent pathway. The SM accumulation in ASM KO cells does not distinguish between localized alterations within the cells. If this pathway can be compensated, how central is it likely to be ?

The authors allude to lower phagosomal escape rate in ASM KO cells compared to inhibitor treatment, which appears to contradict the notion of uptake and intracellular trafficking phenotype being tightly linked. As they point out, these results might be hard to interpret. Could an inducible KD system recapitulate (some of) the phenotype of inhibitor treatment ? If S. aureus does not escape phagosome in macrophages, could it provide a system to potentially decouple the uptake and intracellular trafficking effects by ASM (or its inhibitor treatment) ?

The role of ASM on cell surface remains unclear. The hypothesis proposed by the authors that the localized generation of Cer on the surface by released ASM leads to generation of Cer-enriched platforms could be plausible, but is not backed by data, technical challenges to visualize these platforms notwithstanding. These results do not rule out possible SM independent effects of ASM on the cell surface, if indeed the role of ASM is confirmed by controlled genetic depletion studies.

The reviewer acknowledges technical challenges in directly visualizing lysosomal Ca2+ using the methods outlined. Genetically encoded lysosomal Ca2+ sensor such as Gcamp3-ML1 might provide better ways to directly visualize this during inhibitor treatment, or S. aureus infection.

https://doi.org/10.7554/eLife.102810.2.sa1

Author response:

The following is the authors’ response to the original reviews

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The manuscript by Rühling et al analyzes the mode of entry of S. aureus into mammalian cells in culture. The authors propose a novel mechanism of rapid entry that involves the release of calcium from lysosomes via NAADP-stimulated activation of TPC1, which in turn causes lysosomal exocytosis; exocytic release of lysosomal acid sphingomyelinase (ASM) is then envisaged to convert exofacial sphingomyelin to ceramide. These events not only induce the rapid entry of the bacteria into the host cells but are also described to*

> *alter the fate of the intracellular S. aureus, facilitating escape from the endocytic vacuole to the cytosol.*
>
> *Strengths:*
>
> *The proposed mechanism is novel and could have important biological consequences.*
>
> *Weaknesses:*
>
> *Unfortunately, the evidence provided is unconvincing and insufficient to document the multiple, complex steps suggested. In fact, there appear to be numerous internal inconsistencies that detract from the validity of the conclusions, which were reached mostly based on the use of pharmacological agents of imperfect specificity.*

We thank the reviewer for the detailed evaluation of our manuscript. We will address the criticism below.

We agree with the reviewer that many of the experiments presented in our study rely on the usage of inhibitors. However, we want to emphasize that the main conclusion (invasion pathway affects the intracellular fate/phagosomal escape) was demonstrated without the use of inhibitors or genetic ablation in two key experiments (Figure5 D/E). These experiments were in line with the results we obtained with inhibitors (amitriptyline [Figure 4D], ARC39, PCK310, [Figure 4C] and Vacuolin-1 [Figure4E]). Importantly, the hypothesis was also supported by another key experiment, in which we showed the intracellular fate of bacteria is affected by removal of SM from the plasma membrane before invasion, but not by removal of SM from phagosomal membranes after bacteria internalization (Figure5A-C). Taken together, we thus believe that the main hypothesis is strongly supported by our data.

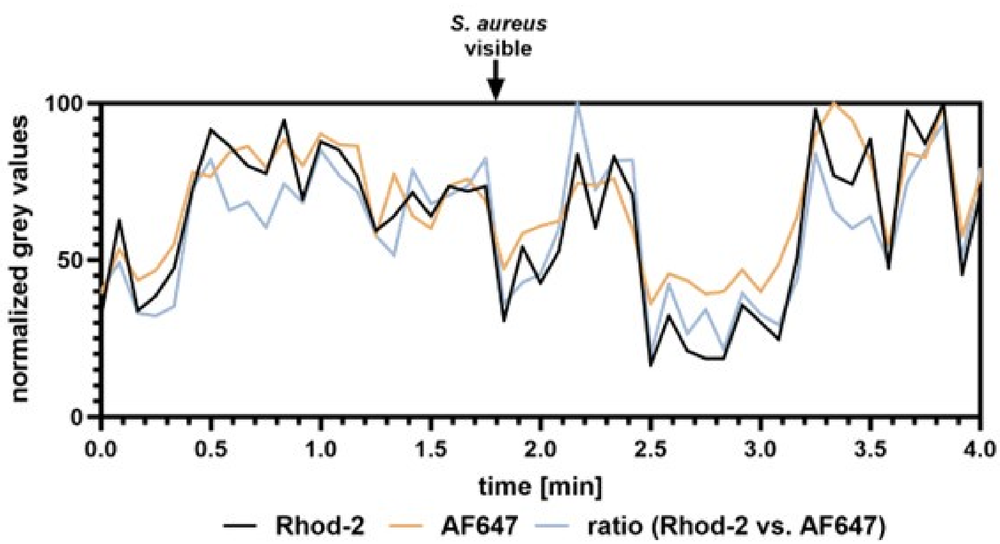
Moreover, we either used different inhibitors for the same molecule (ASM was inhibited by ARC39, amitriptyline and PCK310 with similar outcome) or supported our hypothesis with gene-ablated cell pools (TPC1, Syt7, SARM1), as we will point out in more detail below.

> *Firstly, the release of calcium from lysosomes is not demonstrated. Localized changes in the immediate vicinity of lysosomes need to be measured to ascertain that these organelles are the source of cytosolic calcium changes. In fact, 9-phenantrol, which the authors find to be the most potent inhibitor of invasion and hence of the putative calcium changes, is not a blocker of lysosomal calcium release but instead blocks plasmalemmal TRPM4 channels. On the other hand, invasion is seemingly independent of external calcium. These findings are inconsistent with each other and point to non-specific effects of 9-phenantrol. The fact that ionomycin decreases invasion efficiency is taken as additional evidence of the importance of lysosomal calcium release. It is not clear how these observations support involvement of lysosomal calcium release and exocytosis; in fact treatment with the ionophore should itself have induced lysosomal exocytosis and stimulated, rather than inhibited invasion. Yet, manipulations that increase and others that decrease cytosolic calcium both inhibited invasion.*

With respect to lysosomal Ca^{2+} release, we agree with the reviewer that direct visual demonstration of lysosomal Ca^{2+} release upon infection will improve the manuscript. We therefore performed live cell imaging to visualize lysosomal Ca^{2+} release by a previously published method.1 The approach is based on two dextran-coupled fluorophores that were incubated with host cells. The dyes are endocytosed and eventually stain the lysosomes. One of the dyes, Rhod-2, is Ca^{2+}-sensitive and can be used to estimate the lysosomal Ca^{2+} content. The second dye, AF647, is Ca^{2+}-insensitive and is used to visualize the lysosomes. If the ratio Rhod-2/AF647 within the lysosomes is decreasing, lysosomal Ca^{2+} release is indicated. We monitored lysosomal Ca^{2+} content during *S. aureus* infection with this method (Author response image 1 and Author response video 1). However, the lysosomes are very dynamic, and it is challenging to monitor the fluorescence intensities over time. Thus, quantitative measurements are not possible with our methodology, and we decided to not include these data in the main manuscript. However, one could speculate that lysosomal Ca^{2+} content in the selected ROI (Author response image 1 and Author response video 1) is decreased upon attachment of *S. aureus* to the host cells as indicated by a decrease in Rhod-2/AF647 ratio.

Author response image 1.

Lysosomal Ca^{2+} imaging during *S. aureus* infection. The lysosomes of HuLEC were stained with two dextran-coupled fluorescent dyes. A Ca^{2+}-sensitive dye Rhod-2 as well as Ca^{2+}insensitive AF647. Cells were infected with fluorescent *S. aureus* JE2 and monitored by live cell imaging (see Author response video 1). The intensity of Rhod-2/AF647 was measured close to a *S. aureus*-host contact site. Ratio of Rhod-2 vs. AF647 fluorescence intensity was calculated



As to the TRPM4 involvement in *S. aureus* host cell internalization, it has been reported that TRPM4 is activated by cytosolic Ca^{2+}. However, the channel conducts monovalent cations such as K^+ or Na^+ but is impermeable for Ca^{2+} [2, 3]. The following of our observations are supporting this:

i) *S. aureus* invasion is dependent on intracellular Ca^{2+}, but is independent from extracellular Ca^{2+} (Figure 1A).

ii) 9-phenantrol treatment reduces *S. aureus* internalization by host cells, illustrating the dependence of this process on TRPM4 (data removed from the manuscript) . We therefore hypothesize that TRPM4 is activated by Ca^{2+} released from lysosomes (see above).

TRPM4 is localized to focal adhesions and is connected to actin cytoskeleton[4, 5] – a requisite of host cell entry of *S. aureus*.[6, 7] This speaks for an important function of TRPM4 in uptake of *S. aureus* in general, but does not necessarily have to be involved exclusively in the rapid uptake pathway.

TRPM4 itself is not permeable for Ca^{2+} but is activated by the cation. Thus, it is unlikely to cause lysosomal exocytosis. The stronger bacterial uptake reduction by treatment with 9-phenantrol when compared to Ned19 thus may be caused by the involvement of TRPM4 in additional pathways of *S. aureus* host cell entry involving that association of TRPM4 with focal adhesions or as pointed out by the reviewer, unspecific side effects of 9-phenantrol that we currently cannot exclude. However, we think that experiments with 9-phenantrol distract from the main story (lysosomal Ca^{2+} and exocytosis) and might be confusing for the reader. We thus removed all data and discussion concerning 9phenantrol in the revised manuscript.

Regarding the reduced *S. aureus* invasion after ionomycin treatment, we agree with the reviewer that ionomycin is known to lead to lysosomal exocytosis as was previously shown by others8 as well as our laboratory[9}.

We hypothesized that pretreatment with ionomycin would trigger lysosomal exocytosis and thus would reduce the pool of lysosomes that can undergo exocytosis before host cells are contacted by *S. aureus*. As a result, we should observe a marked reduction of *S. aureus* internalization in such “lysosome-depleted cells”, if the lysosomal exocytosis is coupled to bacterial uptake. Our observation of reduced bacterial internalization after ionomycin treatment supports this hypothesis.

However, ionomycin treatment and *S. aureus* infection of host cells are distinct processes.

While ionomycin results in strong global and non-directional lysosomal exocytosis of all “releasable” lysosomes (~5-10 % of all lysosomes according to previous observations)8, we hypothesize that lysosomal exocytosis upon contact with *S. aureus* only involves a small proportion of lysosomes at host-bacteria contact sites. This is supported by experiments that demonstrate that ~30% of the lysosomes that are released by ionomycin treatment are exocytosed during *S. aureus* infection (see below and Figure 2, A-C). We added this new data as well as an according section to the discussion (line 563 ff). Moreover, we moved the data obtained with ionomycin to Figure 2E and described our idea behind this experiment more precisely (line 166 ff).

> *The proposed role of NAADP is based on the effects of "knocking out" TPC1 and on the pharmacological effects of Ned-19. It is noteworthy that TPC2, rather than TPC1, is generally believed to be the primary TPC isoform of lysosomes. Moreover, the gene ablation accomplished in the TPC1 "knockouts" is only partial and rather unsatisfactory. Definitive conclusions about the role of TPC1 can only be reached with proper, full knockouts. Even the pharmacological approach is unconvincing because the high doses of Ned-19 used should have blocked both TPC isoforms and presumably precluded invasion. Instead, invasion is reduced by only ≈50%. A much greater inhibition was reported using 9-phenantrol, the blocker of plasmalemmal calcium channels. How is the selective involvement of lysosomal TPC1 channels justified?*

As to partial gene ablation of TPC1: To avoid clonal variances, we usually perform pool sorting to obtain a cell population that predominantly contains cells -here- deficient in TPC1, but also a small proportion of wildtype cells as seen by the residual TPC1 protein on the Western blot. We observe a significant reduction in bacterial uptake in this cell pool suggesting that the uptake reduction in a pure K.O. population may be even more pronounced.

As to the inhibition by Ned19:

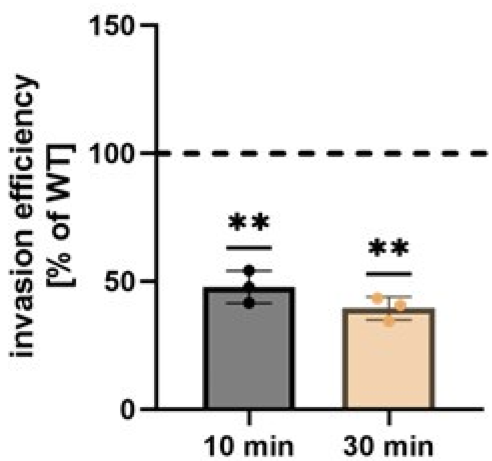
The scale of invasion reduction upon Ned19 treatment (50%, Figure 1B) is comparable with the reduction caused by other compounds that influence the ASM-dependent pathway (such as amitriptyline, ARC39 [Figure 2G], BAPTA-AM [Figure 1A], Vacuolin-1 [Figure 2D], β-toxin [Figure 2L] and ionomycin [Figure 2E]). Further, the partial reduction of invasion is most likely due to the concurrent activity of multiple internalization pathways which are not all targeted by the used compounds and which we briefly discuss in the manuscript.

We agree with the reviewer that Ned19 inhibits TPC1 and TPC2. Since ablation of TPC1 reduced invasion of *S. aureus*, we concluded that TPC1 is important for *S. aureus* host cell invasion. We thus agree with the reviewer that a role for TPC2 cannot be excluded. We clarified this in the revised manuscript (Lines 552). It needs to be noted, however, that deficiency in either TPC1 or TPC2 alone was sufficient to prevent Ebola virus infection10, which is in line with our observations.

In order to address the role of TPC2 for this review process, we kindly were gifted TPCN1/TPCN2 double knock-out HeLa cells by Norbert Klugbauer (Freiburg, Germany), which we tested for *S. aureus* internalization. We found that invasion was reduced in these cell lines supporting a role of lysosomal Ca^{2+} release in *S. aureus* host cell entry and a role for both TPC channels (Author response image 2, see end of the document). Since we did not have a single TPCN2 knock-out available we decided to exclude these data from the main manuscript.

Author response image 2.

Invasion efficiency is reduced in TPC1/TPC2 double K.O. HeLa cells. Invasion efficiency of *S. aureus* JE2 was determined in TPC1/TPC2 double K.O. cells after 10 and 30 min. Results were normalized to the parental HeLa WT cell line (set to 100 %).



> *Invoking an elevation of NAADP as the mediator of calcium release requires measurements of the changes in NAADP concentration in response to the bacteria. This was not performed. Instead, the authors analyzed the possible contribution of putative NAADP-generating systems and reported that the most active of these, CD38, was without effect, while the elimination of SARM1, another potential source of NAADP, had a very modest (≈20%) inhibitory effect that may have been due to clonal variation, which was not ruled out. In view of these data, the conclusion that NAADP is involved in the invasion process seems unwarranted.*

Our results from two independent experimental set-ups (Ned19 [Figure 1B] and TPC1 K.O. [Figure 1C & Figure 2N]) indicate the involvement of NAADP in the process. Together with the metabolomics unit at the Biocenter Würzburg, we attempted to measure cellular NAADP levels, however, this proved to be non-trivial and requires further optimization. However, we can rule out clonal variation in the SARM1 mutant since experiments were conducted with a cell pool as described above in order to avoid clonal variation of single clones.

The mechanism behind biosynthesis of NAADP is still debated. CD38 was the first enzyme discovered to possess the ability of producing NAADP. However, it requires acidic pH to produce NAADP[11] -which does not match the characteristics of a cytosolic NAADP producer. HeLa cells do not express CD38 and hence, it is not surprising that inhibition of CD38 had no effect on *S. aureus* invasion in HeLa cells. However, NAADP production by HeLa cells was observed in absence of CD38[12]. Thus CD38independent NAADP generation is likely. SARM1 can produce NAADP at neutral pH[13] and is expressed in HeLa, thus providing a more promising candidate.

We agree with the reviewer that the reduction of *S. aureus* internalization after ablation of SARM1 is less pronounced than in other experiments of ours. This may be explained by NAADP originating from other enzymes, such as the recently discovered DUOX1, DUOX2, NOX1 and NOX2[14], which – with exception of DUOX2- possess a low expression even in HeLa cells. We add this to the discussion in the revised manuscript (line 579).

We can, however, rule out clonal variation for the inhibitory effect. As stated above we generated K.O. cell pools specifically to avoid inherent problems of clonality. Thus, we also detect some residual wildtype cells within our cell pools.

> *The involvement of lysosomal secretion is, again, predicated largely on the basis of pharmacological evidence. No direct evidence is provided for the insertion of lysosomal components into the plasma membrane, or for the release of lysosomal contents to the medium. Instead, inhibition of lysosomal exocytosis by vacuolin-1 is the sole source of evidence. However, vacuolin-1 is by no means a specific inhibitor of lysosomal secretion: it is now known to act primarily as a PIKfyve inhibitor and to cause massive distortion of the endocytic compartment, including gross swelling of endolysosomes. The modest (20-25%) inhibition observed when using synaptotagmin 7 knockout cells is similarly not convincing proof of the requirement for lysosomal secretion.*

We agree with the reviewer that the manuscript will benefit from a functional analysis of lysosomal exocytosis and therefore conducted assays to investigate exocytosis in the revised manuscript. We previously showed i) by addition of specific antisera that LAMP1 transiently is exposed on the plasma membrane during ionomycin and pore-forming toxin challenge and ii) demonstrated the release of ASM activity into the culture medium under these conditions.[9] However, both measurements are not compatible with *S. aureus* infection, since LAMP1 antibodies also are non-specifically bound by protein A and another IgG-binding proteins on the *S. aureus* surface, which would bias the results. Since protein A also may serve as an adhesin in the investigated pathway, we cannot simply delete the ORF without changing other aspects of staphylococcal virulence. Further, FBS contains a ASM background activity that impedes activity measurements of cell culture medium. We previously removed this background activity by a specific heat-inactivation protocol.[9] However, *S. aureus* invasion is strongly reduced in culture medium containing this heat-inactivated FBS.

We therefore developed a luminescence assay based on split NanoLuc luciferase that enables detection of LAMP1 exposed on the plasma membrane without usage of antibodies (Figure 2, A-C). We added a section on the assay in the revised manuscript. Briefly, we generated reporter cells by fusing a short peptide fragment of NanoLuc called HiBiT between the signal peptide and the mature luminal domain of LAMP1 and stably expressed the resulting protein in HeLa cells by lentiviral transduction. The LgBiT protein domain of NanoLuc luciferase (Promega) as well as the substrate Furimazine are added to the culture medium. HiBiT can reconstitute a functional NanoLuc with LgBiT and process Furimazine when lysosomes are exocytosed thereby generating luminescence measurable in a suitable plate reader.

With this assay we detected that about 30% of lysosomes that were “releasable” by treatment with ionomycin are exocytosed during *S. aureus* infection. Lysosomal exocytosis was strongly reduced (even below the levels of untreated controls), if we treated cells with Vacuolin-1 or Ned19.

We agree with the reviewer that Vacuolin-1 to some extent has unspecific side effects as has been shown by others and which we addressed in the revised version of the manuscript (line

541 ff). However, our new results with the HiBiT reporter cell line clearly demonstrate a reduction of lysosomal exocytosis after Vacuolin-1 treatment. Supported by this and our other results we hypothesize that Vacuolin-1 decreases *S. aureus* internalization due to the inhibition of lysosomal exocytosis.

As to the involvement of synaptotagmin 7: The effect of Syt7 K.O. on invasion was moderate in initial experiments, likely due to a high culture passage and presumably overgrowth of WT cells. However, reduction of invasion in Syt7 K.O.s was more pronounced in experiments with β-toxin complementation (Figure 2, N) and hence, we combined the two data sets (Figure 2, F). This demonstrates the reduction of bacterial invasion by ~40% in Syt7 K.O. cell pools. Moreover, Syt7 is not the only protein possibly involved in Ca^{2+}-dependent exocytosis. For instance, Syt1 has been shown to possess an overlapping function.[15] This may explain the differences between our Vacuolin-1 and Syt7 ablation experiments. We added this information to the discussion.

> *ASM is proposed to play a central role in the rapid invasion process. As above, most of the evidence offered in this regard is pharmacological and often inconsistent between inhibitors or among cell types. Some drugs affect some of the cells, but not others. It is difficult to reach general conclusions regarding the role of ASM. The argument is made even more complex by the authors' use of exogenous sphingomyelinase (beta-toxin). Pretreatment with the toxin decreased invasion efficiency, a seemingly paradoxical result. Incidentally, the effectiveness of the added toxin is never quantified/validated by directly measuring the generation of ceramide or the disappearance of SM.*

Although pharmacological inhibitors can have unspecific side effects, we want to emphasize that the inhibitors used in our study act on the enzyme ASM by completely different mechanisms. Amitriptyline is a so called functional inhibitor of ASM (FIASMA) which induces the detachment of ASM from lysosomal membranes resulting in degradation of the enzyme.[16] By contrast, ARC39 is a competitive inhibitor.[17, 18]

There are no inconsistencies in our data obtained with ASM inhibitors. Amitriptyline and ARC39 both reduce the invasion of *S. aureus* in HuLEC, HuVEC and HeLa cells (Figure 2G). ARC39 needs a longer pre-incubation, since its uptake by host cells is slower (to be published elsewhere). We observe a different outcome in 16HBE14o- and Ea.Hy 926 cells, with 16HBE14o- even demonstrating a slightly increased invasion of *S. aureus* upon ARC39 treatment. Amitriptyline had no effect (Figure 2G).

Thus, the ASM-dependent *S. aureus* internalization is cell type/line specific, which we state in the manuscript. The molecular origin of these differences is unclear and will require further investigation, e.g. in testing cell lines for potential differences in surface receptors. In a separate study we have already developed a biotinylation-based approach to identify potential novel host cell surface interaction partners during *S. aureus* infection.[19]

Moreover, both inhibitors affected the invasion dynamics (Figure 3D), phagosomal escape (Figure 4C and Figure 4D) and Rab7 recruitment (Figure 4A and Supp. Figure 4A-C) in a similar fashion. Proper inhibition of ASM by both compounds in all cell lines used was validated by enzyme assays (Supp. Figure 2H), which again suggests that the ASM-dependent pathway does only exist in specific cell lines and also supports that we do not observe unspecific side effects of the compounds. We clarified this in the revised manuscript.

ASM is a key player for SM degradation and recycling. In clinical context, deficiency in ASM results in the so-called Niemann Pick disease type A/B. The lipid profile of ASM-deficient cells is massively altered[20], which will result in severe side effects. Short-term inhibition by small molecules therefore poses a clear benefit when compared to the usage of ASM K.O. cells. In order to satisfy the query of the reviewer, we generated two ASM K.O. cell pools (generated with two different sgRNAs) and tested these for *S. aureus* invasion efficiency

(Figure 2, I). We did not observe bacterial invasion differences between WT and K.O. cells. However, when we treated the cells additionally with ASM inhibitor, we observed a strongly reduced invasion in WT cells, while invasion efficiency in ASM K.O. was only slightly affected (Figure 2, J). We concluded that the reduced invasion observed in inhibitor-treated WT cells predominantly is due to absence of ASM, while the small reduction observed in ARC39treated ASM K.O.s is likely due to unspecific side effects.

We performed lipidomics on these cells and demonstrated a strongly altered sphingolipid profile in ASM K.O. cells compared to untreated and inhibitor-treated WT cells (Figure 2, K). We speculate that other ASM-independent bacterial invasion pathways are upregulated in ASM K.O.s., thereby obscuring the effect contributed by absence of ASM. We discussed this in the revised manuscript (line 518 ff).

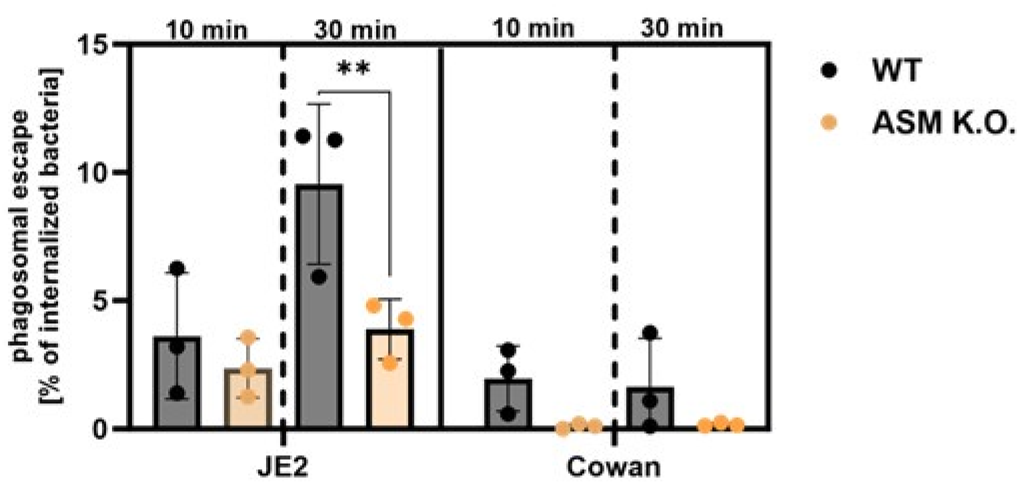
Moreover, we introduced the RFP-CWT escape marker into the ASM K.O. cells and measured phagosomal escape of *S. aureus* JE2 and Cowan I. The latter strain is non-cytotoxic and serves as negative control, since it is known to possess a very low escape rate, due to its inability to produce toxin. Again, we compared early invaders (infection for 10 min) with early^{+}late invaders (infection for 30 min). As observed for JE2, “early invaders” possess lower escape rates than “early^{+}late invaders”.

We did not observe differences between WT and ASM K.O. cells, if we infected for only 10 min. By contrast, we observed a lower escape rate in ASM K.O (Author response image 3, see end of the document). compared to WT cells, when we infected for 30 min.

However, we usually observe an increased phagosomal escape, when we treated host cells with ASM inhibitors (Figure 4C and D). Reduced phagosomal escape of intracellular S. aureus in ASM K.O. cells may be caused by the altered sphingolipid profile(e.g., by interference with binding of bacterial toxins to phagosomal membranes or altered vesicular acidification). We hence think that these data are difficult to interpret, and clarification would require intense additional experimentation. Thus, we did not include this data in the manuscript.

Author response image 3.

Phagosomal escape rates were established in either HeLa wild-type or ASM K.O. cells expressing the phagosomal escape reporter RFP-CWT. Host cells that were infected with the cytotoxic *S. aureus* strain JE2 or the non-cytotoxic strain Cowan I for 10 or 30 minutes and escape rates were determined by microscopy 3h p.i.



As to the treatment with a bacterial sphingomyelinase:

Treatment with the bacterial SMase (bSMase, here: β-toxin) was performed in two different ways:

i) Pretreatment of host cells with β-toxin to remove SM from the host cell surface before infection. This removes the substrate of ASM from the cell surface prior to addition of the bacteria (Figure 2L, Figure 4A-C). Since SM is not present on the extracellular plasma membrane leaflet after treatment, a release of ASM cannot cause localized ceramide formation at the sites of lysosomal exocytosis. Similar observations were made by others.[21]

ii) Addition of bSMase to host cells together with the bacteria to complement for the absence of ASM (Figure 2N).

Removal of the ASM substrate before infection (i) prevents localized ASM-mediated conversion of SM to Cer during infection and resulted in a decreased invasion, while addition of the SMase during infection resulted in an increased invasion in TPC1 and Syt7 ablated cells. Thus, both experiments are consistent with each other and in line with our other observations.

Removal of SM from the plasma membrane by β-toxin was indirectly demonstrated by the absence of Lysenin recruitment to phagosomes/escaped bacteria when host cells were pretreatment with the toxin before infection (Figure5C). We also added another data set that demonstrates degradation of a fluorescence SM derivative upon β-toxin treatment of host cells (Supp Figure 2, M). In another publication, we recently quantified the effectiveness of β-toxin treatment, even though with slightly longer treatment times (75 min vs. 3h).[22]

To clarify our experimental approaches to the readership we added an explanatory section to the revised manuscript (line 287 ff) and we also added a scheme to in Figure 2M describing the experimental settings.

As to the general conclusions regarding the role of ASM: ASM and lysosomal exocytosis has been shown to be involved in uptake of a variety of pathogens[21, 23-27] supporting its role in the process.

> *The use of fluorescent analogs of sphingomyelin and ceramide is not well justified and it is unclear what conclusions can be derived from these observations. Despite the low resolution of the images provided, it appears as if the labeled lipids are largely in endomembrane compartments, where they would presumably be inaccessible to the secreted ASM. Moreover, considering the location of the BODIPY probe, the authors would be unable to distinguish intact sphingomyelin from its breakdown product, ceramide. What can be concluded from these experiments? Incidentally, the authors report only 10% of BODIPY-positive events after 10 min. What are the implications of this finding? That 90% of the invasion events are unrelated to sphingomyelin, ASM, and ceramide?*

During the experiments with fluorescent SM analogues (Figure 3a,b), *S. aureus* was added to the samples immediately before the start of video recording. Hence, bacteria are slowly trickling onto the host cells, and we thus can image the initial contact between them and the bacteria, for instance, the bacteria depicted in Figure 3A contact the host cell about 9 min before becoming BODIPY-FL-positive (see Supp. Video 1, 55 min). Hence, in these cases we see the formation of phagosomes around bacteria rather than bacteria in endomembrane compartments. Since generation of phagosomes happens at the plasma membrane, SM is accessible to secreted ASM.

The “trickling” approach for infection is an experimental difference to our invasion measurements, in which we synchronized the infection by centrifugation. This ensures that all bacteria have contact to host cells and are not just floating in the culture medium. However, live cell imaging of initial bacterialhost contact and synchronization of infection is hard to combine technically.

In our invasion measurements -with synchronization-, we typically see internalization of ~20% of all added bacteria after 30 min. Hence, most bacteria that are visible in our videos likely are still extracellular and only a small proportion was internalized. This explains why only 10% of total bacteria are positive for BODIPY-FL-SM after 10 min. The proportion of internalized bacteria that are positive for BODIPY-FL-SM should be way higher but cannot be determined with this method.

We agree with the reviewer that we cannot observe conversion of BODIPY-FL-SM by ASM. In order to do that, we attempted to visualize the conversion of a visible-range SM FRET probe (Supp. Figure 3), but the structure of the probe is not compatible with measurement of conversion on the plasma membrane, since the FITC fluorophore released into the culture medium by the ASM activity thereby gets lost for imaging. In general, the visualization of SM conversion with subcellular resolution is challenging and even with novel tools developed in our lab[28] visualization of SM on the plasma membrane is difficult.

The conclusions we draw from these experiments are that i.) *S. aureus* invasion is associated with SM and ii.) SM-associated invasion can be very fast, since bacteria are rapidly engulfed by BODIPY-FL-SM containing membranes.

> *It is also unclear how the authors can distinguish lysenin entry into ruptured vacuoles from the entry of RFP-CWT, used as a criterion of bacterial escape. Surely the molecular weights of the probes are not sufficiently different to prevent the latter one from traversing the permeabilized membrane until such time that the bacteria escape from the vacuole.*

We here want to clarify that both Lysenin as well as the CWT reporter have access to ruptured vacuoles (Figure 4B). We used the Lysenin reporter in these experiments for estimation of SM content of phagosomal membranes. If a vacuole is ruptured, both the bacteria and the luminal leaflet of the phagosomal membrane remnants get in contact with the cytosol and hence with the cytosolically expressed reporters YFP-Lysenin as well as RFP-CWT resulting in “Lysenin-positive escape” when phagosomes contained SM (see Figure 5C). By contrast, either β-toxin expression by *S. aureus* or pretreatment with the bSMase resulted in absence of Lysenin recruitment suggesting that the phagosomal SM levels were decreased/undetectable (Figure 5C, Supp Figure 6F, G, I, J).

Although this approach does not enable a quantitative measurement of phagosomal SM, this method is sufficient to show that β-toxin expression and pretreatment result in markedly decreased phagosomal SM levels in the host cells.

The approach we used here to analyze “Lysenin-positive escape” can clearly be distinguished from Lysenin-based methods that were used by others.29 There Lysenin was used to show trans-bilayer movement of SM before rupture of bacteria-containing phagosomes.

To clarify the function of Lysenin in our approach we added additional figures (Figure 4F, Supp. Figure 5) and a movie (Supp. Video 4) to the revised manuscript.

> *Both SMase inhibitors (Figure 4C) and SMase pretreatment increased bacterial escape from the vacuole. The former should prevent SM hydrolysis and formation of ceramide, while the latter treatment should have the exact opposite effects, yet the end result is the*

same. What can one conclude regarding the need and role of the SMase products in the escape process?

As pointed out above, pretreatment of host cells with SMase removes SM from the plasma membrane and hence, ASM does not have access to its substrate. Hence, both treatment with either ASM inhibitors or pretreatment with bacterial SMase prevent ASM from being active on the plasma membrane and hence block the ASM-dependent uptake (Figure 2 G, L). Although overall less bacteria were internalized by host cells under these conditions, the bacteria that invaded host cells did so in an ASM-independent manner.

Since blockage of the ASM-dependent internalization pathway (with ASM inhibitor [Figure 4C, D], SMase pretreatment [Figure 5B] and Vacuolin-1[Figure.4E]) always resulted in enhanced phagosomal escape, we conclude that bacteria that were internalized in an ASM-independent fashion cause enhanced escape. Vice versa, bacteria that enter host cells in an ASM-dependent manner demonstrate lower escape rates.

This is supported by comparing the escape rates of “early” and “late” invaders [Figure 5D, E], which in our opinion is a key experiment that supports this hypothesis. The “early” invaders are predominantly ASM-dependent (see e.g. Figure 3E) and thus, bacteria that entered host cell in the first 10 min of infection should have been internalized predominantly in an ASM-dependent fashion, while slower entry pathways are active later during infection. The early ASM dependent invaders possessed lower escape rates, which is in line with the data obtained with inhibitors (e.g. Figure 4C, D).

We hypothesize that the activity of ASM on the plasma membrane during invasion mediates the recruitment of a specific subset of receptors, which then influences downstream phagosomal maturation and escape. This hypothesis is supported by the fact that the subset of receptors interacting with *S. aureus* is altered upon inhibition of the ASM-dependent uptake pathway. We describe this in another study that is currently under evaluation elsewhere.

Reviewer #2 (Public review):

Summary:

In this manuscript, Ruhling et al propose a rapid uptake pathway that is dependent on lysosomal exocytosis, lysosomal Ca^{2^+} and acid sphingomyelinase, and further suggest that the intracellular trafficking and fate of the pathogen is dictated by the mode of entry.

The evidence provided is solid, methods used are appropriate and results largely support their conclusions, but can be substantiated further as detailed below. The weakness is a reliance on chemical inhibitors that can be non-specific to delineate critical steps.

Specific comments:

A large number of experiments rely on treatment with chemical inhibitors. While this approach is reasonable, many of the inhibitors employed such as amitriptyline and vacuolin1 have other or nondefined cellular targets and pleiotropic effects cannot be ruled out. Given the centrality of ASM for the manuscript, it will be important to replicate some key results with ASM KO cells.

We thank the reviewer for the critical evaluation of our manuscript and plenty of constructive comments.

We agree with the reviewer, that ASM inhibitors such as functional inhibitors of ASM (FIASMA) like amitriptyline used in our study have unspecific side effects given their mode-of-action. FIASMAs induce the detachment of ASM from lysosomal membranes resulting in degradation of the enzyme.[16] However, we want to emphasize that we also used the competitive inhibitor ARC39 in our study[17, 18] which acts on the enzyme by a completely different mechanism. All phenotypes (reduced invasion [Figure 2G], effect on invasion dynamics [Figure 3D], enhanced escape [Figure 4C, D] and differential recruitment of Rab7 [Supp. Figure 4A-C]) were observed with both inhibitors thereby supporting the role of ASM in the process.

We further agree that experiments with genetic evidence usually support and improve scientific findings. However, ASM is a cellular key player for SM degradation and recycling. In a clinical context, deficiency in ASM results in a so-called Niemann Pick disease type A/B. The lipid profile of ASMdeficient cells is massively altered[20], which in itself will result in severe side effects. Thus, the usage of inhibitors provides a clear benefit when compared to ASM K.O. cells, since ASM activity can be targeted in a short-term fashion thereby preventing larger alterations in cellular lipid composition.

We nevertheless generated two ASM K.O. cell pools (generated with two different sgRNAs) and tested for invasion efficiency (Figure 2, I). Here, we did not observe differences between WT and mutants. However, if we treated the cells additionally with ASM inhibitor, we observed a strongly reduced invasion in WT cells, while invasion efficiency in ASM K.O. was only slightly affected (Figure 2, J). We concluded that the reduced invasion observed in WT cells upon inhibitor treatment predominantly is due to inhibition of ASM, whereas the small reduction observed in ARC39-treated ASM K.O.s is likely due to unspecific side effects. We also demonstrated a strongly altered sphingolipid profile in ASM K.O. cells when compared to untreated and inhibitor-treated WT cells (new Figure 2, K). We speculate that other ASM-independent invasion pathways are upregulated in ASM K.O.s., thereby making up for the absence of ASM. We discuss this in the revised manuscript (line 518 ff).

We introduced the RFP-CWT escape marker into the ASM K.O. cells and measured phagosomal escape of *S. aureus* JE2 and Cowan I (Author response image 3). The latter serves as negative control, since it is known to possess a very low escape rate, due to its inability of toxin production. Again, we compared early invaders (infection for 10 min) with early^{+}late invaders (infection for 30 min). As seen before for JE2, early invaders possess lower escape rates than early^{+}late invaders. We did not observe differences between WT and K.O. cells, if we infected for 10 min. By contrast, we observed a lower escape rate in ASM K.O. compared to WT cells, when we infected for 30 min. However, we usually observe an increased phagosomal escape, when we treated host cells with ASM inhibitors (Figure 4C and D). We think that the reduced phagosomal escape in ASM K.O. is caused by the altered sphingolipid profile, which could have versatile effects (e.g., inference with binding of bacterial toxins to phagosomal membranes or changes in acidification). We hence think that these data are difficult to interpret, and clarification would require intense additional experimentation. Thus, we did not include this data in the manuscript.

> *Most experiments are done in HeLa cells. Given the pathway is projected as generic, it will be important to further characterize cell type specificity for the process. Some evidence for a similar mechanism in other cell types S. aureus infects, perhaps phagocytic cell type, might be good.*

Whenever possible we performed the experiments not only in HeLa but also in HuLECs. For example, we refer to experiments concerning the role of Ca^{2+} (Figure 1A/Supp.Figure1A), lysosomal Ca^{2+}/Ned19 (Figure1B/Supp Figure 1C), lysosomal exocytosis/Vacuolin-1 (Figure 2D/Supp. Figure2D), ASM/ARC39 and amitriptyline (Figure 2G), surface SM/β-toxin (Figure 2L/Supp. Figure 2L), analysis of invasion dynamics (complete Figure 3) and measurement of cell death during infection (Figure 6C^{+}E, Supp. Figure 8A^{+}B).

HuLECs, however, are not really genetically amenable and hence we were not able to generate gene deletions in these cells and upon introduction of the fluorescence escape reporter the cells are not readily growing.

As to ASM involvement in phagocytic cells: a role for ASM during the uptake of *S. aureus* by macrophages was previously reported by others.[25] However, in professional phagocytes *S. aureus* does not escape from the phagosome and replicates within the phagosome.[30]

> *I'm a little confused about the role of ASM on the surface. Presumably, it converts SM to ceramide, as the final model suggests. Overexpression of b-toxin results in the near complete absence of SM on phagosomes (having representative images will help appreciate this), but why is phagosomal SM detected at high levels in untreated conditions? If bacteria are engulfed by SM-containing membrane compartments, what role does ASM play on the surface? If surface SM is necessary for phagosomal escape within the cell, do the authors imply that ASM is tuning the surface SM levels to a certain optimal range? Alternatively, can there be additional roles for ASM on the cell surface? Can surface SM levels be visualized (for example, in Figure 4 E, F)?*

We initially hypothesized that we would detect higher phagosomal SM levels upon inhibition of ASM, since our model suggests SM cleavage by ASM on the host cell surface during bacterial cell entry. However, we did not detect any changes in our experiments (Supp. Figure 4F). We currently favor the following explanation: SM is the most abundant sphingolipid in human cells.[31] If peripheral lysosomes are exocytosed and thereby release ASM, only a localized and relative small proportion of SM may get converted to Cer, which most likely is below our detection limit. In addition, the detection of cytosolically exposed phagosomal SM by YFP-Lysenin is not quantitative and provides a “Yes or No” measurement. Hence, we think that the rather limited SM to Cer conversion in combination with the high abundance of SM in cellular membranes does not visibly affect the recruitment of the Lysenin reporter.

In our experiments that employ BODIPY-FL-SM (Figure 3a^{+}b), we cannot distinguish between native SM and downstream metabolites such as Cer. Hence, again we cannot make any assumptions on the extent to which SM is converted on the surface during bacterial internalization. Although our laboratory recently used trifunctional sphingolipid analogs to analyze the SM to Cer conversion[22], the visualization of this process on the plasma membrane is currently still challenging.

Overall, we hypothesize that the localized generation of Cer on the surface by released ASM leads to generation of Cer-enriched platforms. Subsequently, a certain subset of receptors may be recruited to these platforms and influence the uptake process. These platforms are supposed to be very small, which also would explain that we did not detect changes in Lysenin recruitment.

> *Related to that, why is ASM activity on the cell surface important? Its role in non-infectious or other contexts can be discussed.*

ASM release by lysosomal exocytosis is implied in plasma membrane repair upon injury. We added a short description of the role of extracellular ASM in the introduction (line 35).

> *If SM removal is so crucial for uptake, can exocytosis of lysosomes alone provide sufficient ASM for SM removal? How much or to what extent is lysosomal exocytosis enhanced by initial signaling events? Do the authors envisage the early events in their model happening in localized confines of the PM, this can be discussed.*

Ionomycin treatment led to a release of ~10 % of all lysosomes and also increased extracellular ASM activity.[8, 9] In the revised manuscript, we developed an assay to determine lysosomal exocytosis during *S. aureus* infection (Figure 2, A-C). We detected lysosomal exocytosis of ~30% when compared to ionomycin treatment during infection. Since this is only a fraction of the "releasable lysosomes", we assume that the effects (lysosomal Ca^{2+} liberation, lysosomal exocytosis and ASM activity) are very localized and take place only at host-pathogen contact sites (see also above). We discuss this in the revised manuscript (line 563 ff). To our knowledge it is currently unclear to which extent the released ASM affects surface SM levels. We attempted to visualize the local ASM activity on the cell surface by using a visible range FRET probe (Supp. Fig. 3). Cleavage of the probe by ASM on the surface leads to release of FITC into the cell culture medium, which does not contribute a measurable signal at the surface.

> *How are inhibitor doses determined? How efficient is the removal of extracellular bacteria at 10 min? It will be good to substantiate the cfu experiments for infectivity with imaging-based methods. Are the roles of TPC1 and TPC2 redundant? If so, why does silencing TPC1 alone result in a decrease in infectivity? For these and other assays, it would be better to show raw values for infectivity. Please show alterations in lysosomal Ca^{2+} at the doses of inhibitors indicated. Is lysosomal Ca^{2+} released upon S. aureus binding to the cell surface? Will be good to directly visualize this.*

Concerning the inhibitor concentrations, we either used values established in published studies or recommendations of the suppliers (e.g. 2-APB, Ned19, Vacuolin-1). For ASM inhibitors, we determined proper inhibition of ASM by activity assays. Concentrations of ionomycin resulting in Ca^{2+} influx and lysosomal exocytosis was determined in earlier studies of our lab.[9, 32]

As to the removal of bacteria at 10 min p.i.: Lysostaphin is very efficient for removal of extracellular *S. aureus* and sterilizes the tissue culture supernatant. It significantly lyses bacteria within a few minutes, as determined by turbidity assays.[33]

As to imaging-based infectivity assays: We performed imaging-based invasion assays to show reduced invasion efficiency with two ASM inhibitors in the revised manuscript with similar results as obtained by CFU counts (Supp. Figure 2, J).

Regarding the roles of TPC1 and TPC2: from our data we cannot conclude whether the roles of TPC1 and TPC2 are redundant. One could speculate that since blockage of TPC1 alone is sufficient to reduce internalization of bacteria, that both channels may have distinct roles. On the other hand, there might be a Ca^{2+} threshold in order to initiate lysosomal exocytosis that can only be attained if TPC1 and TPC2 are activated in parallel. Thus, our observations are in line with another study that shows reduced Ebola virus infection in absence of either TPC1 or TPC2.[34] In order to address the role of TPC2 for this review process, we kindly were gifted TPCN1/TPCN2 double knock-out HeLa cells by Norbert Klugbauer (Freiburg, Germany), which we tested for *S. aureus* internalization. We found that invasion was reduced in these double KO cell lines even further supporting a role of lysosomal Ca^{2+} release in *S. aureus* host cell entry (Author response image 2, see end of the document). Since we did not have a single TPCN2 knockout available, we decided to exclude these data from the main manuscript.

As to raw CFU counts: whereas the observed effects upon blocking the invasion of *S. aureus* are stable, the number of internalized bacteria varies between individual biological replicates, for instance, by differences in host cell fitness or growth differences in bacterial cultures, which are prepared freshly for each experiment.

With respect to visualization of lysosomal Ca^{2+} release: we agree with the reviewer that direct visual demonstration of lysosomal Ca^{2+} release upon infection would improve the manuscript. We therefore performed live cell imaging to visualize lysosomal Ca^{2+} release by a previously published method.[1] The approach is based on two dextran-coupled fluorophores that were incubated with host cells. The dyes are endocytosed and eventually stain the lysosomes. One of the dyes, Rhod-2, is Ca^{2+}-sensitive and can be used to estimate the lysosomal Ca^{2+} content. The second dye, AF647, is Ca^{2+}-insensitive and is used to visualize the lysosomes. If the ratio Rhod-2/AF647 within the lysosomes is decreasing, lysosomal Ca^{2+} release is indicated. We monitored lysosomal Ca^{2+} content during *S. aureus* infection with this method (Author response image 1 and Author response video 1). However, the lysosomes are very dynamic, and it is challenging to monitor the fluorescence intensities over time. Thus, quantitative measurements are not possible with our methodology, and we decided to not include these data in the final manuscript. However, one could speculate that lysosomal Ca^{2+} content in the selected ROI (Author response image 1 and Author response video 1) is decreased upon attachment of *S. aureus* to the host cells as indicated by a decrease in Rhod-2/AF647 ratio.

> *The precise identification of cytosolic vs phagosomal bacteria is not very easy to appreciate. The methods section indicates how this distinction is made, but how do the authors deal with partial overlaps and ambiguities generally associated with such analyses? Please show respective images.*
>
> *The number of events (individual bacteria) for the live cell imaging data should be clearly mentioned.*

We apologize for not having sufficiently explained the technology to detect escaped *S. aureus*. The cytosolic location of *S. aureus* is indicated by recruitment of RFP-CWT.[35] CWT is the cell wall targeting domain of lysostaphin, which efficiently binds to the pentaglycine cross bridge in the peptidoglycan of *S. aureus*. This reporter is exclusively and homogenously expressed in the host cytosol. Only upon rupture of phagoendosomal membranes, the reporter can be recruited to the cell wall of now cytosolically located bacteria. *S. aureus* mutants, for instance in the agr quorum sensing system, cannot break down the phagosomal membrane in non-professional phagocytes and thus stay unlabeled by the CWT-reporter.[35] We include several images (Figure 4, F, Supp. Figure 5) /movies (Supp. Video 4) of escape events in the revised manuscript. The bacteria numbers for live cell experiments are now shown in Supp. Figure 7.

> *In the phagosome maturation experiments, what is the proportion of bacteria in Rab5 or Rab7 compartments at each time point? Will the decreased Rab7 association be accompanied by increased Rab5? Showing raw values and images will help appreciate such differences. Given the expertise and tools available in live cell imaging, can the authors trace Rab5 and Rab7 positive compartment times for the same bacteria?*

We included the proportion of Rab7-associated bacteria in the revised manuscript (Supp. Figure 4A and C) and also shortly mention these proportions in the text (line 353). Usually, we observe that Rab5 is only transiently (for a few minutes) present on phagosomes and only afterwards the phagosomes become positive for Rab7. We do not think that a decrease in Rab7-positive phagosomes would increase the proportion of Rab5-positive phagosomes. However, we cannot exclude this hypothesis with our data.

We can achieve tracing of individual bacteria for recruitment of Rab5/Rab7 only manually, which impedes a quantitative evaluation. However, we included a Video (Supp. Video 3) that illustrates the consecutive recruitment of the GTPases.

> *The results with longer-term infection are interesting. Live cell imaging suggests that ASM-inhibited cells show accelerated phagosomal escape that reduces by 6 hpi. Where are the bacteria at this time point ? Presumably, they should have reached lysosomes. The relationship between cytosolic escape, replication, and host cell death is interesting,*

> *but the evidence, as presented is correlative for the populations. Given the use of live cell imaging, can the authors show these events in the same cell?*

We think that most bacteria-containing phagoendosomes should have fused with lysosomes 6 h p.i. as we have previously shown by acidification to pH of 5 and LAMP1 decoration.[36]

The correlation between phagosomal escape and replication in the cytosol of non-professional phagocytes has been observed by us and others. In the revised manuscript we also provide images (Supp. Figure 5)/videos (Supp. Video 4) to show this correlation in our experiments.

> *Given the inherent heterogeneity in uptake processes and the use of inhibitors in most experiments, the distinction between ASM-dependent and independent pathways might not be as clear-cut as the authors suggest. Some caution here will be good. Can the authors estimate what fraction of intracellular bacteria are taken up ASM-dependent?*

We agree with the reviewer that an overlap between internalization pathways is likely. A clear distinction is therefore certainly non-trivial. Alternative to ASM-dependent and ASM-independent pathways, the ASM activity may also accelerate one or several internalization pathways. We address this limitation in the discussion of the revised manuscript (line 596 ff).

Early in infection (~10 min after contact with the cells), the proportion of bacteria that enter host cells ASM-dependently is relatively high amounting to roughly 75-80% in HuLEC. After 30 min, this proportion is decreasing to about 50%. We included a paragraph in the discussion of the revised manuscript (line 593 ff).

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) The experiment in Figure 4H is interesting. Details on what proportion of the cell is double positive, and if only this fraction was used for analysis will be good.*

We did use all bacteria found in the images independently from whether host cells were infected with only one or both strains. We unfortunately cannot properly determine the proportion of cells that are double infected, since i) we record the samples with CLSM and hence, cannot exclude that there are intracellular bacteria found in higher or lower optical sections. ii) we visualized cells by staining Nuclei and did not stain the cell borders, thus we cannot precisely tell to which host cell the bacteria localize.

> *(2) Data is sparse for steps 5 and 6 of the model (line 330).*

We apologize for the inconvenience. There is a related study published elsewhere[19], in which we identified NRCAM and PTK7 as putative receptors involved in this invasion pathway. We included a section in the discussion with the corresponding citation (line 569).

> *(3) Data for the reduced number of intracellular bacteria upon blocking ASM-dependent uptake (line 235) is not clear. Do they mean decreased invasion efficiency? These two need not be the same.*

We changed “reduced number of intracellular bacteria” to “invasion efficiency”.

> *(4) b-toxin added to the surface can get endocytosed. Can its surface effect be delineated from endo/phagosomal effect?*

We attempted to delineate effects contributed by the toxin activity on the surface vs. within phagosomes (Figure 5 A-C). We see an increased phagosomal escape, when we pretreated host cells with β-toxin (removal of SM form the surface) and infected either in presence (toxin will be taken up together with the bacteria into the phagosome) or in absence (toxin was washed away shortly before infection) of β-toxin. By contrast, overexpression of β-toxin by *S. aureus* did not affect phagosomal escape rates. The proper activity of β-toxin was confirmed by absence of Lysenin recruitment during phagosomal escape in all three conditions. We concluded that the activity on the surface and not the activity in the phagosome is important.

> *(5) The potential role(s) of bacterial factors in the uptake and subsequent intracellular stages can be discussed.*

There are multiple bacterial adhesins known in *S. aureus*. These usually are either covalently attached to the bacterial cell wall such as the sortase-dependently anchored Fibronectin-binding Proteins A and B but also secreted and “cell wall binding” proteins as well at non proteinaceous factor such as wall-teichoic acids. A discussion of these factors would thus be out of the scope of this manuscript, and we here suggest reverting to specialized reviews on that topic.

> *(6) The manuscript is not very easy to read. The abstract could be rephrased for better clarity and succinctness, with a clearly stated problem statement. The introduction is somewhat haphazard, I feel it can be better structured.*

We apologize for the inconvenience. We stated the problem/research question in the abstract and tried to improve the introduction without adding too much unnecessary detail. In general, we tried to improve the readability of the manuscript and hope that our results and conclusions can be easier understood by the reader in the revised version.

> *(7) Typo in Figure 5F. Step 6 should read "accessory receptors"*

The typo was corrected.

References

(1) Lloyd-Evans, E. et al. Niemann-Pick disease type C1 is a sphingosine storage disease that causes deregulation of lysosomal calcium. Nature Medicine 14, 1247-1255 (2008).

(2) Launay, P. et al. TRPM4 Is a Ca^{2+}-Activated Nonselective Cation Channel Mediating Cell Membrane Depolarization. Cell 109, 397-407 (2002).

(3) Nilius, B. et al. The Ca^{2+}-activated cation channel TRPM4 is regulated by phosphatidylinositol 4,5-biphosphate. The EMBO Journal 25, 467-478-478 (2006).

(4) Cáceres, M. et al. TRPM4 Is a Novel Component of the Adhesome Required for Focal Adhesion Disassembly, Migration and Contractility. PLoS One 10, e0130540 (2015).

(5) Silva, I., Brunett, M., Cáceres, M. & Cerda, O. TRPM4 modulates focal adhesion-associated calcium signals and dynamics. Biophysical Journal 123, 390a (2024).

(6) Schlesier, T., Siegmund, A., Rescher, U. & Heilmann, C. Characterization of the Atl-mediated staphylococcal internalization mechanism. International Journal of Medical Microbiology 310, 151463 (2020).

(7) Jevon, M. et al. Mechanisms of Internalization ofStaphylococcus aureus by Cultured Human Osteoblasts. Infection and Immunity 67, 2677-2681 (1999).

(8) Rodriguez, A., Webster, P., Ortego, J. & Andrews, N.W. Lysosomes behave as Ca^{2+}-regulated exocytic vesicles in fibroblasts and epithelial cells. J Cell Biol 137, 93-104 (1997).

(9) Krones & Rühling et al. Staphylococcus aureus alpha-Toxin Induces Acid Sphingomyelinase Release From a Human Endothelial Cell Line. Front Microbiol 12, 694489 (2021).

(10) Sakurai, Y. et al. Two-pore channels control Ebola virus host cell entry and are drug targets for disease treatment. Science 347, 995-998 (2015).

(11) Aarhus, R., Graeff, R.M., Dickey, D.M., Walseth, T.F. & Lee, H.C. ADP-ribosyl cyclase and CD38 catalyze the synthesis of a calcium-mobilizing metabolite from NADP. J Biol Chem 270, 3032730333 (1995).

(12) Schmid, F., Fliegert, R., Westphal, T., Bauche, A. & Guse, A.H. Nicotinic acid adenine dinucleotide phosphate (NAADP) degradation by alkaline phosphatase. J Biol Chem 287, 32525-32534 (2012).

(13) Angeletti, C. et al. SARM1 is a multi-functional NAD(P)ase with prominent base exchange activity, all regulated bymultiple physiologically relevant NAD metabolites. iScience 25, 103812 (2022).

(14) Gu, F. et al. Dual NADPH oxidases DUOX1 and DUOX2 synthesize NAADP and are necessary for Ca(2^+) signaling during T cell activation. Sci Signal 14, eabe3800 (2021).

(15) Schonn, J.-S., Maximov, A., Lao, Y., Südhof, T.C. & Sørensen, J.B. Synaptotagmin-1 and -7 are functionally overlapping Ca^{2+} sensors for exocytosis in adrenal chromaffin cells. Proceedings of the National Academy of Sciences 105, 3998-4003 (2008).

(16) Kornhuber, J. et al. Functional Inhibitors of Acid Sphingomyelinase (FIASMAs): a novel pharmacological group of drugs with broad clinical applications. Cell Physiol Biochem 26, 9-20 (2010).

(17) Naser, E. et al. Characterization of the small molecule ARC39, a direct and specific inhibitor of acid sphingomyelinase in vitro. J Lipid Res 61, 896-910 (2020).

(18) Roth, A.G. et al. Potent and selective inhibition of acid sphingomyelinase by bisphosphonates. Angew Chem Int Ed Engl 48, 7560-7563 (2009).

(19) Rühling, M., Schmelz, F., Kempf, A., Paprotka, K. & Fraunholz Martin, J. Identification of the Staphylococcus aureus endothelial cell surface interactome by proximity labeling. mBio 0, e03654-03624 (2025).

(20) Schuchman, E.H. & Desnick, R.J. Types A and B Niemann-Pick disease. Mol Genet Metab 120, 27-33 (2017).

(21) Miller, M.E., Adhikary, S., Kolokoltsov, A.A. & Davey, R.A. Ebolavirus Requires Acid Sphingomyelinase Activity and Plasma Membrane Sphingomyelin for Infection. Journal of Virology 86, 7473-7483 (2012).

(22) M. Rühling, L.K., F. Wagner, F. Schumacher, D. Wigger, D. A. Helmerich, T. Pfeuffer, R. Elflein, C. Kappe, M. Sauer, C. Arenz, B. Kleuser, T. Rudel, M. Fraunholz, J. Seibel Trifunctional sphingomyelin derivatives enable nanoscale resolution of sphingomyelin turnover in physiological and infection processes via expansion microscopy. Nat Commun accepted in principle (2024).

(23) Peters, S. et al. Neisseria meningitidis Type IV Pili Trigger Ca(2^+)-Dependent Lysosomal Trafficking of the Acid Sphingomyelinase To Enhance Surface Ceramide Levels. Infect Immun 87 (2019).

(24) Grassmé, H. et al. Acidic sphingomyelinase mediates entry of N. gonorrhoeae into nonphagocytic cells. Cell 91, 605-615 (1997).

(25) Li, C. et al. Regulation of Staphylococcus aureus Infection of Macrophages by CD44, Reactive Oxygen Species, and Acid Sphingomyelinase. Antioxid Redox Signal 28, 916-934 (2018).

(26) Fernandes, M.C. et al. Trypanosoma cruzi subverts the sphingomyelinase-mediated plasma membrane repair pathway for cell invasion. J Exp Med 208, 909-921 (2011).

(27) Luisoni, S. et al. Co-option of Membrane Wounding Enables Virus Penetration into Cells. Cell Host & Microbe 18, 75-85 (2015).

(28) Rühling, M. et al. Trifunctional sphingomyelin derivatives enable nanoscale resolution of sphingomyelin turnover in physiological and infection processes via expansion microscopy. Nature Communications 15, 7456 (2024).

(29) Ellison, C.J., Kukulski, W., Boyle, K.B., Munro, S. & Randow, F. Transbilayer Movement of Sphingomyelin Precedes Catastrophic Breakage of Enterobacteria-Containing Vacuoles. Curr Biol 30, 2974-2983 e2976 (2020).

(30) Moldovan, A. & Fraunholz, M.J. In or out: Phagosomal escape of Staphylococcus aureus. Cell Microbiol 21, e12997 (2019).

(31) Slotte, J.P. Biological functions of sphingomyelins. Progress in Lipid Research 52, 424-437 (2013).

(32) Stelzner, K. et al. Intracellular Staphylococcus aureus Perturbs the Host Cell Ca(2^+) Homeostasis To Promote Cell Death. mBio 11 (2020).

(33) Kunz, T.C. et al. The Expandables: Cracking the Staphylococcal Cell Wall for Expansion Microscopy. Front Cell Infect Microbiol 11, 644750 (2021).

(34) Sakurai, Y. et al. Ebola virus. Two-pore channels control Ebola virus host cell entry and are drug targets for disease treatment. Science 347, 995-998 (2015).

(35) Grosz, M. et al. Cytoplasmic replication of Staphylococcus aureus upon phagosomal escape triggered by phenol-soluble modulin alpha. Cell Microbiol 16, 451-465 (2014).

(36) Giese, B. et al. Staphylococcal alpha-toxin is not sufficient to mediate escape from phagolysosomes in upper-airway epithelial cells. Infect Immun 77, 3611-3625 (2009).

https://doi.org/10.7554/eLife.102810.2.sa0